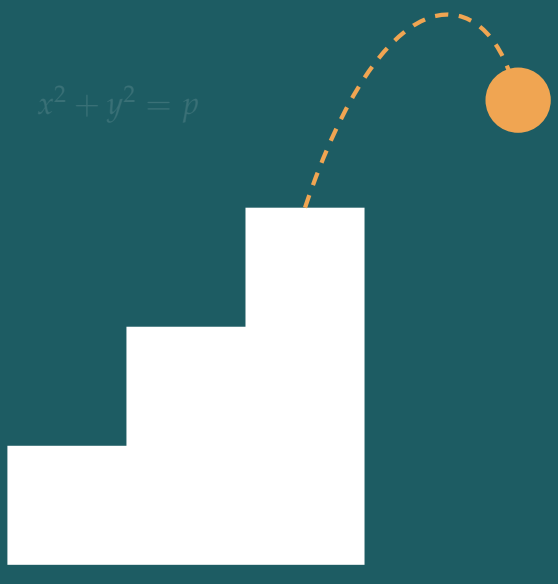
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

1998 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

1998 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

1998 Worked Solutions

Worked solutions to all 42 questions of the 1998 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 1998, Question 1

Pure • Counting Digit Patterns

> **STEP I 1998, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/98-s1-q1

The key idea

Don't try to list numbers. Instead, first choose *which* two digits appear, then count the five-digit strings that use both of them. The only snag is that a number can't start with 0, so we split the count by whether 0 is one of the two digits. And don't forget to check both ends of the range.

Solution

Every integer from 10 000 to 99 999 has exactly five digits, with a non-zero first digit. The one remaining integer in the range is 100 000, which we deal with at the end.

A five-digit number contains exactly two different digits when its digits all come from some pair $\{a, b\}$ and both a and b actually appear. Each such number determines its pair, so we can count pair by pair and add.

Case 1: neither digit is 0

There are $\binom{9}{2} = 36$ ways to choose two different digits from $1, 2, \ldots, 9$. For a fixed pair $\{a, b\}$, each of the five positions can be a or b, giving $2^5 = 32$ strings. Two of these use only one digit ($aaaaa$ and $bbbbb$), so $32 - 2 = 30$ use both. None of them starts with 0, so all 30 are genuine five-digit numbers. This case gives

$$36 \times 30 = 1080.$$

? **Ask yourself** Why can't we treat the pair $\{0, a\}$ in exactly the same way?

Because half of the 32 strings made from 0 and a start with 0, and those aren't five-digit numbers (for example, $0a0aa$ is really a four-digit number). So we have to fix the first digit before counting.

Case 2: one of the digits is 0

The other digit a can be any of $1, \ldots, 9$: 9 choices. The first digit can't be 0, so it must be a. Each of the remaining four positions can be 0 or a, giving $2^4 = 16$ possibilities. Exactly one of these, $aaaa$, contains no 0, which would leave only one distinct digit. So there are $16 - 1 = 15$ numbers for each a, and this case gives

$$9 \times 15 = 135.$$

The endpoint 100 000

The range is inclusive, and 100 000 uses just the digits 1 and 0, so it counts too.

Total

$$1080 + 135 + 1 = 1216.$$

So there are **1216** such integers.

Checks

- A computer check that runs through every integer from 10 000 to 100 000 and counts those whose digits form a set of size 2 gives 1216. It also confirms the split: 1080 without a 0, 135 with one, plus 100 000.
- Try the same method on a smaller case you can list by hand. For two-digit numbers it gives $\binom{9}{2}(2^2 - 2) + 9(2^1 - 1) = 72 + 9 = 81$. That matches the direct count: 90 two-digit numbers, minus the 9 repdigits $11, 22, \ldots, 99$.

Common slips

- Forgetting 100 000, giving 1215. Always look at the endpoints of an "inclusive" range.
- Counting $\binom{10}{2} \times 30 = 1350$, which includes numbers that start with 0.
- Subtracting 2 instead of 1 in Case 2: the all-zero string was already ruled out when we fixed the first digit as a.

Beyond STEP

The trick of subtracting the one-digit strings from all strings over $\{a, b\}$ is the simplest case of the inclusion–exclusion principle. In general, the number of strings of length n that use *all* of k given symbols is $\sum_{j=0}^{k} (-1)^j \binom{k}{j} (k - j)^n$, which is $k!$ times a Stirling number of the second kind. These counts appear throughout combinatorics and probability, for instance in the coupon collector problem. For more of this kind of

counting, try Grimmett and Welsh, *Probability: An Introduction*, or the combinatorics questions in Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 1998, Question 2

Pure • Integration by Substitution

STEP I 1998, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q2

The key idea

The awkward expression $(x^2+1)^{1/2}+x$ sits inside f, so the natural move is to call the whole thing t. The surprise is that this is easy to invert: squaring after isolating the root kills the x^2 terms and leaves x as a simple function of t. Once the first identity is in hand, the second integral is just a special case with $\mathrm{f}(t)=t^{-3}$.

Solution

The change of variable

Put

$$t=(x^2+1)^{1/2}+x.$$

As x runs from 0 to ∞, both $(x^2+1)^{1/2}$ and x are increasing, so t is strictly increasing, starting at $t=1$ when $x=0$ and tending to ∞ as $x\to\infty$. So the substitution is a one-to-one map of $[0,\infty)$ onto $[1,\infty)$, and the new limits are 1 and ∞.

? **Ask yourself** How do we get x back in terms of t?

Isolate the square root and square: $t-x=(x^2+1)^{1/2}$ gives

$$t^2-2tx+x^2=x^2+1 \quad\Longrightarrow\quad x=\frac{t^2-1}{2t}=\frac{1}{2}\left(t-t^{-1}\right),$$

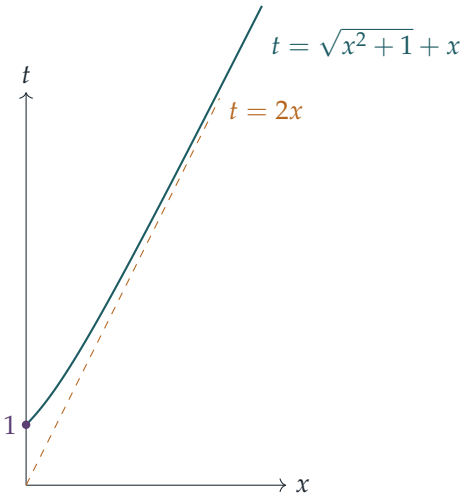
where dividing by t is fine because $t\geq 1$. (Squaring can introduce spurious solutions, but here $t-x=(t+t^{-1})/2>0$, so the root we squared really is positive.) Differentiating,

$$\frac{\mathrm{d}x}{\mathrm{d}t}=\frac{1}{2}\left(1+t^{-2}\right).$$

Therefore

$$\int_0^\infty \mathrm{f}\left((x^2+1)^{1/2}+x\right)\mathrm{d}x=\int_1^\infty \mathrm{f}(t)\cdot\frac{1}{2}\left(1+t^{-2}\right)\mathrm{d}t=\frac{1}{2}\int_1^\infty\left(1+t^{-2}\right)\mathrm{f}(t)\,\mathrm{d}t,$$

which is the required identity.



The sketch shows t rising steadily from 1, so no part of the range is covered twice.

The final part

Take $f(t) = t^{-3}$. The left-hand side of the identity becomes exactly the integral we want, so

$$\begin{aligned}\int_0^\infty \left((x^2+1)^{1/2} + x\right)^{-3} \mathrm{d}x &= \frac{1}{2}\int_1^\infty \left(t^{-3} + t^{-5}\right) \mathrm{d}t \\ &= \frac{1}{2}\left[-\tfrac{1}{2}t^{-2} - \tfrac{1}{4}t^{-4}\right]_1^\infty \\ &= \frac{1}{2}\left(\frac{1}{2} + \frac{1}{4}\right) = \frac{3}{8},\end{aligned}$$

as required. Both improper integrals converge, since t^{-3} and t^{-5} are integrable on $[1, \infty)$.

? **Ask yourself** Is there a cleaner way to see the substitution?

Yes: put $x = \sinh\theta$. Then $(x^2+1)^{1/2} = \cosh\theta$, so $t = \cosh\theta + \sinh\theta = \mathrm{e}^\theta$. Our formula $x = \frac{1}{2}(t - t^{-1})$ is just $\sinh\theta = \frac{1}{2}(\mathrm{e}^\theta - \mathrm{e}^{-\theta})$ in disguise.

Checks

- At $x = 0$, $t = 1$, and $x = \frac{1}{2}(1-1) = 0$: the inverse formula agrees. At $x = \frac{3}{4}$, $t = \frac{5}{4} + \frac{3}{4} = 2$ and $\frac{1}{2}(2 - \frac{1}{2}) = \frac{3}{4}$.
- For large x, $t \approx 2x$, so the integrand in the final part behaves like $(2x)^{-3}$ and the integral certainly converges.

- Numerically, Simpson's rule on the original integral gives 0.375000, matching $\frac{3}{8}$. The identity in the first part also checks numerically for $\mathrm{f}(t) = \mathrm{e}^{-t}$ (both sides 0.258187) and $\mathrm{f}(t) = 1/(1+t^4)$ (both sides 0.188387).

Common slips

- Forgetting to change the limits: the lower limit becomes $t = 1$, not $t = 0$.
- Writing $\mathrm{d}t$ in terms of x and then struggling to eliminate x; it is far easier to express x in terms of t first and differentiate that.

Beyond STEP

The identity is a statement about changing variables under a monotonic map, which in analysis is justified for improper integrals by taking limits of integrals over $[0, X]$. The substitution itself is the inverse of sinh: $\ln\left(x + \sqrt{x^2+1}\right) = \operatorname{arsinh} x$, which is why t turns out to be $\mathrm{e}^{\operatorname{arsinh} x}$. For a careful treatment of improper integrals and substitution, see Bartle and Sherbert, *Introduction to Real Analysis*.

WORKED SOLUTION

STEP I 1998, Question 3

Pure • True or False Proofs

STEP I 1998, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q3

The key idea

Each part needs a different tool, and the first job is to decide which way it goes. A "true" needs a proof that covers every case; a "false" needs just one counterexample, or an argument that no object of the required kind can exist. Taking logarithms settles (i), a single well-chosen value of θ settles (ii), the behaviour of polynomials for large θ settles (iii), and a perfect square settles (iv).

Solution

Part (i): true

Since $a > 0$, we can write $a = \mathrm{e}^{\ln a}$. Then

$$a^{\ln b} = \left(\mathrm{e}^{\ln a}\right)^{\ln b} = \mathrm{e}^{\ln a \ln b},$$
$$b^{\ln a} = \left(\mathrm{e}^{\ln b}\right)^{\ln a} = \mathrm{e}^{\ln b \ln a}.$$

The exponents are equal because multiplication commutes, so $a^{\ln b} = b^{\ln a}$ for all $a, b > 0$. (Equivalently: both sides are positive and have the same logarithm, $\ln a \ln b$.)

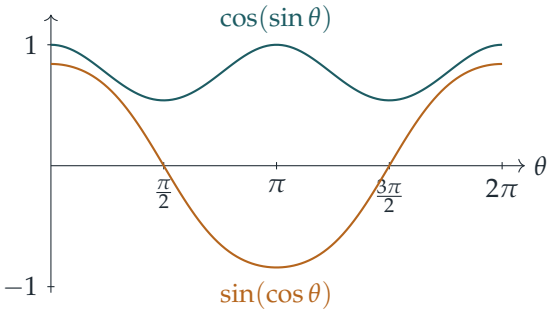
Part (ii): false

One counterexample is enough. Take $\theta = 0$:

$$\cos(\sin 0) = \cos 0 = 1, \qquad \sin(\cos 0) = \sin 1.$$

Since $0 < 1 < \frac{\pi}{2}$, we have $0 < \sin 1 < 1$ (in fact $\sin 1 \approx 0.841$), so the two sides differ and the statement is false.

In fact the two sides are *never* equal; the graph shows $\cos(\sin\theta)$ always above $\sin(\cos\theta)$.



To prove this (not needed for the question), note $\sin(\cos\theta) \leq \sin|\cos\theta| = \cos\left(\frac{\pi}{2} - |\cos\theta|\right)$ and $\cos(\sin\theta) = \cos|\sin\theta|$. Both $|\sin\theta|$ and $\frac{\pi}{2} - |\cos\theta|$ lie in $[0, \frac{\pi}{2}]$, where cosine is decreasing, and $|\sin\theta| < \frac{\pi}{2} - |\cos\theta|$ because $|\sin\theta| + |\cos\theta| \leq \sqrt{2} < \frac{\pi}{2}$. Hence $\cos(\sin\theta) > \sin(\cos\theta)$ for every θ.

Part (iii): false

Suppose P is any polynomial. We show that $|\mathrm{P}(\theta) - \cos\theta| > 10^{-6}$ for some real θ, splitting into two cases.

P *constant*, say $\mathrm{P}(\theta) = c$. Using $\theta = 0$ and $\theta = \pi$, where $\cos\theta$ is 1 and -1, the triangle inequality gives

$$|c-1| + |c+1| \geq |(c+1) - (c-1)| = 2,$$

so at least one of $|c-1|$, $|c+1|$ is at least 1, which is far bigger than 10^{-6}.

P *of degree* $n \geq 1$, say $\mathrm{P}(\theta) = a_n\theta^n + \cdots + a_0$ with $a_n \neq 0$. For $\theta \neq 0$,

$$\mathrm{P}(\theta) = a_n\theta^n\left(1 + \frac{a_{n-1}}{a_n\theta} + \cdots + \frac{a_0}{a_n\theta^n}\right),$$

and the bracket tends to 1 as $\theta \to \infty$, while $|a_n\theta^n| \to \infty$. So $|\mathrm{P}(\theta)| \to \infty$, and we can choose θ with $|\mathrm{P}(\theta)| > 2$. For that θ, since $|\cos\theta| \leq 1$,

$$|\mathrm{P}(\theta) - \cos\theta| \geq |\mathrm{P}(\theta)| - |\cos\theta| > 2 - 1 = 1 > 10^{-6}.$$

In both cases the inequality fails somewhere, so no such polynomial exists.

Ask yourself But the Maclaurin series of $\cos\theta$ is a polynomial-like sum that gets as close as we like. Why doesn't a truncation work?

A truncated Maclaurin series *is* within 10^{-6} of $\cos\theta$ on any fixed bounded interval, provided we take enough terms. But the statement asks for one polynomial that works for *all real* θ, and every non-constant polynomial eventually runs off to $\pm\infty$ while $\cos\theta$ stays trapped in $[-1, 1]$. The phrase "for all real θ" is what makes the statement false.

Part (iv): true

For $x > 0$,

$$x^4 + 3 + x^{-4} - 5 = x^4 - 2 + x^{-4} = \left(x^2 - x^{-2}\right)^2 \geq 0,$$

since a square of a real number is non-negative. Hence $x^4 + 3 + x^{-4} \geq 5$ for all $x > 0$, with equality exactly when $x^2 = x^{-2}$, that is, when $x = 1$. (Alternatively, the AM–GM inequality gives $x^4 + x^{-4} \geq 2\sqrt{x^4 \cdot x^{-4}} = 2$.)

Summary: (i) true, (ii) false, (iii) false, (iv) true.

Checks

- (i): with $a = 2$, $b = 3$, both $2^{\ln 3}$ and $3^{\ln 2}$ equal about 2.141; random positive a, b agreed to about 10^{-15} relative error.
- (ii): sampling θ over a full period, $\cos(\sin\theta) - \sin(\cos\theta)$ never fell below about 0.107.
- (iii): the degree-20 Maclaurin polynomial for cos is accurate at $\theta = 1$, but is out by about 7.5 at $\theta = 10$ and by about 10^{11} at $\theta = 30$.
- (iv): on a fine grid the minimum of $x^4 + 3 + x^{-4}$ is 5, at $x = 1$.

Common slips

- Trying to "prove" a false statement by checking a few values, or "disproving" a true one by an algebra slip: decide which way each part goes before writing.
- In (iii), forgetting the constant polynomials: the "tends to infinity" argument needs degree at least 1.
- In (iii), quoting the Taylor series as a polynomial that works everywhere; it doesn't.

Beyond STEP

Part (iii) contrasts with the Weierstrass approximation theorem: on any *closed bounded* interval, every continuous function can be approximated uniformly by polynomials to any accuracy you like. Uniform approximation on the whole real line is impossible, exactly because of the growth argument above. Abbott's *Understanding Analysis* discusses the Weierstrass theorem and uniform convergence.

WORKED SOLUTION

STEP I 1998, Question 4

Pure • Constrained Optimisation

STEP I 1998, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q4

The key idea

A rectangle inscribed in a circle has its diagonals as diameters, so its sides a and b are tied together by Pythagoras: $a^2 + b^2$ is fixed. Everything then becomes a question about how $a + b$, $a^2 + b^2$ and $a^3 + b^3$ behave when $a^2 + b^2$ is held constant. The case $n = 2$ is the hinge: below it the square wins, at it everything ties, and above it the square comes last.

Solution

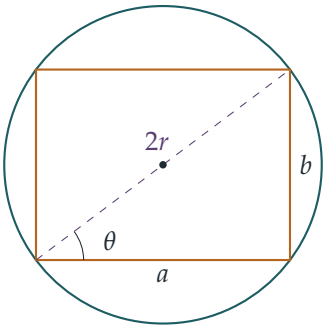
Let the circle have radius r and let the rectangle have sides $a, b > 0$. Each angle of the rectangle is a right angle, so it subtends a diameter (the angle in a semicircle); hence each diagonal is a diameter and

$$a^2 + b^2 = (2r)^2 = 4r^2.$$

Conversely, any $a, b > 0$ with $a^2 + b^2 = 4r^2$ give a rectangle that fits in the circle with its vertices on it. So it is convenient to write

$$a = 2r\cos\theta, \qquad b = 2r\sin\theta, \qquad 0 < \theta < \tfrac{\pi}{2},$$

where θ is the angle between a diagonal and a side of length a.



The perimeter

The perimeter is $P = 2(a+b)$. Since $(a-b)^2 \geq 0$, we have $2ab \leq a^2 + b^2$, so

$$(a+b)^2 = a^2 + b^2 + 2ab \leq 2(a^2+b^2) = 8r^2,$$

with equality if and only if $a = b$. As $a + b > 0$, this gives $a + b \leq 2\sqrt{2}\,r$, that is

$$P \leq 4\sqrt{2}\,r,$$

and the value $4\sqrt{2}\,r$ is attained exactly when $a = b = \sqrt{2}\,r$. So the rectangle of greatest perimeter is the square, as required.

The case $n = 2$

The sum of the squares of the four sides is

$$2(a^2+b^2) = 8r^2,$$

whatever the rectangle. So every inscribed rectangle gives the same value: there is nothing to choose between them, and in particular the square is no better than any other.

The case $n = 3$

Now we look at $S = 2(a^3+b^3)$. Since each side is shorter than the diameter, $0 < a < 2r$ and $0 < b < 2r$, so $a^3 < 2r\,a^2$ and $b^3 < 2r\,b^2$. Adding,

$$S = 2(a^3+b^3) < 2 \cdot 2r(a^2+b^2) = 16r^3.$$

On the other hand, as $\theta \to 0$ (a long thin rectangle, with $a \to 2r$ and $b \to 0$), $S \to 2(2r)^3 = 16r^3$. So S can be made as close to $16r^3$ as we like but never reaches it: *there is no rectangle of greatest S*. The only "shape" achieving $16r^3$ is the degenerate one, a diameter traced twice, which is not a rectangle.

? **Ask yourself** So what has happened to the square?

It has gone from best to worst. With $f(\theta) = \cos^3\theta + \sin^3\theta$, we have $S = 16r^3 f(\theta)$ and

$$f'(\theta) = -3\cos^2\theta\sin\theta + 3\sin^2\theta\cos\theta = 3\sin\theta\cos\theta\,(\sin\theta - \cos\theta).$$

On $0 < \theta < \frac{\pi}{2}$ the factor $\sin\theta\cos\theta$ is positive, and $\sin\theta - \cos\theta$ is negative for $\theta < \frac{\pi}{4}$ and positive for $\theta > \frac{\pi}{4}$. So f decreases and then increases, and the square ($\theta = \frac{\pi}{4}$) gives the *least* value,

$$S_{\min} = 16r^3\left(2 \cdot \frac{1}{2\sqrt{2}}\right) = 8\sqrt{2}\,r^3.$$

In summary, for $n = 3$ the values of S fill the interval $[8\sqrt{2}\,r^3,\ 16r^3)$: the square minimises, and no rectangle maximises.

The same happens for every $n > 2$: $\cos^n\theta + \sin^n\theta < \cos^2\theta + \sin^2\theta = 1$ on $0 < \theta < \frac{\pi}{2}$, because both $\cos\theta$ and $\sin\theta$ lie strictly between 0 and 1, so the supremum $2(2r)^n$ is approached only by flattening the rectangle.

Checks

- With $r = 1.7$ we scanned θ over $(0, \frac{\pi}{2})$ numerically: the perimeter peaks at $\theta = \frac{\pi}{4}$ with value $4\sqrt{2}\,r \approx 9.617$; the sum of squares is constantly $8r^2 = 23.12$; and the sum of cubes falls from near $16r^3 \approx 78.61$ to $8\sqrt{2}\,r^3 \approx 55.58$ at the square and rises again, never reaching 78.61.
- A quick sanity check: a $\sqrt{2} \times \sqrt{2}$ square in a unit circle has perimeter $4\sqrt{2} \approx 5.66$, whereas a 1.9×0.62 rectangle (nearly a diameter) has perimeter about 5.04.

Common slips

- Saying the maximum for $n = 3$ is $16r^3$ "at $\theta = 0$". That is a line segment, not a rectangle, so the maximum does not exist; you must say so and justify the strict inequality.
- Finding the stationary point $\theta = \frac{\pi}{4}$ for $n = 3$ and calling it a maximum without checking its nature.

Beyond STEP

The $n = 3$ case is a first taste of the difference between a *supremum* and a *maximum*: a continuous function on an open interval need not attain its bounds, which is exactly why the extreme value theorem insists on a closed, bounded interval. The switch at $n = 2$ is an instance of the power-mean inequalities, which compare $\left(\frac{a^p + b^p}{2}\right)^{1/p}$ for different p. Abbott's *Understanding Analysis* treats suprema and the extreme value theorem carefully.

WORKED SOLUTION

STEP I 1998, Question 5

Pure • Complex Numbers and Rotation

STEP I 1998, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q5

The key idea

Both parts are about using the algebra of complex numbers to do the work for us. In part (i), going from one vertex of a regular hexagon to the next is a rotation about the centre through $60°$, and a rotation about the origin is just multiplication by a complex number of modulus 1, so we work relative to Q. In part (ii), a number is real exactly when its imaginary part is zero, so we realise each denominator and set the imaginary part to zero.

Solution

Part (i)

Work relative to the centre. The vector from Q to A is

$$A - Q = (10 + 2i) - (4 + 6i) = 6 - 4i.$$

The six vertices of a regular hexagon are equally spaced round a circle centred on Q, so the angle AQB is $360°/6 = 60°$, and QB has the same length as QA. Moving *clockwise* means turning through $-60°$, that is, multiplying by

$$\mathrm{e}^{-i\pi/3} = \cos\tfrac{\pi}{3} - i\sin\tfrac{\pi}{3} = \tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\,i.$$

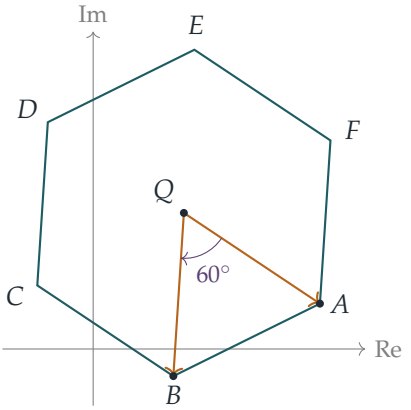
? **Ask yourself** Why multiply $A - Q$ rather than A itself?

Multiplying by $\mathrm{e}^{-i\pi/3}$ rotates about the *origin*. To rotate about Q we shift Q to the origin (subtract Q), rotate, and shift back (add Q). So

$$\begin{aligned} B - Q &= (6 - 4i)\left(\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\,i\right) \\ &= 3 - 3\sqrt{3}\,i - 2i + 2\sqrt{3}\,i^2 \\ &= (3 - 2\sqrt{3}) - (2 + 3\sqrt{3})\,i, \end{aligned}$$

and adding $Q = 4 + 6i$,

$$\boxed{B = (7 - 2\sqrt{3}) + (4 - 3\sqrt{3})\,i} \approx 3.536 - 1.196\,i.$$



The picture confirms the direction: going clockwise from A, which lies down and to the right of Q, the next vertex lies almost directly below Q, as our B does.

Part (ii)

Realise each denominator by multiplying top and bottom by the conjugate, using $(1 + ki)(1 - ki) = 1 + k^2$:

$$\frac{1}{1+i} = \frac{1-i}{2}, \qquad \frac{1}{1+2i} = \frac{1-2i}{5}, \qquad \frac{1}{1+3i} = \frac{1-3i}{10}.$$

Since a, b, c are real, the imaginary part of the whole expression is

$$-\frac{a}{2} - \frac{2b}{5} - \frac{3c}{10} = -\frac{5a + 4b + 3c}{10}.$$

The expression is real exactly when this imaginary part is zero, so the condition is

$$\boxed{5a + 4b + 3c = 0}.$$

(Any non-zero integer multiple, such as $-5a - 4b - 3c = 0$, is the same condition.) In fact this condition is both sufficient and necessary, since we found the imaginary part exactly.

Checks

- In part (i), $|A - Q| = |6 - 4i| = \sqrt{52} = 2\sqrt{13}$. Our B gives $|B - Q|^2 = (3 - 2\sqrt{3})^2 + (2 + 3\sqrt{3})^2 = 21 - 12\sqrt{3} + 31 + 12\sqrt{3} = 52$, so B is on the right circle, and numerically $|B - A| \approx 7.211 = |A - Q|$ too, as it must be for an equilateral triangle QAB.

- In part (ii), try $a = 3$, $b = 0$, $c = -5$ (which satisfies $5a + 4b + 3c = 0$): $\frac{3(1-i)}{2} - \frac{5(1-3i)}{10} = \frac{3}{2} - \frac{1}{2} + \left(-\frac{3}{2} + \frac{3}{2}\right) i = 1$, which is real.
- A numerical check over 10 000 random triples with $c = -(5a + 4b)/3$ gave an imaginary part of zero every time.

Common slips

- Rotating anticlockwise (multiplying by $e^{i\pi/3}$) gives $(7 + 2\sqrt{3}) + (4 + 3\sqrt{3})i$, which is vertex F, not B.
- Rotating A about the origin instead of about Q.
- In part (ii), setting the *real* part to zero, or forgetting that a, b, c must be real for the imaginary part to be read off like this.

Beyond STEP

Multiplication by $e^{i\theta}$ is a rotation, and a map $z \mapsto \alpha z + \beta$ is a rotation and enlargement followed by a translation; these are the simplest examples of the Möbius transformations studied in a first course on complex analysis. The fact that the vertices of a regular polygon centred at the origin are z_0 times the roots of unity is the starting point for a great deal of algebra and geometry. Needham's *Visual Complex Analysis* develops this geometric view of complex multiplication beautifully.

WORKED SOLUTION

STEP I 1998, Question 6

Pure • Trigonometric Recurrence Relations

STEP I 1998, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q6

The key idea

Every step of the recurrence averages something with a cosine, and the half-angle identity $\frac{1}{2}(1+\cos\theta) = \cos^2\frac{\theta}{2}$ turns that average into a perfect square. The square root in the formula for b_{n+1} then removes the square again. Once we notice that each term contains the previous one as a factor, the general pattern is easy to guess, and checking it is an induction step.

Solution

Throughout, $0 < x < \frac{\pi}{2}$, so every angle $x/2^k$ with $k \geq 1$ lies in $(0, \frac{\pi}{4})$ and its cosine is positive. We take the square root in the definition of b_{n+1} to be the positive one.

Finding a_2 and b_2

Using $1+\cos\theta = 2\cos^2\frac{\theta}{2}$,

$$a_2 = \tfrac{1}{2}(\cos x + 1) = \cos^2\frac{x}{2}, \qquad b_2 = (a_2 b_1)^{1/2} = \left(\cos^2\frac{x}{2}\right)^{1/2} = \cos\frac{x}{2},$$

where the last step uses $\cos\frac{x}{2} > 0$.

Finding a_3 and b_3

Take out the common factor $\cos\frac{x}{2}$ and then use the half-angle identity again, this time with $\theta = \frac{x}{2}$:

$$a_3 = \tfrac{1}{2}\left(\cos^2\frac{x}{2} + \cos\frac{x}{2}\right) = \cos\frac{x}{2}\cdot\tfrac{1}{2}\left(1+\cos\frac{x}{2}\right) = \cos\frac{x}{2}\cos^2\frac{x}{4}.$$

Then

$$b_3 = (a_3 b_2)^{1/2} = \left(\cos^2\frac{x}{2}\cos^2\frac{x}{4}\right)^{1/2} = \cos\frac{x}{2}\cos\frac{x}{4},$$

since both cosines are positive. These are the two required expressions.

The general pattern

Comparing the terms so far,

$$b_2 = \cos\frac{x}{2}, \quad b_3 = \cos\frac{x}{2}\cos\frac{x}{4}, \qquad a_2 = b_2\cos\frac{x}{2}, \quad a_3 = b_3\cos\frac{x}{4}.$$

Each b gains one more cosine, with the angle halved, and each a is the matching b times its last cosine once more. So we guess that, for $n \geq 2$,

$$b_n = \cos\frac{x}{2}\cos\frac{x}{4}\cdots\cos\frac{x}{2^{n-1}} = \prod_{k=1}^{n-1}\cos\frac{x}{2^k},$$
$$a_n = b_n\cos\frac{x}{2^{n-1}} = \cos\frac{x}{2}\cos\frac{x}{4}\cdots\cos\frac{x}{2^{n-2}}\,\cos^2\frac{x}{2^{n-1}}.$$

Ask yourself What exactly has to be checked to verify that a guess satisfies a recurrence?

We check two things: the guess matches the starting values, and if the formulae hold for a_n and b_n then the recurrence produces the formulae for a_{n+1} and b_{n+1}. Together these show by induction that the formulae hold for every $n \geq 2$.

Starting values. For $n = 2$ the guess gives $b_2 = \cos\frac{x}{2}$ and $a_2 = \cos^2\frac{x}{2}$, which match what we found above.

Inductive step. Suppose the formulae hold for some $n \geq 2$. Write $\theta = x/2^{n-1}$, so that $x/2^n = \theta/2$. Then

$$a_{n+1} = \tfrac{1}{2}(a_n + b_n) = b_n\cdot\tfrac{1}{2}\left(1+\cos\theta\right) = b_n\cos^2\frac{\theta}{2}.$$

The guess for b_{n+1} is $b_n\cos\frac{\theta}{2}$ (one more cosine, with angle $x/2^n$), so this says $a_{n+1} = b_{n+1}\cos\frac{x}{2^n}$, which is the guessed form of a_{n+1}. Next,

$$b_{n+1} = \left(a_{n+1}b_n\right)^{1/2} = \left(b_n^2\cos^2\frac{\theta}{2}\right)^{1/2} = b_n\cos\frac{x}{2^n} = \prod_{k=1}^{n}\cos\frac{x}{2^k},$$

taking the positive root, which is legitimate because b_n and $\cos\frac{x}{2^n}$ are both positive. That is the guessed form of b_{n+1}, so both equations of the recurrence are satisfied and the formulae hold for all $n \geq 2$.

In fact, with the empty product equal to 1, the formulae also give $b_1 = 1$ and $a_1 = \cos x$, so they hold from $n = 1$.

Checks

- Take $x = \frac{\pi}{3}$. Running the recurrence directly gives $a_3 \approx 0.808013$ and $b_3 \approx 0.836516$, and $\cos\frac{\pi}{6}\cos^2\frac{\pi}{12}$ and $\cos\frac{\pi}{6}\cos\frac{\pi}{12}$ give the same values.
- For 2000 random x in $(0, \frac{\pi}{2})$ we compared the recurrence with the product formulae up to $n = 24$: they agreed to within 10^{-15}.

- As $x \to 0$, every cosine tends to 1, and so do a_1 and b_1. All the terms tend to 1, as they should.

Common slips

- Writing $\sqrt{\cos^2\theta} = \cos\theta$ without saying why $\cos\theta > 0$. Here it holds because every angle lies in $(0, \frac{\pi}{4})$.
- Guessing the pattern from $n = 2$ and 3 and stopping there. The question asks us to verify the recurrence, so the inductive step is needed.

Beyond STEP

Multiplying b_n by $\sin\frac{x}{2^{n-1}}$ and using $2\sin\phi\cos\phi = \sin 2\phi$ repeatedly gives $b_n = \dfrac{\sin x}{2^{n-1}\sin(x/2^{n-1})}$. This tends to $\dfrac{\sin x}{x}$ as $n \to \infty$ (with $x = 1$ the recurrence gives 0.841471 after a few dozen steps). Taking $x = \frac{\pi}{2}$ gives Viète's infinite product for $2/\pi$. This averaging-then-geometric-mean scheme is a cousin of Gauss's arithmetic–geometric mean, and makes a good first example of proving that a sequence converges and finding its limit. For a careful treatment of limits of sequences, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP I 1998, Question 7

Pure • Geometric Series

STEP I 1998, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q7

The key idea

Don't try to track the balance year by year. Follow each deposit separately instead: one payment of £a that sits in the bank for k years of interest grows to £ar^k, and the total is a geometric series. For the withdrawals, turn the same idea round. Each withdrawal is *paid for* by a piece of the starting balance that has been earning interest until it is taken out, so we compare values at a single moment in time.

Solution

The amount after the quiet period

Interest of $\rho\%$ multiplies the balance by $r = 1 + \frac{\rho}{100}$ at the end of each year.

Number the deposit years $1, 2, \ldots, m$. The deposit made at the beginning of year k earns interest at the end of years $k, k+1, \ldots, m$, which is $m - k + 1$ times. So at the end of year m it is worth ar^{m-k+1}. Adding over k, the balance at the end of year m is

$$\begin{aligned} a\left(r^m + r^{m-1} + \cdots + r\right) &= ar\left(1 + r + \cdots + r^{m-1}\right) \\ &= ar\,\frac{r^m - 1}{r - 1}, \end{aligned}$$

by the formula for the sum of a geometric series (we have $r \neq 1$ because $\rho > 0$).

Over the next l years nothing goes in or out, so this is multiplied by r another l times. The total at the end of the period is therefore

$$£a\,\frac{r^{l+1}(r^m - 1)}{r - 1},$$

as required. Call this amount P.

The withdrawals

The first withdrawal comes at the beginning of the year immediately after the quiet period, which is the moment the balance is P. Let B_k be the balance just after the kth withdrawal. Then

$$B_1 = P - b, \qquad B_k = rB_{k-1} - b \quad (k = 2, \ldots, n),$$

because between withdrawals the money earns one year's interest. Unwinding this recurrence,

$$\begin{aligned} B_n &= r^{n-1}P - b\left(r^{n-1} + r^{n-2} + \cdots + r + 1\right) \\ &= r^{n-1}P - b\,\frac{r^n - 1}{r - 1}. \end{aligned}$$

? **Ask yourself** Why is there a factor r^{n-1} on P, and not r^n?

Between the first and the nth withdrawal there are only $n - 1$ year-ends, so the money earns interest $n - 1$ times. Interest paid after the last withdrawal doesn't matter, because the account is then empty. In the same way, the jth withdrawal (counting from the last, with $j = 0$ for the last one) gets multiplied by r^j before the nth withdrawal.

The account is empty after the nth withdrawal, so $B_n = 0$. This gives

$$r^{n-1}P = b\,\frac{r^n - 1}{r - 1}.$$

Substituting for P:

$$a\,\frac{r^{l+n}(r^m - 1)}{r - 1} = b\,\frac{r^n - 1}{r - 1},$$

and so

$$\boxed{\frac{a}{b} = \frac{r^n - 1}{r^{l+n}\,(r^m - 1)}}$$

An equivalent way to see this equation is to take "present values". Measured at the moment of the first withdrawal, the kth withdrawal is worth b/r^{k-1}, and these values must add up to P:

$$P = b\left(1 + r^{-1} + \cdots + r^{-(n-1)}\right).$$

Checks

- If $m = n$ the answer simplifies to $a/b = r^{-(l+n)}$, so $b = ar^{l+n}$. This makes sense. Pair the kth deposit with the kth withdrawal. They are exactly $m + l = l + n$ years apart, so each deposit pays for exactly one withdrawal.
- As $r \to 1$ (no interest), $\dfrac{r^n - 1}{r^m - 1} \to \dfrac{n}{m}$, so $a/b \to n/m$, that is, $ma = nb$. The money in equals the money out.

- For $\rho = 5$, $m = n = 10$, $l = 5$ and $a = 1000$, a year-by-year simulation gives $P \approx 16855.58$. The formula gives $a/b = 1.05^{-15} \approx 0.48102$, so $b \approx 2078.93$, and with this b the simulated account ends at exactly zero. A computer also confirms both formulas exactly, using fractions, on 500 random choices of r, l, m and n.

Common slips

- Getting r^m in place of r^{m+1} in the first sum. Because deposits are made at the *beginning* of each year, even the last deposit earns one year's interest.
- Writing $r^n P$ in place of $r^{n-1}P$. The interest at the end of the nth year comes after the account has been emptied.

Beyond STEP

This is the mathematics of annuities. Comparing payments by discounting them to a common date (present value) is the basic principle of actuarial and financial mathematics. The recurrence $B_k = rB_{k-1} - b$ is a first-order linear difference equation. Its solution, a particular constant plus a multiple of r^k, mirrors exactly the way we solve first-order linear differential equations, a parallel taken up in first-year courses on difference and differential equations. For more problems of this kind, see Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 1998, Question 8

Pure • Second-Order Differential Equations

STEP I 1998, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q8

The key idea

Despite the fluid-dynamics dressing, this is a pure differential equations question. The left-hand side is a derivative of a derivative, so we just integrate twice, remembering that dividing by r in between turns a constant into a logarithm. Physical sense, a finite velocity on the axis, then pins down one constant, and the no-slip condition at the wall pins down the other.

Solution

The general solution

Multiplying the equation by r gives

$$\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}v}{\mathrm{d}r}\right) = -kr.$$

Integrating once with respect to r,

$$r\frac{\mathrm{d}v}{\mathrm{d}r} = -\tfrac{1}{2}kr^2 + A,$$

where A is a constant. We divide by r (fine, since $r > 0$ inside the tube away from the axis):

$$\frac{\mathrm{d}v}{\mathrm{d}r} = -\tfrac{1}{2}kr + \frac{A}{r}.$$

Integrating again, and using $\ln r$ rather than $\ln|r|$ because $r > 0$,

$$\boxed{v = -\tfrac{1}{4}kr^2 + A\ln r + B}$$

with A and B arbitrary constants. Two constants is what we expect for a second-order equation.

Behaviour near the axis

As $r \to 0^+$, the terms $-\frac{1}{4}kr^2$ and B stay bounded (they tend to 0 and B), while $\ln r \to -\infty$. So if $A \neq 0$, then $A \ln r$ tends to $-\infty$ or $+\infty$ (according to the sign of A), and hence $|v| \to \infty$. The only way to avoid this is to take $A = 0$, in which case $v \to B$, which is finite.

Ask yourself Why should we insist that $A = 0$?

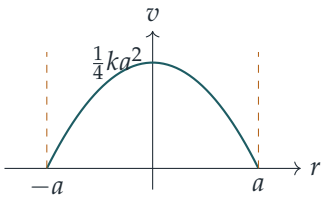
The axis $r = 0$ is inside the fluid, so the speed there must be finite: real fluid doesn't flow infinitely fast along the centre of a pipe. The $\ln r$ solution is a perfectly good solution of the equation for $r > 0$, but it's physically unacceptable here. (It would be allowed for flow between two coaxial cylinders, where the axis isn't part of the fluid.)

The velocity profile

With $A = 0$, we have $v = B - \frac{1}{4}kr^2$. The condition $v = 0$ at $r = a$ gives $B = \frac{1}{4}ka^2$, so

$$\boxed{v = \tfrac{1}{4}k\left(a^2 - r^2\right).}$$

The profile is a parabola: the speed is greatest, $\frac{1}{4}ka^2$, on the axis and falls to zero at the wall.



(The diagram shows a slice across a diameter, with the walls dashed.)

The flow rate

Substituting into the given formula,

$$F = 2\pi \int_0^a r \cdot \tfrac{1}{4}k\left(a^2 - r^2\right) \mathrm{d}r = \frac{\pi k}{2} \int_0^a \left(a^2 r - r^3\right) \mathrm{d}r$$
$$= \frac{\pi k}{2}\left[\frac{a^2 r^2}{2} - \frac{r^4}{4}\right]_0^a = \frac{\pi k}{2} \cdot \frac{a^4}{4},$$

so

$$\boxed{F = \frac{\pi k a^4}{8}.}$$

Checks

- Substitute $v = \frac{1}{4}k(a^2 - r^2)$ back: $r\,v' = -\frac{1}{2}kr^2$, whose derivative is $-kr$, and dividing by r gives $-k$, as required.
- The average speed is $F/(\pi a^2) = \frac{1}{8}ka^2$, exactly half the maximum speed $\frac{1}{4}ka^2$, which is the well-known property of a parabolic profile.
- Numerically, with $k = 2$ and $a = 1$, Simpson's rule gives $F \approx 0.785398$, matching $\pi/4$.

Common slips

- Forgetting the constant at the first integration, which loses the $A \ln r$ term and makes the "unless" part meaningless.
- Integrating $-kr$ to get $-\frac{1}{2}kr^2$ and then forgetting to divide by r before integrating again.
- Dropping the factor r inside the integral for F: the flow through a thin ring of radius r is proportional to its circumference $2\pi r$, not to a constant.

Beyond STEP

The operator $\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}}{\mathrm{d}r}\right)$ is the Laplacian ∇^2 in cylindrical coordinates for a function of r alone, so this question is really Poisson's equation $\nabla^2 v = -k$; the $\ln r$ solution is the two-dimensional analogue of the $1/r$ potential of a point charge. The result $F \propto a^4$ is Poiseuille's law: halving a pipe's radius cuts the flow by a factor of sixteen, which is why narrowed arteries matter so much. For the differential equations side, see Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP I 1998, Question 9

Mechanics • Collisions of Pendulums

STEP I 1998, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q9

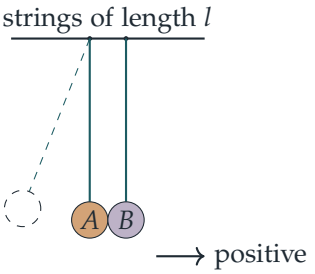
The key idea

Because the displacement is very small, each sphere moves as a simple pendulum in simple harmonic motion, and the period $2\pi\sqrt{l/g}$ doesn't depend on the amplitude or the speed. The two pendulums have the same length, so after any collision at the lowest point both spheres come back to the lowest point at the same moment, half a period later, with their velocities reversed. Once we know where and when the second impact happens, it's two lines of momentum and Newton's experimental law each time.

Solution

The model

We treat A and B as particles of mass m on light inextensible strings of the same length l, ignore air resistance, and take the collisions as instantaneous, with impulses along the horizontal line of centres. Take the positive direction to be the direction A is moving at the first impact (from A towards B), and let $\omega = \sqrt{g/l}$.



Why every collision is at the lowest point

For small oscillations each sphere moves with simple harmonic motion of angular frequency ω, whatever its amplitude. So a sphere that leaves the lowest point with

speed v has horizontal displacement $\frac{v}{\omega}\sin\omega t$, and it is back at the lowest point, moving with velocity $-v$, after a time π/ω (half a period) — for every v.

The first impact is at the lowest point: B hangs there at rest, and A reaches it with speed u. Suppose a collision happens at the lowest point and afterwards A and B have velocities $p \le q$ (they can't move towards each other just after impact). Measured from the lowest point, the gap between the spheres at time t later is

$$\frac{q}{\omega}\sin\omega t - \frac{p}{\omega}\sin\omega t = \frac{q-p}{\omega}\sin\omega t,$$

which is positive for $0 < t < \pi/\omega$ and zero at $t = \pi/\omega$. So they can't meet earlier, and they meet again exactly when both are back at the lowest point. That's the hint: the two pendulums have the same period, so they always return to the lowest point together. By induction, every collision is at the lowest point, and consecutive collisions are half a period apart.

The first impact

Let A and B have velocities v_A and v_B just after the first impact. Conservation of momentum (the string tensions are vertical at the lowest point, so they give no horizontal impulse) and Newton's experimental law give

$$\begin{aligned} mv_A + mv_B &= mu, \\ v_B - v_A &= eu. \end{aligned}$$

Adding and subtracting,

$$v_A = \tfrac{1}{2}u(1-e), \qquad v_B = \tfrac{1}{2}u(1+e).$$

Both are ≥ 0, so both spheres swing out on B's side (if $e = 1$, A stops dead and stays hanging at the lowest point).

Ask yourself What are the velocities just before the second impact?

Half a period later each sphere is back at the lowest point with its velocity reversed: A has velocity $-\frac{1}{2}u(1-e)$ and B has velocity $-\frac{1}{2}u(1+e)$. B is the faster, so it catches A from behind. Its speed of approach relative to A is $\frac{1}{2}u(1+e) - \frac{1}{2}u(1-e) = eu$.

The second impact

Let A and B have velocities w_A and w_B just after the second impact. Conservation of momentum gives

$$mw_A + mw_B = -\tfrac{1}{2}mu(1-e) - \tfrac{1}{2}mu(1+e) = -mu,$$

and Newton's experimental law (the speed of separation is e times the speed of approach, eu) gives

$$w_B - w_A = e \cdot eu = e^2u.$$

Subtracting and adding,

$$w_A = -\tfrac{1}{2}u(1+e^2), \qquad w_B = -\tfrac{1}{2}u(1-e^2).$$

So immediately after the second impact A has speed $\frac{1}{2}u(1+e^2)$, as required, and B has speed $\frac{1}{2}u(1-e^2)$, both moving towards the side from which A was originally released.

Checks

- Perfectly elastic, $e = 1$: A ends with speed u and B is at rest. That's right: after the first impact A stops and B takes speed u, and B returns with speed u and gives it all back to A.
- $e = 0$: both speeds are $\frac{1}{2}u$. The spheres coalesce in effect, moving together with speed $\frac{1}{2}u$ after the first impact, so this is right too.
- Momentum after the second impact is $-\frac{1}{2}mu(1+e^2) - \frac{1}{2}mu(1-e^2) = -mu$, the reverse of the original mu.
- We simulated two real (nonlinear) pendulums with $l = 1\,\text{m}$ and an initial angle of 0.01 rad. The second impact came 1.0001 half-periods after the first, within 3×10^{-6} rad of the lowest point, and with $e = 0.7$ the speeds after it were $0.745u$ and $0.255u$, matching $\frac{1}{2}(1+e^2)$ and $\frac{1}{2}(1-e^2)$.

Common slips

- Forgetting that both velocities are reversed before the second impact, and treating B as hitting a stationary A.
- Using eu in place of $e \cdot (\text{speed of approach})$ at the second impact. Here the approach speed happens to be eu, so the separation speed is e^2u.
- Saying the collisions are at the lowest point "because that is where the spheres hang". The real reason is that the period doesn't depend on the amplitude, and that's why the drawing aside has to be small.

Beyond STEP

Isochronism (a period that doesn't depend on amplitude) holds only approximately for a pendulum: for larger swings the period grows, given exactly by an elliptic integral, and the collisions would drift away from the lowest point. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats both the pendulum and collisions with care.

WORKED SOLUTION

STEP I 1998, Question 10

Mechanics • Projectile Motion

STEP I 1998, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q10

The key idea

We don't know the angle of projection, and we don't want it: the answer is in terms of v, r and t only. So look for a relation that eliminates the angle. The cleanest is to think in vectors: where a particle lands after time t is its initial velocity times t, plus the drop $\frac{1}{2}gt^2$ caused by gravity. These three vectors form a right-angled triangle, and Pythagoras does the rest.

Solution

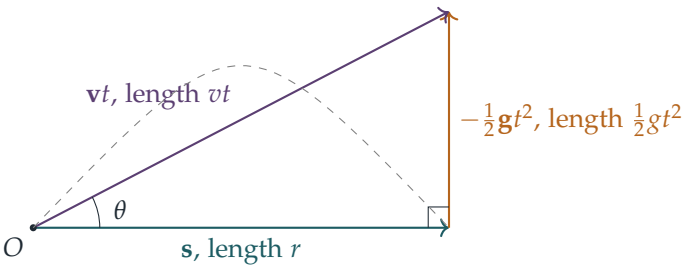
We model each particle of earth as a point particle moving freely under gravity alone (no air resistance), starting from the centre of the explosion O at ground level. A particle that goes straight up lands at O, so we may take $r > 0$ and the projection angle θ above the horizontal with $0 < \theta < \frac{\pi}{2}$.

The landing times

With constant acceleration $\mathbf{g}$ (vertically downwards), the displacement after time t is

$$\mathbf{s} = \mathbf{v}t + \tfrac{1}{2}\mathbf{g}t^2,$$

where $\mathbf{v}$ is the initial velocity, of magnitude v. At the landing time, $\mathbf{s}$ is horizontal with length r. Rearranging, $\mathbf{v}t = \mathbf{s} - \frac{1}{2}\mathbf{g}t^2$, which is the sum of a horizontal vector of length r and a vertical (upward) vector of length $\frac{1}{2}gt^2$.



These two are at right angles, so by Pythagoras

$$v^2t^2 = r^2 + \left(\tfrac{1}{2}gt^2\right)^2 = r^2 + \tfrac{1}{4}g^2t^4.$$

Multiplying by g^2 and writing $u = \frac{1}{2}g^2t^2$ (so that $u^2 = \frac{1}{4}g^4t^4$ and $g^2v^2t^2 = 2v^2u$), this becomes

$$u^2 - 2v^2u + g^2r^2 = 0,$$

a quadratic in u. Its roots are

$$\tfrac{1}{2}g^2t^2 = u = v^2 \pm \sqrt{v^4 - g^2r^2},$$

which is the required result.

Ask yourself Without vectors, how would we get rid of θ?

Use components. Horizontally $r = vt\cos\theta$; vertically, landing at ground level means $0 = vt\sin\theta - \frac{1}{2}gt^2$, so $vt\sin\theta = \frac{1}{2}gt^2$ (as $t \neq 0$). Squaring and adding, using $\cos^2\theta + \sin^2\theta = 1$, gives exactly $v^2t^2 = r^2 + \frac{1}{4}g^2t^4$ again. The vector picture is just this calculation drawn as a triangle.

Both roots are real only when $v^4 \geq g^2r^2$, that is $r \leq v^2/g$: the familiar maximum range. When $r < v^2/g$ both roots are positive (their product is $g^2r^2 > 0$ and their sum $2v^2 > 0$), so there are two landing times, one for a steep shot and one for a shallow one. Since $t = 2v\sin\theta/g$ increases with θ, the $+$ sign belongs to the steeper trajectory, $\theta > \frac{\pi}{4}$, and the $-$ sign to the flatter one.

The greatest height

The flight is symmetric: the vertical velocity $v\sin\theta - gt'$ vanishes at $t' = \frac{1}{2}t$, halfway through the flight. From the top, the particle falls from rest (vertically) for time $\frac{1}{2}t$, so the greatest height is

$$H = \tfrac{1}{2}g\left(\tfrac{1}{2}t\right)^2 = \tfrac{1}{8}gt^2 = \frac{1}{4g}\cdot\tfrac{1}{2}g^2t^2.$$

Using the first part,

$$H = \frac{v^2 \pm \sqrt{v^4 - g^2r^2}}{4g},$$

with the $+$ sign for the steeper trajectory and the $-$ sign for the flatter one (both are possible when $r < v^2/g$; they coincide, at $H = v^2/(4g)$, when $r = v^2/g$).

Checks

- Vertical launch: $r = 0$ gives $\frac{1}{2}g^2t^2 = 2v^2$, so $t = 2v/g$, and $H = v^2/(2g)$, which is right. The other root, $t = 0$, $H = 0$, is the grazing shot along the ground.
- The two heights add up to $v^2/(2g)$. That is as it should be: the two trajectories have complementary angles θ and $\frac{\pi}{2} - \theta$, and $\frac{v^2\sin^2\theta}{2g} + \frac{v^2\cos^2\theta}{2g} = \frac{v^2}{2g}$.
- Numerically, with $g = 9.8$, $v = 20$ and $\theta = 1$ radian, stepping the motion gives $t \approx 3.4346$ and $H \approx 14.450$; the formulae with the $+$ sign give 3.4346 and 14.450. The complementary angle matches the $-$ sign ($t \approx 2.2053$, $H \approx 5.958$).

Common slips

- Giving only one height. Particles with the same speed can reach the same point by two different routes, and the $\pm$ carries through to H.
- Writing H in terms of θ and stopping. The question wants v, r and g only, so the link $H = \frac{1}{8}gt^2$ to the first part is the efficient route.

Beyond STEP

The quadratic in t^2 is the time-domain version of the "envelope of trajectories": the points with $v^4 = g^2r^2$ (and, off the ground, the paraboloid $z = \frac{v^2}{2g} - \frac{gr^2}{2v^2}$) mark the boundary of the region the earth can reach, inside which every point is hit by exactly two trajectories. Envelopes of families of curves are a nice application of partial differentiation in a first-year course. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats projectile motion and vector kinematics at the right level.

WORKED SOLUTION

STEP I 1998, Question 11

Mechanics • Connected Particles Over Pulley

STEP I 1998, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q11

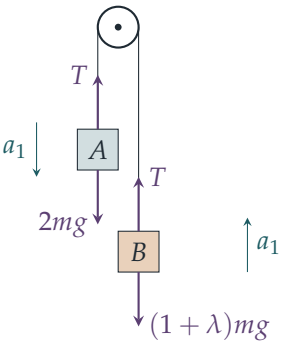
The key idea

Each round trip has two stages, and in each stage the masses are constant, so the acceleration is constant and we can use $s = \frac{1}{2}at^2$. On the way down, the full bucket of water outweighs the ore and bucket, and the pair moves as one Atwood machine. On the way back up, bucket A is empty and massless, so nothing holds B back: it simply falls freely. For the second part, "most ore in a very long time" means "largest ore per cycle divided by time per cycle".

Solution

The time for one cycle

Model: the buckets and their contents are particles, the chain is light and inextensible, and the pulley is small, smooth and light. So the tension T is the same at both ends of the chain, and both buckets move with the same speed. When A is at the top, B is at the bottom (the chain has length l), so each stage moves each bucket a distance l.



Stage 1: A descends. Bucket A carries $2m$ of water; bucket B has total mass $m + \lambda m = (1 + \lambda)m$. Let a_1 be the common acceleration (A down, B up). Newton's second law for

each bucket gives

$$\begin{aligned} A \text{ (downwards)}: &\quad 2mg - T = 2ma_1, \\ B \text{ (upwards)}: &\quad T - (1+\lambda)mg = (1+\lambda)ma_1. \end{aligned}$$

Adding eliminates T:

$$(1-\lambda)mg = (3+\lambda)ma_1 \quad \Longrightarrow \quad a_1 = \frac{(1-\lambda)g}{3+\lambda},$$

which is positive because $\lambda < 1$, so A really does descend. Starting from rest and travelling a distance l with constant acceleration, $l = \frac{1}{2}a_1 t_1^2$, so

$$t_1 = \sqrt{\frac{2l}{a_1}} = \sqrt{\frac{2l}{g}}\sqrt{\frac{3+\lambda}{1-\lambda}}.$$

Ask yourself In stage 2 bucket A has no mass at all. What is the tension in the chain?

Stage 2: B descends. Now A is empty, so its mass is zero, and B has mass m. For the massless bucket A, Newton's law reads $T - 0 \cdot g = 0 \cdot a$, so $T = 0$: a light chain attached to nothing with mass can't pull on anything. So B falls freely with acceleration g from rest through a distance l, and $l = \frac{1}{2}gt_2^2$ gives

$$t_2 = \sqrt{\frac{2l}{g}}.$$

(The same answer comes from the Atwood formula with masses 0 and m: $a = \frac{m-0}{m+0}g = g$.) At the end of this stage A is back at the top for the first time.

Total. The time from releasing A at the top until it first returns to the top is

$$\boxed{t_1 + t_2 = \sqrt{\frac{2l}{g}}\left(1 + \sqrt{\frac{3+\lambda}{1-\lambda}}\right) = \sqrt{\frac{2l}{g}}\,\frac{(1-\lambda)^{1/2} + (3+\lambda)^{1/2}}{(1-\lambda)^{1/2}}}$$

Maximising the ore raised

Write $\tau(\lambda) = t_1 + t_2$ for the cycle time. Each cycle raises ore of mass λm, and the process repeats identically. In a long time S there are $\lfloor S/\tau \rfloor$ complete cycles, so the ore raised is between $\lambda m\,(S/\tau - 1)$ and $\lambda m\, S/\tau$. When S is very large compared with τ, the lost part-cycle is negligible, and the ore raised is essentially

$$\frac{\lambda m S}{\tau(\lambda)} = mS\sqrt{\frac{g}{2l}}\,\frac{\lambda(1-\lambda)^{1/2}}{(1-\lambda)^{1/2} + (3+\lambda)^{1/2}} = mS\sqrt{\frac{g}{2l}}\,\mathrm{f}(\lambda).$$

The factor $mS\sqrt{g/2l}$ doesn't depend on λ, so we must choose $\lambda \in (0,1)$ to make $\mathrm{f}(\lambda)$ as large as possible.

Ask yourself Why must the best λ be a stationary point, and not an end point?

The function f is differentiable on $(0,1)$, since $1-\lambda>0$ and $3+\lambda>0$ there and the denominator is positive. It is positive on $(0,1)$, and it tends to 0 at both ends: as $\lambda\to 0$ the factor λ vanishes (no ore is carried), and as $\lambda\to 1$ the factor $(1-\lambda)^{1/2}$ vanishes (the buckets nearly balance, so stage 1 takes a very long time). So the largest value of f is reached at an interior point, and a maximum of a differentiable function at an interior point has zero derivative. So the best λ satisfies $\mathrm{f}'(\lambda)=0$, as required.

Checks

- Limiting case: as $\lambda\to 1$ the buckets in stage 1 have masses $2m$ and $2m$, so $a_1\to 0$ and $t_1\to\infty$, which matches the formula.
- As $\lambda\to 0$ we get $a_1=g/3$, the usual Atwood value for masses $2m$ and m.
- Numerically, stepping the motion with $g=9.8$ for random λ and l agreed with the formula: for example $\lambda=0.504$, $l=26.9$ gives a cycle of 8.568 s. Maximising f numerically gives $\lambda\approx 0.704$, and counting whole cycles in a long time gives the same best λ.

Common slips

- Forgetting the mass m of bucket B itself, and using λm instead of $(1+\lambda)m$ in stage 1.
- Treating stage 2 as an Atwood machine with some tension: with A massless, B falls freely.
- Maximising ore per cycle, λm, instead of ore per unit time. Bigger loads mean slower cycles.

Beyond STEP

Optimising a rate (output per cycle divided by time per cycle) is the idea behind the *renewal–reward theorem* in probability, where cycle lengths are random and the long-run reward rate is still the mean reward per cycle divided by the mean cycle length. The mechanics here is the constant-acceleration motion that Kleppner and Kolenkow, *An Introduction to Mechanics*, develops into a full treatment of systems of particles.

WORKED SOLUTION

STEP I 1998, Question 12

Probability and Statistics • Counting and Independence

STEP I 1998, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q12

The key idea

Everything here is counting. Once we know how many solutions there are, and how many have X equal to a given value, each probability is a ratio of two counts. For independence, the quickest test is an extreme case: a single pair of values where the joint probability can't possibly equal the product of the marginal probabilities. For the last part, "XYZ divisible by 5" means "at least one of X, Y, Z is divisible by 5", which calls for inclusion–exclusion.

Solution

Counting the solutions

If $X = x$ is fixed (with $0 \le x \le 20$), then Y can be any of $0, 1, \ldots, 20 - x$, and $Z = 20 - x - Y$ is then determined and non-negative. So exactly $21 - x$ solutions have $X = x$, and the total number of solutions is

$$N = \sum_{x=0}^{20} (21 - x) = 21 + 20 + \cdots + 1 = \frac{21 \cdot 22}{2} = 231.$$

(This is the "stars and bars" count $\binom{22}{2}$: arrange 20 stars and 2 bars in a row.) Each solution has probability $1/231$, and

$$\mathrm{P}(X = x) = \frac{21 - x}{231}, \qquad x = 0, 1, \ldots, 20.$$

By symmetry, Y (and Z) has the same distribution.

Are X and Y independent?

No. Take $X = 20$ and $Y = 20$. Each event is possible on its own:

$$\mathrm{P}(X = 20) = \mathrm{P}(Y = 20) = \frac{1}{231} > 0,$$

but they can't happen together, since $X + Y \leq 20$. So

$$\mathrm{P}(X = 20,\ Y = 20) = 0 \neq \frac{1}{231^2} = \mathrm{P}(X = 20)\,\mathrm{P}(Y = 20),$$

and X and Y are not independent. Intuitively, a large value of X leaves less room for Y.

> **Ask yourself** Isn't it enough to say "$X + Y \leq 20$, so they're obviously dependent"?

A constraint like that is the right instinct, but independence is a precise statement about probabilities, so the justification should exhibit one pair of values where $\mathrm{P}(X = x, Y = y) \neq \mathrm{P}(X = x)\mathrm{P}(Y = y)$. A single counterexample is all it takes, and the extreme pair $(20, 20)$ makes the arithmetic trivial.

X divisible by 5

The multiples of 5 in range are $X = 0, 5, 10, 15, 20$, with $21, 16, 11, 6, 1$ solutions respectively. So

$$\mathrm{P}(5 \mid X) = \frac{21 + 16 + 11 + 6 + 1}{231} = \frac{55}{231} = \frac{5}{21},$$

as required.

XYZ divisible by 5

Since 5 is prime, $5 \mid XYZ$ exactly when 5 divides at least one of X, Y, Z. (Remember that 0 counts as divisible by 5.) Let A, B, C be the sets of solutions in which X, Y, Z respectively are divisible by 5. From the previous part, $|A| = |B| = |C| = 55$.

For $A \cap B$: write $X = 5a$, $Y = 5b$ with $a, b \geq 0$ and $a + b \leq 4$. The number of such pairs is $5 + 4 + 3 + 2 + 1 = 15$. In every one of them $Z = 20 - 5a - 5b$ is also a multiple of 5, so

$$|A \cap B| = |B \cap C| = |C \cap A| = |A \cap B \cap C| = 15.$$

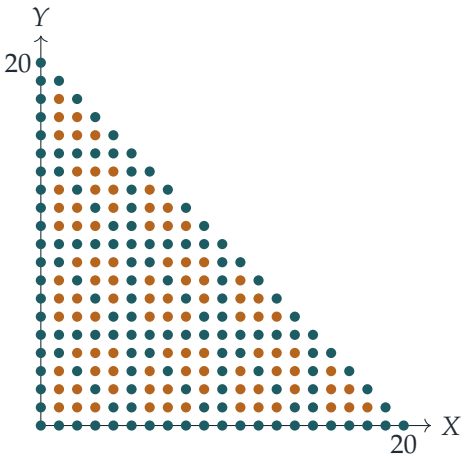
By inclusion–exclusion,

$$\begin{aligned} |A \cup B \cup C| &= 3 \cdot 55 - 3 \cdot 15 + 15 \\ &= 135, \end{aligned}$$

and therefore

$$\mathrm{P}(5 \mid XYZ) = \frac{135}{231} = \frac{45}{77}.$$

The picture shows the solutions as lattice points (X, Y) with $X + Y \leq 20$. The teal points lie on one of the lines $X = 5k$, $Y = 5k$ or $X + Y = 20 - 5k$ (that is, $Z = 5k$), and the amber points are the 96 solutions where none of X, Y, Z is a multiple of 5.



Checks

- The complement should have $231 - 135 = 96$ solutions in which none of X, Y, Z is a multiple of 5: the amber points in the picture. Listing all 231 solutions by computer, with exact fractions, confirms 96, $|A \cap B| = 15$, $5/21$ and $135/231 = 45/77 \approx 0.584$.
- A sanity check on $5/21$: X divisible by 5 is a bit more likely than $1/5$, because small values of X are the most likely ones and $X = 0$ is the most likely of all.

Common slips

- Forgetting that 0 is divisible by 5. Leaving out $X = 0$ gives $34/231$, not $5/21$.
- Adding $3 \times \frac{5}{21}$, which counts the overlaps several times. Here the overlaps are large: whenever two of the variables are multiples of 5, so is the third.
- Arguing for dependence without giving a specific pair of values where the product rule fails.

Beyond STEP

Choosing a solution uniformly at random is choosing a random *composition* of 20 into three parts. If 20 is replaced by a total n and n grows, $(X/n, Y/n)$ becomes uniformly distributed on a triangle, the simplest case of the Dirichlet distribution. Inclusion–exclusion is the standard tool for "at least one" events, and it appears again in derangements and in Euler's totient function. Grimmett and Welsh, *Probability: An Introduction*, covers both the counting and the independence ideas.

WORKED SOLUTION

STEP I 1998, Question 13

Probability and Statistics • Sampling With Replacement

STEP I 1998, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q13

The key idea

This is a conditional probability: the event we condition on is "exactly one pastille eaten in n draws", and we split it up according to *when* that pastille was eaten. Each piece is a product of simple probabilities, because the bag's contents only change at the moment a red is eaten. Adding the pieces is a geometric series, and that's where the difference of two nth powers comes from.

Solution

Write $N = r + b$ for the number of pastilles at the start. We assume $r \geq 1$ and $b \geq 1$ (if $b = 0$ and $n \geq 2$, eating exactly one pastille is impossible and there is nothing to condition on).

The bag at each draw. A non-red pastille goes back, so the bag changes only when a red is eaten. Before the red is eaten the bag holds N pastilles, b of them non-red; afterwards it holds $N - 1$ pastilles, still b of them non-red. Because the bag is shaken thoroughly before each draw, each draw is a fresh uniform choice from whatever is in the bag at that moment, so we can multiply the probabilities draw by draw.

Splitting by the time of eating. For $k = 1, 2, \ldots, n$ let E_k be the event that the only pastille eaten in the n draws was eaten on draw k. Then draws $1, \ldots, k-1$ are non-red from the full bag, draw k is red from the full bag, and draws $k+1, \ldots, n$ are non-red from the reduced bag:

$$\mathrm{P}(E_k) = \left(\frac{b}{N}\right)^{k-1} \frac{r}{N} \left(\frac{b}{N-1}\right)^{n-k} = \frac{r\, b^{n-1}}{N^k\, (N-1)^{n-k}}.$$

The events $E_1, \ldots, E_n$ are disjoint, and their union is the event E that exactly one pastille was eaten. So

$$\mathrm{P}(E) = r\,b^{n-1} \sum_{k=1}^{n} \frac{1}{N^k (N-1)^{n-k}} = \frac{r\,b^{n-1}}{N^n (N-1)^n} \sum_{k=1}^{n} N^{n-k}(N-1)^k,$$

where we multiplied top and bottom of each term by $N^n(N-1)^n$.

Ask yourself How do we add up $\sum_{k=1}^{n} N^{n-k}(N-1)^k$?

It is a geometric series with first term $N^{n-1}(N-1)$ and common ratio $\frac{N-1}{N}$. Even quicker, it is the familiar factorisation

$$x^n - y^n = (x-y)\left(x^{n-1} + x^{n-2}y + \cdots + y^{n-1}\right)$$

with $x = N$, $y = N-1$, so that $x - y = 1$:

$$\sum_{k=1}^{n} N^{n-k}(N-1)^k = (N-1) \sum_{j=0}^{n-1} N^{n-1-j}(N-1)^j = (N-1)\left(N^n - (N-1)^n\right).$$

Hence

$$\mathrm{P}(E) = \frac{r\,b^{n-1}\left(N^n - (N-1)^n\right)}{N^n (N-1)^{n-1}}.$$

The conditional probability. The event "I ate it on the last draw, and ate only one" is E_n, with $\mathrm{P}(E_n) = \dfrac{r\,b^{n-1}}{N^n}$. Since $E_n \subseteq E$,

$$\begin{aligned}\mathrm{P}(E_n \mid E) = \frac{\mathrm{P}(E_n)}{\mathrm{P}(E)} &= \frac{r\,b^{n-1}}{N^n} \cdot \frac{N^n (N-1)^{n-1}}{r\,b^{n-1}\left(N^n - (N-1)^n\right)} \\ &= \frac{(N-1)^{n-1}}{N^n - (N-1)^n} = \frac{(r+b-1)^{n-1}}{(r+b)^n - (r+b-1)^n},\end{aligned}$$

which is the required result.

Notice that r and b cancel completely, apart from their total. In fact the working shows more: given E, the probability that the pastille was eaten on draw k is proportional to $\left(\frac{N-1}{N}\right)^k$, so earlier draws are *more* likely than later ones. That makes sense: eating early leaves a bag in which non-reds are a larger fraction, which makes the remaining run of "no more reds" more probable.

Checks

- $n = 1$: the formula gives $\dfrac{1}{N - (N-1)} = 1$, correct, since the only draw is the last one.

- $n = 2$: it gives $\dfrac{N-1}{2N-1}$, which is a little less than $\frac{1}{2}$, consistent with earlier draws being slightly more likely.
- With $r = 3$, $b = 5$, $n = 4$ the formula gives $343/1695 \approx 0.2024$, against $\frac{1}{4}$ if all four draws were equally likely. A simulation of 4×10^5 runs gave 0.2028, and exact enumeration of every draw sequence, tracking the bag, reproduced the formula exactly for all $1 \leq r, b \leq 4$ and $1 \leq n \leq 8$.

Common slips

- Using b/N for the non-red draws *after* the red has been eaten. The bag now has $N-1$ pastilles, and this is the whole point of the question: with b/N throughout every E_k would have the same probability and the answer would be $1/n$.
- Forgetting to condition: $\mathrm{P}(E_n) = rb^{n-1}/N^n$ on its own is not the answer, because we are told that exactly one pastille was eaten.

Beyond STEP

The bag here is a simple Markov chain: its state is the number of reds left, and the chance of moving down depends only on the current state. Conditioning on the end state of such a chain and asking about the path it took is the idea behind "hidden Markov models" and Bayesian inference about histories. Grimmett and Welsh, *Probability: An Introduction*, covers conditional probability and Markov chains at exactly this level, and Blitzstein and Hwang, *Introduction to Probability*, has many problems in the same spirit.

WORKED SOLUTION

STEP I 1998, Question 14

Probability and Statistics • Geometric Probability

STEP I 1998, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s1-q14

The key idea

Everything about where the coin lands is captured by where its *centre* lands. By the regular pattern of the board, we may take the centre to be uniformly distributed over a single 2×2 square, and then each outcome corresponds to a region of that square. Turning "crosses a line" and "covers an intersection" into conditions on the centre's distances from the lines and corners turns the probability into an area, which is a quadratic in a that we can maximise.

Solution

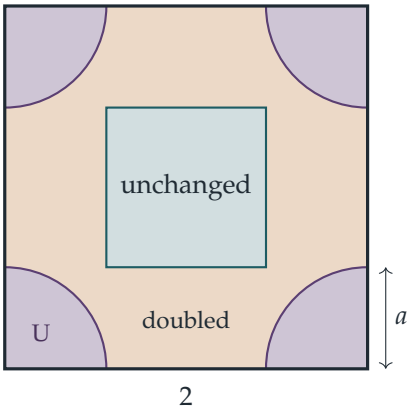
The probability that the tax is doubled

Model the coin's centre as landing uniformly at random in one square of the grid; the board is large, so edge effects are negligible, and every square looks the same. Put coordinates on this square so that it is $[0,2] \times [0,2]$, with the grid lines along its edges.

Let U be the distance from the centre to the nearest vertical line and V the distance to the nearest horizontal line. Since the x-coordinate is uniform on $[0,2]$, $U = \min(x, 2-x)$ is uniform on $[0,1]$; similarly V is uniform on $[0,1]$, and U, V are independent because x and y are. So (U,V) is uniform on the unit square $[0,1]^2$, and probabilities are just areas in that square.

- The coin crosses no line exactly when $U > a$ and $V > a$. This has probability $(1-a)^2$.
- The coin covers an intersection exactly when the nearest corner of the square is within a of the centre, that is, when $U^2 + V^2 < a^2$. Since $a < 1$, this quarter-disc lies wholly inside $[0,1]^2$, so the probability is $\frac{1}{4}\pi a^2$.

These two events are disjoint: if the coin covers an intersection, it certainly crosses the lines through it. (Boundary cases such as the coin just touching a line have probability zero, so it doesn't matter whether we use $<$ or $\leq$.)



The figure shows the possible positions of the centre in one square: teal (no line crossed) and plum (an intersection covered, marked U at one corner) leave the tax unchanged; amber doubles it.

So the probability that the tax is doubled is

$$\begin{aligned} p(a) &= 1-(1-a)^2-\tfrac{1}{4}\pi a^2 \\ &= 2a-\left(1+\frac{\pi}{4}\right)a^2. \end{aligned}$$

Maximising the tax

To raise most tax we want the doubling to be as likely as possible, so we maximise $p(a)$ over $0<a<1$. It is a quadratic with negative leading coefficient, and

$$p'(a)=2-2\left(1+\frac{\pi}{4}\right)a=0 \iff a=\left(1+\frac{\pi}{4}\right)^{-1}.$$

This value is $\frac{4}{4+\pi}\approx 0.560$, which does lie in $(0,1)$, and since $p''(a)<0$ it gives the maximum, as required. Writing a^* for this value, we have $(1+\frac{\pi}{4})a^{*2}=a^*$, so the maximum value is

$$p_{\max}=2a^*-a^*=a^*=\frac{4}{4+\pi}.$$

The expected tax after n years

? **Ask yourself** Why is the expected growth factor each year $1+p$ and not $2p$?

In a year when the tax is not doubled it stays as it was, so it is multiplied by 1, not by 0. Let M_k be the factor for year k: $M_k=2$ with probability p and $M_k=1$ with probability $1-p$. Then

$$\mathrm{E}(M_k)=2p+(1-p)=1+p=1+\frac{4}{4+\pi}=\frac{8+\pi}{4+\pi}.$$

Assume each year's throw is independent of the others. The tax after n years is $T_n=M_1M_2\cdots M_n$ Slihings per yak (starting from 1), and the expectation of a product

of independent random variables is the product of their expectations, so

$$\mathrm{E}(T_n) = \mathrm{E}(M_1)\,\mathrm{E}(M_2)\cdots\mathrm{E}(M_n) = \left(\frac{8+\pi}{4+\pi}\right)^n,$$

as required.

Checks

- At $a \to 0$ the coin almost never touches a line, and at $a = 1$ it always crosses a line, so $p(1) = 1 - \frac{\pi}{4} \approx 0.215$, the chance of avoiding the corner discs. Both agree with the formula, and the maximum lies between them.
- A simulation with 400 000 random centres gave $p \approx 0.5620$ at $a = 0.56$, against the formula's 0.5601. A fine search over a puts the maximum at $a \approx 0.5601 = 4/(4+\pi)$.
- Simulating the yearly doubling with $p = 4/(4+\pi)$ gave a mean tax after 3 years of about 3.805, against $\left(\frac{8+\pi}{4+\pi}\right)^3 \approx 3.797$.

Common slips

- Subtracting the "covers an intersection" area from the "no line" area, or counting it twice. The two events are disjoint, and both are removed from 1.
- Using the full disc πa^2 at each corner and forgetting that only a quarter of it lies in the square (or, equivalently, that each corner disc is shared by four squares).
- Writing the expected tax as $(2p)^n$, forgetting that an undoubled year leaves the tax unchanged.

Beyond STEP

This is a cousin of Buffon's needle problem: a randomly placed object is reduced to a random point in a fundamental region, and probabilities become areas. That idea is the start of *geometric probability* and integral geometry, which gives surprising results such as estimating π by throwing needles. The final part uses the fact that expectations of independent random variables multiply, which is the basis of multiplicative growth models in finance and population biology. Grimmett and Welsh, *Probability: An Introduction*, covers both ideas.

WORKED SOLUTION

STEP II 1998, Question 1

Pure • Diophantine Equations and Parity

STEP II 1998, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q1

The key idea

Expanding the cubes makes most of the terms cancel, leaving a cubic in n with a small constant term. Comparing parities on the two sides forces n to be even. Writing the cubic as $n^2(\text{something}) = 54$ then turns the question into one about divisors. For $(**)$, the even substitution $n = 2m$ takes us straight back to $(*)$.

Solution

The equation $(*)$

Using $(a+b)^3 - (a-b)^3 = 6a^2b + 2b^3$ with $a = n$, $b = 3$, we get

$$(n+3)^3 - (n-3)^3 = 18n^2 + 54,$$

so $(*)$ is equivalent to

$$n^3 = 18n^2 + 54, \qquad \text{that is,} \qquad n^2(n-18) = 54. \tag{1}$$

n is even. The right-hand side $18n^2 + 54 = 2(9n^2 + 27)$ is even for every integer n. If n were odd, then n^3 would be odd, contradicting $n^3 = 18n^2 + 54$. Hence n is even.

n^2 divides 54. In the second form of (1), $n - 18$ is an integer, so 54 is n^2 times an integer. Hence n^2 is a factor of 54.

Ask yourself How do these two facts together rule out every integer?

Since n is even, write $n = 2k$. Then $n^2 = 4k^2$, which is a multiple of 4 (and $n \neq 0$, since $n = 0$ gives $0 = 54$ in (1)). But $54 = 2 \times 27$ is not divisible by 4, so no multiple of 4 other than 0 can divide it. This contradiction shows that no integer n satisfies $(*)$.

(Another way to see it: the only squares dividing $54 = 2 \cdot 3^3$ are 1 and 9, so $n \in \{\pm 1, \pm 3\}$, all of which are odd.)

The equation $(**)$

Now take $b = 6$:

$$(n+6)^3 - (n-6)^3 = 36n^2 + 432,$$

so $(**)$ is equivalent to

$$n^3 = 36n^2 + 432.$$

The right-hand side is even for every integer n, so, exactly as before, n^3 is even. Since the cube of an odd number is odd, n is even.

? **Ask yourself** Why can't we just copy the divisor argument from the first part?

We could write $n^2(n-36) = 432$, but $432 = 2^4 \cdot 3^3$ has plenty of even square factors (4, 16, 36, 144), so a divisibility argument alone doesn't finish. Instead we use the evenness. Put $n = 2m$ with m an integer. Then

$$8m^3 = 36 \cdot 4m^2 + 432 \quad \Longleftrightarrow \quad m^3 = 18m^2 + 54,$$

which is precisely equation (1) for the integer m. In other words, m satisfies $(*)$. Indeed, $(**)$ with $n = 2m$ is just $(*)$ with $n = m$, multiplied through by 8:

$$(2m-6)^3 + (2m)^3 - (2m+6)^3 = 8\big[(m-3)^3 + m^3 - (m+3)^3\big].$$

We showed that no integer satisfies $(*)$, so no such m exists. Hence no integer n satisfies $(**)$.

Checks

- Try $n = 1$ in the expansion: $(4)^3 - (-2)^3 = 64 + 8 = 72 = 18 + 54$. ✓
- The cubic $n^3 - 18n^2 - 54$ has exactly one real root, at about $n \approx 18.164$, which is not an integer. (A direct computer search over $|n| < 10^5$ finds no solution to either equation.)
- For $(**)$, the real solution is $n \approx 2 \times 18.164 = 36.33$, which matches the scaling $n = 2m$.

Common slips

- Forgetting that $n = 0$ needs a separate word: 0 is even, but 0 is not a factor of 54. Here (1) rules it out at once.
- Expanding the cubes in full and making an arithmetic error. The identity $(a+b)^3 - (a-b)^3 = 6a^2b + 2b^3$ halves the work.
- In $(**)$, trying to show $n^2 \mid 432$ leads nowhere. The point of "n is even" is the substitution $n = 2m$.

Beyond STEP

The final step, where a solution of $(**)$ produces a solution of an equation of the same shape with smaller numbers, is a simple form of Fermat's *method of descent*. Applied repeatedly, this method shows that $x^3 + y^3 = z^3$ has no solutions in positive integers, the cubic case of Fermat's Last Theorem first proved by Euler. Here the three terms form an arithmetic progression, which makes things much easier. In fact, $(n - 3k)^3 + n^3 = (n + 3k)^3$ says that $t = n/k$ is a root of $t^3 - 18t^2 - 54$, and by the rational root theorem this cubic has no rational roots, so there are no integer solutions for any $k \neq 0$. Davenport's *The Higher Arithmetic* is a lovely introduction to divisibility, parity arguments and Diophantine equations.

WORKED SOLUTION

STEP II 1998, Question 2

Pure • Binomial Series Approximation

STEP II 1998, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q2

The key idea

A binomial series converges fast when its argument is small, so we want to write $\sqrt{2}$ (or $\sqrt[3]{2}$) as a simple rational number times $(1+x)^a$ with x tiny. For square roots, $49/50$ is a near miss for 1 that contains a factor of 2; for cube roots we look for a cube that is almost $2\times$ another cube, and $128 = 2\times 4^3$ sits right next to $125 = 5^3$.

Solution

The square root of 2

First we connect the given expression to $\sqrt{2}$. Since $1-\frac{1}{50} = \frac{49}{50}$,

$$\left(1-\frac{1}{50}\right)^{1/2} = \frac{7}{\sqrt{50}} = \frac{7}{5\sqrt{2}} = \frac{7\sqrt{2}}{10}, \qquad \text{so} \qquad \sqrt{2} = \frac{10}{7}\left(1-\frac{1}{50}\right)^{1/2}.$$

The binomial series for $(1+x)^{1/2}$ begins

$$(1+x)^{1/2} = 1 + \tfrac{1}{2}x - \tfrac{1}{8}x^2 + \tfrac{1}{16}x^3 - \cdots,$$

valid for $|x|<1$. Put $x = -\frac{2}{100} = -2y$ with $y = \frac{1}{100}$. The powers of 2 cancel nicely:

$$\begin{aligned}\left(1-\frac{2}{100}\right)^{1/2} &\approx 1 - \tfrac{1}{2}(2y) - \tfrac{1}{8}(2y)^2 - \tfrac{1}{16}(2y)^3 \\ &= 1 - y - \tfrac{1}{2}y^2 - \tfrac{1}{2}y^3 \\ &= 1 - 0.01 - 0.00005 - 0.0000005 \\ &= 0.9899495.\end{aligned}$$

This is why writing $1/50$ as $2/100$ helps: every term becomes a power of 10 times 1 or $\frac{1}{2}$, so the arithmetic is exact decimals.

Hence

$$\sqrt{2} \approx \frac{10\times 0.9899495}{7} = \frac{9.899495}{7} = 1.41421357\ldots,$$

and to six decimal places $\sqrt{2} \approx 1.414214$, as required.

The cube root of 2

Ask yourself Which N and a? What makes $(1+N/125)^a$ related to $\sqrt[3]{2}$?

Since $125 = 5^3$, a cube root of $(125+N)/125$ will be $\sqrt[3]{125+N}/5$. We therefore want $125+N$ to be 2 times a perfect cube, with N small. The cubes 2×1^3, 2×2^3, 2×3^3, $2\times 4^3 = 2, 16, 54, 128$ show that 128 is only 3 away from 125. So take $N=3$ and $a=\frac{1}{3}$:

$$\left(1+\frac{3}{125}\right)^{1/3} = \frac{\sqrt[3]{128}}{5} = \frac{4\sqrt[3]{2}}{5}, \qquad \text{so} \qquad \sqrt[3]{2} = \frac{5}{4}\left(1+\frac{3}{125}\right)^{1/3}.$$

The binomial series for $(1+x)^{1/3}$ begins

$$(1+x)^{1/3} = 1+\tfrac{1}{3}x+\frac{\frac{1}{3}\cdot(-\frac{2}{3})}{2!}x^2+\frac{\frac{1}{3}\cdot(-\frac{2}{3})\cdot(-\frac{5}{3})}{3!}x^3+\cdots = 1+\tfrac{1}{3}x-\tfrac{1}{9}x^2+\tfrac{5}{81}x^3-\cdots.$$

Just as before, a substitution makes the arithmetic clean. Put $x=3y$ with $y=\frac{1}{125}=0.008$; then $\frac{1}{3}x=y$, $\frac{1}{9}x^2=y^2$ and $\frac{5}{81}x^3=\frac{5}{3}y^3$. So

$$\begin{aligned}\left(1+\frac{3}{125}\right)^{1/3} &\approx 1+y-y^2+\tfrac{5}{3}y^3\\ &= 1+0.008-0.000064+\tfrac{5}{3}(0.000000512)\\ &= 1.007936+0.00000085333\ldots\\ &= 1.00793685333\ldots.\end{aligned}$$

Multiplying by $\frac{5}{4}$,

$$\sqrt[3]{2} \approx 1.25\times 1.00793685333\ldots = 1.25992106\ldots,$$

so to six decimal places $\sqrt[3]{2}\approx 1.259921$.

Checks

A calculator gives $\sqrt{2}=1.41421356\ldots$ and $\sqrt[3]{2}=1.25992105\ldots$, so both approximations are correct to six decimal places; our values are 1.41421357 and 1.25992107. We checked both series with exact fractions by computer. The first omitted terms give a sense of the error: $-\frac{5}{128}x^4$ contributes about -9×10^{-9} to the $\sqrt{2}$ estimate, and $-\frac{10}{243}x^4$ contributes about -1.7×10^{-8} to the $\sqrt[3]{2}$ estimate. Both are far below the 5×10^{-7} that would change the sixth decimal place.

Common slips

- Forgetting to undo the scaling: the series gives $\frac{7\sqrt{2}}{10}$ and $\frac{4\sqrt[3]{2}}{5}$, not $\sqrt{2}$ and $\sqrt[3]{2}$ themselves.

- Sign errors in the series. With x negative in the first part, all four terms after the 1 are negative. In the second part the signs alternate.
- Rounding too early. Keep at least eight decimal places until the final division or multiplication.

Beyond STEP

Behind this question is Taylor's theorem with a remainder term, which turns "the next term is tiny" into a rigorous bound on the error. It explains why a series with ratio about 0.02 gains roughly one and a half to two decimal places per term. The same trick, finding a rational number p/q with p^n/q^n close to a simple multiple of the target, links to continued fractions and rational approximation in number theory. Davenport's *The Higher Arithmetic* is a good place to read about these, and Abbott's *Understanding Analysis* covers Taylor's remainder carefully.

WORKED SOLUTION

STEP II 1998, Question 3

Pure • Telescoping Series

STEP II 1998, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q3

The key idea

A general term with a product of consecutive factors underneath begs to be split into partial fractions. The pieces then cancel across the sum (they telescope), leaving only a few terms at the two ends. In the second part, a ratio a_n/a_{n-1} means a_n/a_1 is a product of ratios, and that product telescopes too. With luck it turns a_n back into the series from the first part, which it does.

Solution

The sum S_N

Write

$$\frac{2n-1}{n(n+1)(n+2)} = \frac{A}{n} + \frac{B}{n+1} + \frac{C}{n+2}.$$

Using the cover-up rule: at $n=0$, $A = \frac{-1}{1\cdot 2} = -\frac{1}{2}$; at $n=-1$, $B = \frac{-3}{(-1)(1)} = 3$; at $n=-2$, $C = \frac{-5}{(-2)(-1)} = -\frac{5}{2}$. So

$$\frac{2n-1}{n(n+1)(n+2)} = -\frac{1}{2n} + \frac{3}{n+1} - \frac{5}{2(n+2)}.$$

Notice that $A+B+C=0$. That is what makes the sum telescope.

Let $H_N = 1 + \frac{1}{2} + \cdots + \frac{1}{N}$. Then

$$\sum_{n=1}^{N} \frac{1}{n+1} = H_N - 1 + \frac{1}{N+1}, \qquad \sum_{n=1}^{N} \frac{1}{n+2} = H_N - 1 - \frac{1}{2} + \frac{1}{N+1} + \frac{1}{N+2}.$$

Hence

$$\begin{aligned} S_N &= -\tfrac{1}{2}H_N + 3\left(H_N - 1 + \frac{1}{N+1}\right) - \tfrac{5}{2}\left(H_N - \tfrac{3}{2} + \frac{1}{N+1} + \frac{1}{N+2}\right) \\ &= \left(-\tfrac{1}{2} + 3 - \tfrac{5}{2}\right)H_N + \left(-3 + \tfrac{15}{4}\right) + \left(3 - \tfrac{5}{2}\right)\frac{1}{N+1} - \frac{5}{2(N+2)} \\ &= \frac{3}{4} + \frac{1}{2(N+1)} - \frac{5}{2(N+2)} = \frac{1}{2}\left(\frac{3}{2} + \frac{1}{N+1} - \frac{5}{N+2}\right), \end{aligned}$$

as required. The coefficient of H_N vanishes because $A + B + C = 0$.

As $N \to \infty$, both $\frac{1}{N+1}$ and $\frac{5}{N+2}$ tend to 0, so

$$S_N \to \frac{3}{4}.$$

The sequence a_n

For $n \geq 2$, multiply the given ratios together:

$$\frac{a_n}{a_1} = \prod_{k=2}^{n} \frac{a_k}{a_{k-1}} = \prod_{k=2}^{n} \frac{k-1}{k+2} \times \prod_{k=2}^{n} \frac{2k-1}{2k-3}.$$

Ask yourself How do these two products simplify?

Write them out. The second product telescopes directly, because each numerator cancels the next denominator:

$$\prod_{k=2}^{n} \frac{2k-1}{2k-3} = \frac{3}{1} \cdot \frac{5}{3} \cdot \frac{7}{5} \cdots \frac{2n-1}{2n-3} = 2n - 1.$$

In the first product the numerators run through $1, 2, \ldots, n-1$ and the denominators through $4, 5, \ldots, n+2$. Everything from 4 to $n-1$ cancels, so

$$\prod_{k=2}^{n} \frac{k-1}{k+2} = \frac{(n-1)!}{(n+2)!/3!} = \frac{6}{n(n+1)(n+2)}.$$

The factorial form is valid for every $n \geq 2$, even when $n < 5$ and there is nothing to cancel. Therefore

$$\frac{a_n}{a_1} = \frac{6(2n-1)}{n(n+1)(n+2)}.$$

At $n = 1$ the right-hand side is $\frac{6}{6} = 1$, so the formula holds for all $n \geq 1$. As a check, $n = 2$ gives $\frac{18}{24} = \frac{3}{4}$, which agrees with the given ratio $\frac{1 \cdot 3}{4 \cdot 1}$.

When $a_1 = \frac{2}{9}$,

$$a_n = \frac{2}{9} \cdot \frac{6(2n-1)}{n(n+1)(n+2)} = \frac{4}{3} \cdot \frac{2n-1}{n(n+1)(n+2)}.$$

So a_n is $\frac{4}{3}$ times the nth term of the first series. Its partial sums are $\frac{4}{3}S_N \to \frac{4}{3} \cdot \frac{3}{4}$, and so

$$\sum_{n=1}^{\infty} a_n = 1.$$

Checks

- $N = 1$: the formula gives $\frac{1}{2}\left(\frac{3}{2} + \frac{1}{2} - \frac{5}{3}\right) = \frac{1}{6}$, which is the first term $\frac{1}{1\cdot 2\cdot 3}$. $N = 2$ gives $\frac{1}{2}\left(\frac{3}{2} + \frac{1}{3} - \frac{5}{4}\right) = \frac{7}{24} = \frac{1}{6} + \frac{1}{8}$. Both are correct.
- Numerically, the first 1999 terms sum to $0.74900\ldots$, approaching $\frac{3}{4}$ from below as the formula predicts ($S_N \approx \frac{3}{4} - \frac{2}{N}$).
- Generating a_n from the recurrence with $a_1 = \frac{2}{9}$, the first 299 terms sum to $0.9911\ldots$. The tail is about $\frac{4}{3}\cdot\frac{2}{300} \approx 0.009$, so this is consistent with a total of 1.

Common slips

- Mishandling the ends of the telescoping sum. Writing it in terms of H_N, as above, keeps the bookkeeping honest.
- Getting the factorial product wrong by one place. Write out the first and last few factors before cancelling.
- Forgetting to check that the formula for a_n/a_1 also holds at $n = 1$, where the product is empty.

Beyond STEP

Telescoping is the discrete version of the fundamental theorem of calculus: $\sum(f(n+1) - f(n)) = f(N+1) - f(1)$, just as $\int f' = f(b) - f(a)$. The second part is a *hypergeometric* sequence, in which the ratio of consecutive terms is a rational function of n. Gosper's algorithm decides exactly when such a sum has a closed form, and this is how computer algebra systems sum series. For the analysis of limits of partial sums, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 1998, Question 4

Pure • Integration by Parts

STEP II 1998, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q4

The key idea

The awkward factor $\operatorname{cosec}(x/2)$ can't be integrated directly, but the *difference* of two consecutive integrands is much kinder: the sine difference formula produces a factor $\sin(x/2)$ that cancels the cosecant exactly. What is left is a polynomial times a cosine, which is a standard integration by parts. Then the differences telescope down to a starting value we can compute by hand.

Solution

The difference $I_n - I_{n-1}$

Since $nx + x/2 = (n + \frac{1}{2})x$, the two integrands differ by

$$\left(\tfrac{\pi}{2} - x\right)\left[\sin(n+\tfrac{1}{2})x - \sin(n-\tfrac{1}{2})x\right]\operatorname{cosec}\tfrac{x}{2}.$$

By the factor formula $\sin A - \sin B = 2\cos\frac{A+B}{2}\sin\frac{A-B}{2}$, with $\frac{A+B}{2} = nx$ and $\frac{A-B}{2} = \frac{x}{2}$,

$$\sin(n+\tfrac{1}{2})x - \sin(n-\tfrac{1}{2})x = 2\cos nx\,\sin\tfrac{x}{2}.$$

The $\sin\frac{x}{2}$ cancels the cosecant, so

$$I_n - I_{n-1} = 2\int_0^{\pi}\left(\tfrac{\pi}{2} - x\right)\cos nx\,\mathrm{d}x.$$

Ask yourself Is the cancellation legitimate at $x = 0$, where $\operatorname{cosec}(x/2)$ is infinite?

Yes. Each original integrand has a finite limit as $x \to 0$, because $\sin(n+\frac{1}{2})x \big/ \sin\frac{x}{2} \to 2n+1$, so the integrals are perfectly ordinary. For $0 < x \leq \pi$ the cancellation is exact algebra, and changing an integrand at the single point $x = 0$ does not change the integral.

Now integrate by parts with $u = \frac{\pi}{2} - x$ and $\frac{\mathrm{d}v}{\mathrm{d}x} = \cos nx$, so $v = \frac{1}{n}\sin nx$ (valid as $n \geq 1$):

$$\int_0^{\pi} \left(\tfrac{\pi}{2} - x\right)\cos nx \,\mathrm{d}x = \left[\left(\tfrac{\pi}{2} - x\right)\frac{\sin nx}{n}\right]_0^{\pi} + \frac{1}{n}\int_0^{\pi} \sin nx \,\mathrm{d}x$$
$$= 0 + \frac{1}{n}\left[-\frac{\cos nx}{n}\right]_0^{\pi} = \frac{1-\cos n\pi}{n^2} = \frac{1-(-1)^n}{n^2},$$

where the boundary term vanishes because $\sin 0 = \sin n\pi = 0$. Therefore

$$I_n - I_{n-1} = \frac{2\big(1-(-1)^n\big)}{n^2} = \begin{cases} \dfrac{4}{n^2}, & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}$$

Evaluating I_n

To telescope we need a starting value. The integral defines I_0 just as well: with $n = 0$ the integrand is $(\frac{\pi}{2} - x)\sin\frac{x}{2}\operatorname{cosec}\frac{x}{2} = \frac{\pi}{2} - x$, so

$$I_0 = \int_0^{\pi}\left(\tfrac{\pi}{2} - x\right)\mathrm{d}x = \left[\tfrac{\pi}{2}x - \tfrac{1}{2}x^2\right]_0^{\pi} = 0,$$

and the difference formula above holds for $n = 1$ too (the working used only $n \geq 1$). Adding the differences for $1, 2, \ldots, n$:

$$I_n = I_0 + \sum_{k=1}^{n}\big(I_k - I_{k-1}\big) = \sum_{k=1}^{n}\frac{2\big(1-(-1)^k\big)}{k^2}.$$

Only odd k contribute, each giving $4/k^2$, so

$$\boxed{I_n = 4\sum_{\substack{k=1 \\ k \text{ odd}}}^{n}\frac{1}{k^2} = 4\Big(1 + \frac{1}{3^2} + \frac{1}{5^2} + \cdots\Big),}$$

the last term being $1/n^2$ if n is odd and $1/(n-1)^2$ if n is even. Equivalently, $I_n = 4\sum_{r=1}^{m}\frac{1}{(2r-1)^2}$ with $m = \lceil n/2 \rceil$.

If you are uneasy about using $n = 0$ (the question says n is positive), start from I_1 instead. Since $\sin\frac{3x}{2} = \sin\frac{x}{2}(1 + 2\cos x)$, we get $I_1 = \int_0^{\pi}(\frac{\pi}{2} - x)(1 + 2\cos x)\,\mathrm{d}x = 0 + 2\cdot 2 = 4$, using the integration by parts above with $n = 1$. This agrees with the formula.

Checks

- $I_1 = I_2 = 4$ and $I_3 = I_4 = 4 + \frac{4}{9} = \frac{40}{9} \approx 4.4444$. Simpson's rule on the original integral (with the integrand's limit $\frac{\pi}{2}(2n+1)$ at $x = 0$) gives $4.444444\ldots$ for I_3, and agrees with the sum to 10^{-8} for every $n \leq 12$.
- The integrand for n is $\frac{\pi}{2} - x$ times $1 + 2\cos x + \cdots + 2\cos nx$, and $\frac{\pi}{2} - x$ is odd about $x = \frac{\pi}{2}$. For even k (and for the constant 1), $\cos kx$ is even about $x = \frac{\pi}{2}$, so the product is odd and integrates to zero over $[0, \pi]$. That is why $I_0 = 0$ and even k contribute nothing.

Common slips

- Getting the sign or the factor 2 wrong in the sine difference formula; test it at a value such as $x = \pi$, $n = 1$.
- Writing $\cos n\pi = 1$ instead of $(-1)^n$, which makes every difference zero.
- Forgetting the starting value, or writing the answer as a sum over all k without noticing that the even terms vanish.

Beyond STEP

The function $\sin(n + \frac{1}{2})x \big/ \sin\frac{x}{2} = 1 + 2\sum_{k=1}^{n} \cos kx$ is the *Dirichlet kernel*, the central object in proving that Fourier series converge. As $n \to \infty$ our sum tends to $4 \cdot \frac{\pi^2}{8} = \frac{\pi^2}{2}$, using $\sum_{\text{odd } k} 1/k^2 = \pi^2/8$, a close relative of Euler's $\sum 1/k^2 = \pi^2/6$ (numerically, $I_{200} \approx 4.925$ against $\pi^2/2 \approx 4.935$). Fourier series and these sums are covered well in Boyce and DiPrima, *Elementary Differential Equations*, and the analysis behind the convergence in Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 1998, Question 5

Pure • Complex Triangle Inequality

STEP II 1998, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q5

The key idea

$|z_1 - z_2|$ is the distance between two points in the Argand diagram, so the inequality for $|z_1 + z_2|$ is just the statement that one side of a triangle is no longer than the other two put together. Once we have that for n terms, the final part is a matter of bounding each term by its modulus and recognising a geometric series: a small $|z|$ makes the left-hand side too small ever to reach 1.

Solution

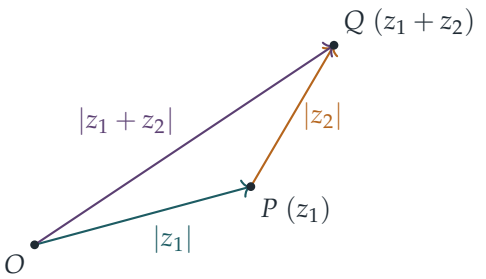
Modulus, distance and the triangle inequality

If $z = x + \mathrm{i}y$ with x, y real, the modulus of z is

$$|z| = \sqrt{x^2 + y^2},$$

the distance from the origin to the point (x, y) representing z. If $z_1 = x_1 + \mathrm{i}y_1$ and $z_2 = x_2 + \mathrm{i}y_2$, then $|z_1 - z_2| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$, which is the distance between the points representing z_1 and z_2.

Now let O, P and Q be the points representing 0, z_1 and $z_1 + z_2$.



By the interpretation above,

$$OP = |z_1 - 0| = |z_1|, \qquad PQ = |(z_1 + z_2) - z_1| = |z_2|, \qquad OQ = |z_1 + z_2|.$$

The straight segment OQ is the shortest path from O to Q, so it is no longer than the path $O \to P \to Q$; that is, $OQ \le OP + PQ$, with equality exactly when P lies on the segment OQ (including the degenerate cases where the points coincide). Therefore

$$|z_1 + z_2| \le |z_1| + |z_2|.$$

The inequality for n terms

We prove by induction that for every $n \ge 1$ and all complex $z_1, \dots, z_n$,

$$|z_1 + z_2 + \cdots + z_n| \le |z_1| + |z_2| + \cdots + |z_n|.$$

For $n = 1$ both sides equal $|z_1|$. Suppose the result holds for some $n = k \ge 1$. Given $z_1, \dots, z_{k+1}$, apply the two-term inequality to the two complex numbers $w = z_1 + \cdots + z_k$ and z_{k+1}:

$$\begin{aligned} |z_1 + \cdots + z_k + z_{k+1}| = |w + z_{k+1}| &\le |w| + |z_{k+1}| \\ &\le |z_1| + \cdots + |z_k| + |z_{k+1}|, \end{aligned}$$

using the inductive hypothesis for the last step. So the result holds for $n = k + 1$, and by induction for all $n \ge 1$.

The final part

Suppose $|z| \le \frac{1}{4}$ and write $r = |z|$, so $0 \le r \le \frac{1}{4}$. Each term satisfies $|a_k z^k| = |a_k|\,|z|^k \le 3r^k$, so by the n-term inequality

$$\begin{aligned} |a_1 z + a_2 z^2 + \cdots + a_n z^n| &\le |a_1|\, r + |a_2|\, r^2 + \cdots + |a_n|\, r^n \\ &\le 3\big(r + r^2 + \cdots + r^n\big). \end{aligned}$$

Ask yourself How do we turn $3(r + r^2 + \cdots + r^n)$ into something we can compare with 1, and how do we make sure the inequality is *strict*?

It is a finite geometric series, and since $r \le \frac{1}{4} < 1$ we may use the sum formula:

$$3\big(r + r^2 + \cdots + r^n\big) = \frac{3r(1 - r^n)}{1 - r}.$$

If $r = 0$ this is 0. If $0 < r \le \frac{1}{4}$, then $0 < 1 - r^n < 1$, so

$$\frac{3r(1 - r^n)}{1 - r} < \frac{3r}{1 - r} \le \frac{3 \cdot \frac{1}{4}}{1 - \frac{1}{4}} = 1,$$

where the last step holds because $\dfrac{3r}{1-r}$ is increasing in r on $[0,1)$ (its numerator increases and its positive denominator decreases). In either case

$$|a_1 z + a_2 z^2 + \cdots + a_n z^n| < 1,$$

so the left-hand side cannot equal 1. Hence the equation has no solution with $|z| \le \frac{1}{4}$.

Checks

- The bound is sharp: with every $a_k = 3$ and $z = \frac{1}{4}$ the left-hand side is $3\left(\frac{1}{4} + \cdots + \frac{1}{4^n}\right) = 1 - 4^{-n}$, for example 0.75, 0.9375, 0.984375 for $n = 1, 2, 3$. It gets as close to 1 as we like but never reaches it, which is why the strict inequality matters.
- In a random test of 200 000 choices of $n \le 15$, a_k with $|a_k| \le 3$ and z with $|z| \le \frac{1}{4}$, the largest value of $|a_1 z + \cdots + a_n z^n|$ found was about 0.996, always below 1.

Common slips

- Using the infinite sum $3r/(1-r) \le 1$ alone gives only $|\text{LHS}| \le 1$, which does not rule out a solution. The finite sum (or the fact that $r^n > 0$ when $r > 0$) supplies the strict inequality.
- In the induction, the two-term inequality must be applied to $z_1 + \cdots + z_k$ treated as a *single* complex number; that is the whole point of the step.

Beyond STEP

The triangle inequality is the defining property of a *metric*, and $|z_1 - z_2|$ makes $\mathbb{C}$ a metric space; almost every convergence argument in analysis rests on it. The final part is a baby version of results that locate the zeros of polynomials and power series: here $f(z) = a_1 z + \cdots + a_n z^n - 1$ has no zeros in the disc $|z| \le \frac{1}{4}$, in the spirit of Rouché's theorem. For the geometry of complex numbers, read Needham, *Visual Complex Analysis*; for metric ideas in analysis, Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 1998, Question 6

Pure • Cycloids and Normals

STEP II 1998, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q6

The key idea

Both curves are cycloids, the paths traced by a point on the rim of a rolling circle, and everything about them gets simpler with the half-angle formulae $1+\cos\theta = 2\cos^2\frac{\theta}{2}$, $1-\cos\theta = 2\sin^2\frac{\theta}{2}$ and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$. The gradients collapse to single tangents or cotangents of $\frac{\theta}{2}$, and the normal's equation shrinks to one tidy line. To show a line is a tangent to curve (2), find a point of (2) that lies on the line and where the curve's gradient equals the line's.

(The question's second curve has y_1 where it should say y_2; it's a misprint.)

Solution

Gradients

For curve (1), $\dfrac{\mathrm{d}x_1}{\mathrm{d}\theta} = 1+\cos\theta$ and $\dfrac{\mathrm{d}y_1}{\mathrm{d}\theta} = -\sin\theta$, so

$$\frac{\mathrm{d}y_1}{\mathrm{d}x_1} = \frac{-\sin\theta}{1+\cos\theta} = \frac{-2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}} = -\tan\frac{\theta}{2},$$

valid wherever $\cos\frac{\theta}{2} \neq 0$, that is, $\theta \neq \pi$.

For curve (2), $\dfrac{\mathrm{d}x_2}{\mathrm{d}\theta} = 1-\cos\theta$ and $\dfrac{\mathrm{d}y_2}{\mathrm{d}\theta} = \sin\theta$, so

$$\frac{\mathrm{d}y_2}{\mathrm{d}x_2} = \frac{\sin\theta}{1-\cos\theta} = \frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\sin^2\frac{\theta}{2}} = \cot\frac{\theta}{2},$$

valid for $\theta \neq 0, 2\pi$.

At $\theta = \pi/2$, $\frac{\theta}{2} = \frac{\pi}{4}$: curve (1) has gradient -1 and curve (2) has gradient 1.

At $\theta = 3\pi/2$, $\frac{\theta}{2} = \frac{3\pi}{4}$, where tan $= -1$ and cot $= -1$: curve (1) has gradient 1 and curve (2) has gradient -1.

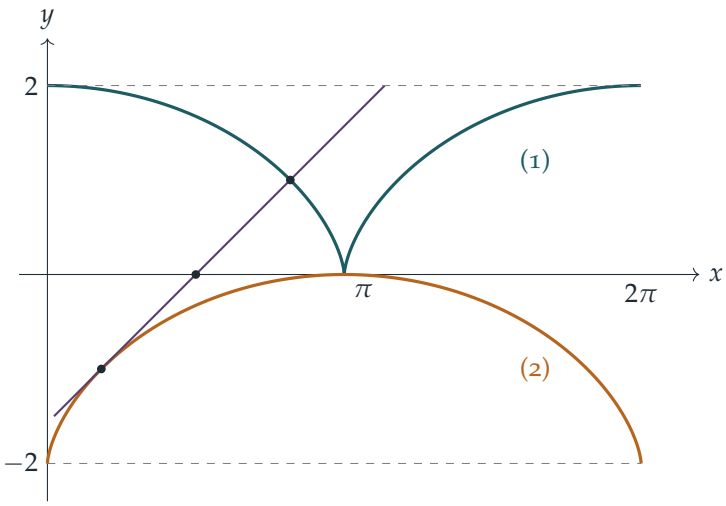
Sketch

Both x_1 and x_2 are non-decreasing in θ (their derivatives $1 \pm \cos\theta$ are never negative), so each curve runs left to right from $x = 0$ to $x = 2\pi$.

Curve (1) starts at $(0, 2)$ with gradient 0, falls to $(\pi, 0)$, then rises symmetrically to $(2\pi, 2)$. At $\theta = \pi$ both derivatives vanish and $|\mathrm{d}y_1/\mathrm{d}x_1| = |\tan\frac{\theta}{2}| \to \infty$, so there is a cusp at $(\pi, 0)$ with a vertical tangent. Since $0 \le y_1 \le 2$, the curve lies between $y = 0$ and $y = 2$.

Curve (2) has a cusp at $(0, -2)$ (vertical tangent, since $\cot\frac{\theta}{2} \to \infty$), rises to $(\pi, 0)$ where its gradient is $\cot\frac{\pi}{2} = 0$, then falls to the cusp $(2\pi, -2)$. It's an ordinary cycloid arch standing on the line $y = -2$.

The two curves touch at $(\pi, 0)$: curve (1) arrives there vertically, curve (2) passes over it horizontally. The picture is symmetric about $x = \pi$, since $\theta \mapsto 2\pi - \theta$ sends $x \mapsto 2\pi - x$ and leaves y unchanged on both curves. The plum line is the normal to (1) at $\theta = \pi/2$, which we meet below.



The normal to curve (1)

The tangent to (1) has gradient $-\tan\frac{\theta}{2}$, so the normal has gradient $\cot\frac{\theta}{2}$ (for $\theta \neq 0, \pi, 2\pi$; we deal with those below). The normal at the point $(\theta + \sin\theta,\ 1 + \cos\theta)$ is

$$y - (1 + \cos\theta) = \frac{\cos\frac{\theta}{2}}{\sin\frac{\theta}{2}}\left(x - \theta - \sin\theta\right).$$

Multiply through by $\sin\frac{\theta}{2}$ and use the half-angle forms:

$$y\sin\tfrac{\theta}{2} - 2\cos^2\tfrac{\theta}{2}\sin\tfrac{\theta}{2} = (x - \theta)\cos\tfrac{\theta}{2} - 2\sin\tfrac{\theta}{2}\cos^2\tfrac{\theta}{2}.$$

The last terms on each side are equal and cancel, leaving

$$\boxed{y\sin\tfrac{\theta}{2} = (x - \theta)\cos\tfrac{\theta}{2}}$$

as the equation of the normal.

This form also covers the exceptional values. At $\theta = 0$ it reads $x = 0$, the vertical line through $(0, 2)$, which is right because the tangent there is horizontal. At $\theta = 2\pi$ it gives $x = 2\pi$ in the same way. At $\theta = \pi$ it reads $y = 0$, the horizontal line through the cusp, perpendicular to the vertical cusp tangent.

Notice that the normal always passes through $(\theta, 0)$ on the x-axis. That's the point where the rolling circle touches the line it rolls on, a classic property of cycloids.

The normal is a tangent to curve (2)

Ask yourself Which point of curve (2) should the normal touch?

The tangent to (2) at parameter ϕ has gradient $\cot\frac{\phi}{2}$, and the normal has gradient $\cot\frac{\theta}{2}$. For ϕ, θ in $(0, 2\pi)$ these agree only when $\phi = \theta$, since cot is one-to-one on $(0, \pi)$. So the only candidate is the point of (2) with the same parameter θ.

That point is $(x_2, y_2) = (\theta - \sin\theta,\ -(1 + \cos\theta))$. Substituting into the normal's equation:

$$\text{LHS} = y_2 \sin\tfrac{\theta}{2} = -2\cos^2\tfrac{\theta}{2}\,\sin\tfrac{\theta}{2},$$
$$\text{RHS} = (x_2 - \theta)\cos\tfrac{\theta}{2} = -\sin\theta\cos\tfrac{\theta}{2} = -2\sin\tfrac{\theta}{2}\cos^2\tfrac{\theta}{2}.$$

These are equal, so the normal to (1) at θ passes through the point of (2) with parameter θ. At that point the gradient of (2) is $\cot\frac{\theta}{2}$, the same as the normal's. A line through a point of a curve with the curve's gradient there is the tangent at that point, so the normal to (1) is a tangent to (2), as required.

The special values work too. For $\theta = \pi$ the normal $y = 0$ touches (2) at its top $(\pi, 0)$, where the gradient of (2) is 0. For $\theta = 0$ (or 2π) the normal $x = 0$ (or $x = 2\pi$) is the vertical tangent at the cusp of (2).

Checks

- At $\theta = \pi/2$ the normal is $y = x - \frac{\pi}{2}$. It passes through $(\frac{\pi}{2} + 1, 1)$ on (1), through $(\frac{\pi}{2}, 0)$, and through $(\frac{\pi}{2} - 1, -1)$ on (2), where (2) has gradient 1, as found in the first part. That's the plum line in the sketch.
- A numerical check of the gradients by finite differences agreed with $-1, 1$ and $1, -1$ to six decimal places. At 1000 random values of θ, the normal passed through both points and was parallel to the tangent of (2), to within 10^{-12}.

Common slips

- Leaving the gradients as $-\sin\theta/(1 + \cos\theta)$ and then grinding through messy algebra for the normal. The half-angle forms make the normal a two-line calculation.

- Showing only that the normal meets curve (2). Meeting isn't enough: you also need the gradients to match at the meeting point.

Beyond STEP

The curve traced out by the centres of curvature of a curve is its *evolute*, and it is also the envelope of the curve's normals. This question shows that the evolute of a cycloid is an identical cycloid shifted by half an arch. The distance along the normal from (1) to its point of contact with (2) is $4|\cos\frac{\theta}{2}|$, which is exactly the arc length of (2) from that point to its top. So (1) is traced by the end of a string unwinding from (2). Huygens used this to build a pendulum whose period doesn't depend on its amplitude. For curvature, evolutes and involutes, try Pressley, *Elementary Differential Geometry*.

WORKED SOLUTION

STEP II 1998, Question 7

Pure • Inequalities via Derivatives

STEP II 1998, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q7

The key idea

Each inequality is proved the same way: the function is zero at $x = 0$ and its derivative is positive, so the function goes up from zero. The neat part is that the functions are chained. The derivative of g turns out to involve f, and the derivative of h turns out to be g evaluated at $2x$. So each part gives you the positive derivative you need for the next one. In part (iv), taking logarithms turns the quotient F into a sum, and the sign of F'/F is the sign of the expression from part (iii).

Solution

The tool used throughout is this: if ϕ is differentiable, $\phi(0) = 0$ and $\phi'(x) > 0$ for $0 < x < a$, then ϕ is strictly increasing on $[0, a)$, so $\phi(x) > \phi(0) = 0$ for $0 < x < a$. (By the mean value theorem, $\phi(x) = \phi(x) - \phi(0) = x\,\phi'(c)$ for some $c \in (0, x)$, and this is positive.)

Part (i)

We have $\mathrm{f}(0) = \tan 0 - 0 = 0$, and

$$\mathrm{f}'(x) = \sec^2 x - 1 = \tan^2 x.$$

For $0 < x < \frac{1}{2}\pi$, $\tan x$ is defined and positive, so $\mathrm{f}'(x) > 0$. Hence $\mathrm{f}(x) > 0$ for $0 < x < \frac{1}{2}\pi$; that is, $\tan x > x$ there.

Part (ii)

Again $\mathrm{g}(0) = 2 - 2 - 0 = 0$. Differentiating, using the product rule on $x \sin x$,

$$\begin{aligned}\mathrm{g}'(x) &= 2\sin x - (\sin x + x\cos x)\\ &= \sin x - x\cos x\\ &= \cos x\,(\tan x - x) = \cos x\,\mathrm{f}(x).\end{aligned}$$

For $0 < x < \frac{1}{2}\pi$ we have $\cos x > 0$, and $f(x) > 0$ by part (i). So $g'(x) > 0$, and since $g(0) = 0$ it follows that $g(x) > 0$ for $0 < x < \frac{1}{2}\pi$.

Part (iii)

Once more $h(0) = 0$. Differentiating,

$$\begin{aligned} h'(x) &= 2 + \cos 2x - 2x \sin 2x - 3\cos 2x \\ &= 2 - 2\cos 2x - 2x \sin 2x. \end{aligned}$$

? **Ask yourself** Have you seen this expression before?

It is g with x replaced by $2x$:

$$h'(x) = 2 - 2\cos(2x) - (2x)\sin(2x) = g(2x).$$

If $0 < x < \frac{1}{4}\pi$ then $0 < 2x < \frac{1}{2}\pi$, so $g(2x) > 0$ by part (ii). This is why the range is now $\frac{1}{4}\pi$ rather than $\frac{1}{2}\pi$. So $h'(x) > 0$ on $0 < x < \frac{1}{4}\pi$, and with $h(0) = 0$ this gives $h(x) > 0$ for $0 < x < \frac{1}{4}\pi$.

For the "hence", rewrite the given expression using double angles. Since $\sin^2 x + 3\cos^2 x = 1 + 2\cos^2 x = 2 + \cos 2x$ and $3 \sin x \cos x = \frac{3}{2}\sin 2x$,

$$\begin{aligned} x(\sin^2 x + 3\cos^2 x) - 3\sin x \cos x &= x(2 + \cos 2x) - \tfrac{3}{2}\sin 2x \\ &= 2x + x\cos 2x - \tfrac{3}{2}\sin 2x = h(x). \end{aligned}$$

So the expression is exactly $h(x)$, which is positive for $0 < x < \frac{1}{4}\pi$, as required.

Part (iv)

For $0 < x < \frac{1}{4}\pi$, the quantities x, $\sin x$ and $\cos x$ are all positive, so $F(x) > 0$ and we may take logarithms:

$$\ln F(x) = \ln x + \tfrac{1}{3}\ln\cos x - \ln\sin x.$$

Differentiating both sides with respect to x,

$$\frac{F'(x)}{F(x)} = \frac{1}{x} - \frac{\sin x}{3\cos x} - \frac{\cos x}{\sin x}.$$

? **Ask yourself** How do we see the sign of this?

Put it over the common denominator $3x \sin x \cos x$, which is positive on this interval:

$$\begin{aligned} \frac{F'(x)}{F(x)} &= \frac{3\sin x \cos x - x\sin^2 x - 3x\cos^2 x}{3x \sin x \cos x} \\ &= -\frac{x(\sin^2 x + 3\cos^2 x) - 3\sin x \cos x}{3x \sin x \cos x} = -\frac{h(x)}{3x \sin x \cos x}. \end{aligned}$$

By part (iii) the numerator $\mathrm{h}(x)$ is positive and the denominator is positive, so $\mathrm{F}'(x)/\mathrm{F}(x) < 0$. Since $\mathrm{F}(x) > 0$, multiplying through gives $\mathrm{F}'(x) < 0$ for $0 < x < \frac{1}{4}\pi$, as required.

Checks

- Spot values: $\mathrm{F}(0.1) \approx 0.999998$, $\mathrm{F}(0.5) \approx 0.99849$ and $\mathrm{F}(\frac{1}{4}\pi) \approx 0.98954$, so F is decreasing, but only slowly.
- Near 0 the inequalities are very tight. Expanding in Maclaurin series gives $\mathrm{h}(x) = \frac{4}{15}x^5 + \cdots$, and the x, x^3 terms cancel. That is why you can't simply "see" that $\mathrm{h} > 0$ and need the chain of derivatives. Numerically, $\mathrm{h}(0.01)/0.01^5 \approx 0.26666$.
- We tested all four inequalities, and the identities $\mathrm{g}' = \mathrm{f}\cos x$, $\mathrm{h}'(x) = \mathrm{g}(2x)$ and $\mathrm{F}'/\mathrm{F} = -\mathrm{h}/(3x\sin x\cos x)$, at 20 000 random points.

Common slips

- Arguing only that $\mathrm{f}'(x) \geq 0$, or forgetting to state $\mathrm{f}(0) = 0$. Both facts are needed: the zero start and the strictly positive derivative.
- Using part (ii) in part (iii) without checking that $2x$ lies in $(0, \frac{1}{2}\pi)$. That is exactly where the restriction $x < \frac{1}{4}\pi$ comes from.
- Concluding $\mathrm{F}' < 0$ from $\mathrm{F}'/\mathrm{F} < 0$ without noting that $\mathrm{F} > 0$.

Beyond STEP

The principle "zero at the start and positive derivative means positive" is a consequence of the mean value theorem, one of the first big theorems in a university analysis course. Repeating it, as here, is a standard way of proving inequalities such as $\sin x > x - \frac{1}{6}x^3$, and it is closely related to Taylor's theorem with remainder. Abbott's *Understanding Analysis* covers the mean value theorem and its uses carefully.

WORKED SOLUTION

STEP II 1998, Question 8

Pure • Vector Geometry

STEP II 1998, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q8

The key idea

Everything here is a matter of writing each line in the form "point + parameter × direction" and then choosing the parameter cleverly. The expressions are symmetric in **a**, **b**, **c**, so if one parameter value gives an answer that is symmetric in the three vectors, the same point lies on all three lines. That is exactly what "by inspection" is inviting us to look for.

Solution

The medians meet at R

The mid-point of BC has position vector $\frac{1}{2}(\mathbf{b}+\mathbf{c})$, so the line from A to it is

$$\mathbf{r} = \mathbf{a} + s\left(\tfrac{1}{2}(\mathbf{b}+\mathbf{c}) - \mathbf{a}\right) = (1-s)\mathbf{a} + \tfrac{s}{2}(\mathbf{b}+\mathbf{c}).$$

We want the three coefficients to be equal, so that the result is symmetric: $1 - s = \frac{s}{2}$ gives $s = \frac{2}{3}$, and the point is

$$\mathbf{r} = \tfrac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c}).$$

The lines from B and from C are obtained by permuting $\mathbf{a}, \mathbf{b}, \mathbf{c}$ cyclically, and the same choice $s = \frac{2}{3}$ gives the same symmetric point on each. So all three lines pass through the point R with position vector

$$\mathbf{r}_R = \tfrac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c}),$$

the centroid of the triangle. (Since A, B, C form a genuine triangle, no two of these lines coincide, so R is their single common point.)

The three parallel lines meet at Q

Let P have position vector $\mathbf{p}$. The three lines are

$$\begin{aligned}
&\text{through the mid-point of } BC, \text{ parallel to } PA: && \mathbf{r} = \tfrac{1}{2}(\mathbf{b}+\mathbf{c}) + \lambda(\mathbf{a}-\mathbf{p}),\\
&\text{through the mid-point of } CA, \text{ parallel to } PB: && \mathbf{r} = \tfrac{1}{2}(\mathbf{c}+\mathbf{a}) + \mu(\mathbf{b}-\mathbf{p}),\\
&\text{through the mid-point of } AB, \text{ parallel to } PC: && \mathbf{r} = \tfrac{1}{2}(\mathbf{a}+\mathbf{b}) + \nu(\mathbf{c}-\mathbf{p}).
\end{aligned}$$

Ask yourself What value of λ makes the first expression symmetric in $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$?

In the first line, $\mathbf{b}$ and $\mathbf{c}$ each carry the coefficient $\frac{1}{2}$ while $\mathbf{a}$ carries λ; matching them needs $\lambda = \frac{1}{2}$. Then

$$\mathbf{r} = \tfrac{1}{2}(\mathbf{a}+\mathbf{b}+\mathbf{c}-\mathbf{p}).$$

Taking $\mu = \frac{1}{2}$ and $\nu = \frac{1}{2}$ in the other two lines gives exactly the same vector. Hence all three lines pass through the point Q with

$$\mathbf{q} = \tfrac{1}{2}(\mathbf{a}+\mathbf{b}+\mathbf{c}-\mathbf{p}).$$

PQ meets the plane ABC at R

First, $Q \neq P$: if $\mathbf{q} = \mathbf{p}$ then $\mathbf{a}+\mathbf{b}+\mathbf{c} = 3\mathbf{p}$, so $\mathbf{p} = \mathbf{r}_R$ and P would be the point R, which lies in the plane ABC. So PQ is a genuine line, namely

$$\mathbf{r} = \mathbf{p} + t(\mathbf{q}-\mathbf{p}) = \mathbf{p} + t\left(\tfrac{1}{2}(\mathbf{a}+\mathbf{b}+\mathbf{c}) - \tfrac{3}{2}\mathbf{p}\right).$$

To kill the $\mathbf{p}$ terms we need $1 - \frac{3}{2}t = 0$, that is $t = \frac{2}{3}$, and then

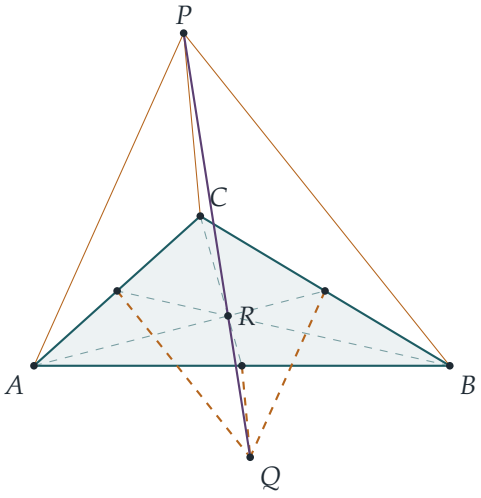
$$\mathbf{r} = \mathbf{p} + \tfrac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c}) - \mathbf{p} = \tfrac{1}{3}(\mathbf{a}+\mathbf{b}+\mathbf{c}) = \mathbf{r}_R.$$

So R lies on the line PQ. It also lies in the plane ABC (it is on the median from A, which lies in that plane).

Ask yourself Could PQ meet the plane somewhere else as well?

No. A line meets a plane in no points, exactly one point, or lies entirely in it. The line PQ does not lie in the plane ABC, because P is on it and P is not in the plane. So it meets the plane in at most one point, and that point is R. This proves that PQ meets the plane ABC at R.

As a bonus, $t = \frac{2}{3}$ tells us that R divides PQ in the ratio $PR : RQ = 2 : 1$; equivalently $\mathbf{q} - \mathbf{r}_R = -\frac{1}{2}(\mathbf{p} - \mathbf{r}_R)$.



Checks

- Each claimed intersection can be checked by substituting back: for instance, $\mathbf{q} - \frac{1}{2}(\mathbf{b}+\mathbf{c}) = \frac{1}{2}(\mathbf{a}-\mathbf{p})$, which is visibly parallel to PA.
- With $A=(0,0,0)$, $B=(3,0,0)$, $C=(0,3,0)$ and $P=(0,0,3)$ we get $R=(1,1,0)$ and $Q=(\frac{3}{2},\frac{3}{2},-\frac{3}{2})$; the point two-thirds of the way from P to Q is $(1,1,0)=R$, on the plane $z=0$.
- We also checked all the results with exact rational arithmetic on 2000 random choices of A, B, C, P: the line PQ always met the plane ABC at R, with parameter $t=\frac{2}{3}$.

Common slips

- Finding a point on two of the lines by solving equations and forgetting to check that it lies on the third.
- Showing that R lies on PQ but not explaining why it is *the* point where PQ meets the plane; this is where the condition that P is not in the plane is used.

Beyond STEP

There is a neat geometric reason behind the result. The enlargement with centre R and scale factor $-\frac{1}{2}$ (the map $\mathbf{x} \mapsto \mathbf{r}_R - \frac{1}{2}(\mathbf{x}-\mathbf{r}_R)$) sends A, B, C to the mid-points of BC, CA, AB, and sends any line to a parallel line. So it carries the lines PA, PB, PC, which meet at P, onto the three new lines, which must therefore meet at the image of P; that image is Q, and P, R, Q are collinear because the centre of an enlargement is collinear with every point and its image. Thinking of geometry through its transformations (affine maps and their fixed points) is a central idea of first-year linear algebra; Strang's

Introduction to Linear Algebra is a good place to see it developed.

WORKED SOLUTION

STEP II 1998, Question 9

Mechanics • Statics of Frameworks

STEP II 1998, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q9

The key idea

The framework is light, so only two external forces act on it: the support at A and the weight at C. Two forces in equilibrium must be equal, opposite and along the same line, and that settles both the support force and how the hexagon hangs. After that it is the method of joints: start at a joint where only two rods meet, resolve, and work round the hexagon one joint at a time, always moving to a joint with at most two unknown forces.

Solution

The supporting force

Model the hexagon's sides and the three diagonal rods as light rods, smoothly pinned at the six vertices, and write W for the weight of the 1 kg mass, so $W = g \approx 9.8\,\text{N}$. The rods are light, so the only external forces on the whole framework are the force $\mathbf{R}$ from the support at A and the weight W acting vertically downwards at C.

? **Ask yourself** Can the framework hang at any angle we like?

No. For the framework as a whole, the forces must sum to zero and have zero total moment. Taking moments about A, the support force has no moment, so the weight at C must also have zero moment about A. That happens only if C is directly below A. So the framework hangs with the short diagonal AC vertical, and resolving gives

$$\mathbf{R} = W \text{ vertically upwards}, \qquad |\mathbf{R}| = g \approx 9.8\ \text{N},$$

the weight of 1 kg acting along CA.

The forces in the rods

Take side length 1 (the forces don't depend on it). With AC vertical, B lies at the mid-height of AC, on one side of it, and the rest of the hexagon lies on the other side:

F level with A, E level with B, D level with C. So AF, BE and CD are horizontal, the sides AB, BC, DE, EF make 60° with the horizontal, and the short diagonals BD, BF make 30° with the horizontal.

Let T_{XY} be the tension in rod XY; at each end it pulls the joint towards the other end. A negative tension means the rod is in thrust.

Ask yourself There are nine unknown rod forces. Where do we start?

At a joint where only two rods meet, because two equations (horizontal and vertical) can then be solved at once. Joint C has only CB and CD, and it carries the load.

Joint C. Rod CB points 60° above the horizontal and CD is horizontal, away from B's side. Resolving,

$$\begin{aligned} \text{vertically:} \quad & T_{BC}\sin 60^\circ = W \;\Rightarrow\; T_{BC} = \tfrac{2W}{\sqrt{3}}, \\ \text{horizontally:} \quad & T_{CD} = T_{BC}\cos 60^\circ = \tfrac{W}{\sqrt{3}}. \end{aligned}$$

Joint D. Now only DE and DB are unknown. From D, rod DC points horizontally towards B's side, DE points 60° above the horizontal away from it, and DB points 30° above the horizontal towards it. Resolving,

$$\begin{aligned} \text{vertically:} \quad & T_{DE}\sin 60^\circ + T_{BD}\sin 30^\circ = 0 \;\Rightarrow\; T_{BD} = -\sqrt{3}\,T_{DE}, \\ \text{horizontally:} \quad & T_{CD} - T_{DE}\cos 60^\circ + T_{BD}\cos 30^\circ = 0 \;\Rightarrow\; T_{CD} - \tfrac{1}{2}T_{DE} - \tfrac{3}{2}T_{DE} = 0, \end{aligned}$$

so $T_{DE} = \frac{1}{2}T_{CD} = \frac{W}{2\sqrt{3}}$ and $T_{BD} = -\frac{W}{2}$.

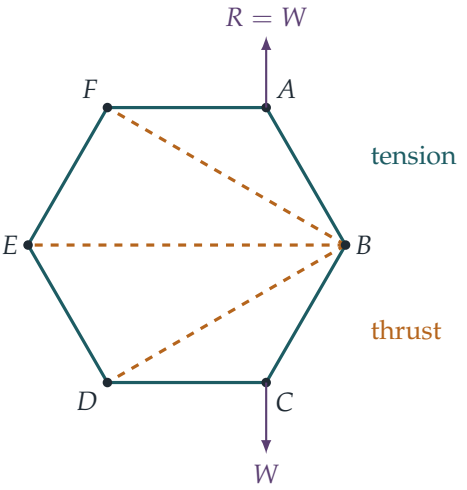
Joint E. Rod ED points 60° below the horizontal, EF points 60° above it, and EB is horizontal, all towards B's side. Resolving vertically gives $T_{EF} = T_{DE} = \frac{W}{2\sqrt{3}}$. Resolving horizontally,

$$\tfrac{1}{2}T_{DE} + \tfrac{1}{2}T_{EF} + T_{BE} = 0 \;\Rightarrow\; T_{BE} = -\tfrac{W}{2\sqrt{3}}.$$

Joint F. This is the mirror image of joint D in the line BE. Resolving vertically, $T_{BF} = -\sqrt{3}\,T_{EF} = -\frac{W}{2}$; horizontally, $T_{FA} = 2T_{EF} = \frac{W}{\sqrt{3}}$.

Joint A. Horizontally, $T_{AB}\cos 60^\circ = T_{FA}$, so $T_{AB} = \frac{2W}{\sqrt{3}}$. Vertically, $R = T_{AB}\sin 60^\circ = W$, agreeing with the first part.

Joint B gives two more equations, but they hold automatically: the framework as a whole is already in equilibrium. Checking them is a good test of the arithmetic.



Summarising, with $W = g$:

- tension in all six sides: AB and BC carry $\frac{2W}{\sqrt{3}}$, CD and FA carry $\frac{W}{\sqrt{3}}$, DE and EF carry $\frac{W}{2\sqrt{3}}$;
- thrust in all three inner rods: BD and BF carry $\frac{W}{2}$, and BE carries $\frac{W}{2\sqrt{3}}$.

So the force in BE is a thrust of magnitude

$$\frac{g}{2\sqrt{3}} = \frac{\sqrt{3}\,g}{6} \approx 2.83 \text{ N}.$$

Checks

- Symmetry: reflecting in the line BE swaps $A \leftrightarrow C$ and $F \leftrightarrow D$, and turns the upward force W at A into the downward force W at C. So the loading is symmetric, and the answers should be too: $T_{AB} = T_{BC}$, $T_{FA} = T_{CD}$, $T_{EF} = T_{DE}$, $T_{BF} = T_{BD}$. They are.
- Joint B: horizontally, the two sides give $2 \cdot \frac{2W}{\sqrt{3}} \cdot \frac{1}{2}$ one way, and BD, BF, BE give $2 \cdot \frac{W}{2} \cdot \frac{\sqrt{3}}{2} + \frac{W}{2\sqrt{3}} = \frac{2W}{\sqrt{3}}$ the other; vertically, AB and BC cancel, as do BD and BF.
- We solved all twelve joint equations numerically with $g = 9.8$: they are consistent, give $R = 9.8\,\text{N}$ upwards and $|T_{BE}| \approx 2.829\,\text{N}$ in thrust, and have no solution if AC is tilted.

Common slips

- Drawing the hexagon with a side or a long diagonal vertical. The hanging position is forced: C below A.

- Mixing up sign conventions between joints. Fix "tension pulls the joint towards the other end of the rod" once, and let negative answers tell you which rods are in thrust.

Beyond STEP

A framework with j joints is rigid and statically determinate when it has exactly $2j - 3$ rods: here $2 \cdot 6 - 3 = 9$, which is why the forces can be found by statics alone. With more rods the structure is "redundant" and you need the elasticity of the rods; this is the start of structural mechanics, and of the mathematics of rigidity of graphs. For the vector mechanics behind it, try Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP II 1998, Question 10

Mechanics • Particle on Moving Wedge

STEP II 1998, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q10

The key idea

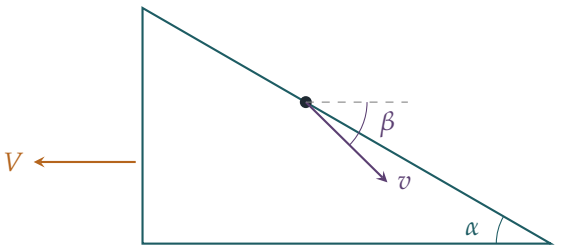
Two facts drive the whole question. Because the particle stays on the face, its velocity *relative to the wedge* must point along the face. And because nothing outside the system pushes horizontally, and nothing does work except gravity, both horizontal momentum and energy are conserved. Each of (i)–(iii) is one of these facts written as an equation. Part (ii) then shows that β never changes, which turns the final part into motion with constant acceleration.

Solution

Model. The particle is a particle; the wedge is a rigid body that slides without rotating; the face and the floor are smooth; the particle stays in contact with the face. Take the face sloping down to the right, so the particle moves down and to the right and the wedge is pushed to the left, as in the diagram.

Taking horizontal components positive to the right and vertical components positive upwards,

$$\text{particle velocity} = (v\cos\beta,\ -v\sin\beta), \qquad \text{wedge velocity} = (-V,\ 0).$$



Part (i)

The velocity of the particle relative to the wedge is

$$(v\cos\beta,\ -v\sin\beta) - (-V,\ 0) = (v\cos\beta + V,\ -v\sin\beta).$$

While the particle is on the face, this relative velocity is along the face, at angle α below the horizontal (pointing down and to the right). So

$$\frac{v\sin\beta}{v\cos\beta + V} = \tan\alpha = \frac{\sin\alpha}{\cos\alpha}.$$

Cross-multiplying,

$$\begin{aligned} & v\sin\beta\cos\alpha = v\cos\beta\sin\alpha + V\sin\alpha \\ \Longrightarrow \quad & V\sin\alpha = v(\sin\beta\cos\alpha - \cos\beta\sin\alpha) = v\sin(\beta-\alpha), \end{aligned}$$

as required. This is pure kinematics: no mechanical principle is needed, only the fact that the particle doesn't leave the face.

Part (ii)

Principle: conservation of linear momentum horizontally. Treat the particle and wedge as one system. The external forces on it are the two weights and the normal reaction from the floor, all vertical, because the floor is smooth. (The reaction between particle and face is internal.) So the total horizontal momentum is constant, and it starts at zero because everything starts from rest:

$$mv\cos\beta - MV = 0, \qquad \text{so} \qquad V = \frac{m}{M}\,v\cos\beta.$$

Substituting into (i), and expanding $\sin(\beta-\alpha)$,

$$\frac{m}{M}\,v\cos\beta\sin\alpha = v(\sin\beta\cos\alpha - \cos\beta\sin\alpha).$$

For $t > 0$ the particle is moving, so $v \neq 0$. Also $\cos\alpha \neq 0$, and $\cos\beta \neq 0$: if β were 90°, momentum would give $V = 0$, and then (i) would give $\sin(\beta-\alpha) = 0$, so $\beta = \alpha < 90^\circ$, a contradiction. Dividing by $v\cos\alpha\cos\beta$,

$$\frac{m}{M}\tan\alpha = \tan\beta - \tan\alpha \quad \Longrightarrow \quad \tan\beta = \left(1 + \frac{m}{M}\right)\tan\alpha.$$

Ask yourself Why isn't β simply equal to α?

Because the face itself is moving away to the left. Relative to the wedge the particle goes down the slope at angle α, but the wedge retreats, so relative to the ground the particle drops more steeply. Notice that the right-hand side of (ii) is a constant: β is the same at every instant, so the particle moves in a straight line. If $M \to \infty$ the wedge stays put and $\beta \to \alpha$, as it should.

Part (iii)

Principle: conservation of energy. All the contacts are smooth, so the only forces that do work on the system as a whole are the weights. (The normal reaction between particle and face does work on each body separately, but the two amounts cancel, because the relative velocity is along the face and so perpendicular to the reaction. The floor's reaction is perpendicular to the wedge's motion.) The wedge's centre of mass doesn't move vertically, so the loss of potential energy is mgy. Starting from rest,

$$mgy = \tfrac{1}{2}mv^2 + \tfrac{1}{2}MV^2.$$

Using $V = \dfrac{m}{M}v\cos\beta$ from (ii),

$$mgy = \tfrac{1}{2}mv^2 + \frac{m^2v^2\cos^2\beta}{2M}.$$

Multiplying by $2/m$,

$$2gy = v^2\Big(1 + \frac{m\cos^2\beta}{M}\Big) = \frac{v^2\,(M + m\cos^2\beta)}{M},$$

as required.

The differential equation for y

The rate at which the particle descends is the downward component of its velocity:

$$\frac{\mathrm{d}y}{\mathrm{d}t} = v\sin\beta.$$

Put $v = \dot{y}/\sin\beta$ into (iii) (note $\sin\beta \neq 0$ since $\beta > \alpha > 0$). This gives the differential equation

$$\Big(\frac{\mathrm{d}y}{\mathrm{d}t}\Big)^2 = k\,y, \qquad \text{where} \qquad k = \frac{2gM\sin^2\beta}{M + m\cos^2\beta}.$$

The key point is that k is a *constant*, because β is constant by (ii).

To solve it, differentiate with respect to t:

$$2\dot{y}\,\ddot{y} = k\,\dot{y}.$$

For $t > 0$ the particle is descending, so $\dot{y} > 0$ and we may divide by it: $\ddot{y} = \tfrac{1}{2}k$. The particle's vertical motion has constant acceleration

$$\ddot{y} = \frac{gM\sin^2\beta}{M + m\cos^2\beta}.$$

Integrating twice with $y = 0$ and $\dot{y} = 0$ at $t = 0$ (it starts from rest),

$$y = \frac{gMt^2\sin^2\beta}{2\,(M + m\cos^2\beta)},$$

which is the required result.

(Alternatively, take the positive square root, $\dot{y} = \sqrt{ky}$, and separate: $\int y^{-1/2}\,\mathrm{d}y = \sqrt{k}\,t$ gives $2\sqrt{y} = \sqrt{k}\,t$ and the same answer. Differentiating first avoids worrying about the solution $y \equiv 0$, which also satisfies $\dot{y}^2 = ky$ with $y(0) = 0$ but isn't the motion here.)

Checks

- *Fixed wedge.* Let $M \to \infty$. Then $\beta \to \alpha$ and $y \to \frac{1}{2}gt^2 \sin^2\alpha$, which is right: on a fixed smooth slope the acceleration along the slope is $g\sin\alpha$, and its vertical component is $g\sin^2\alpha$.
- *A numerical case.* With $M = 2$, $m = 1$, $\alpha = 30^\circ$, (ii) gives $\tan\beta = \frac{3}{2}\cdot\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{2}$, so $\sin^2\beta = \frac{3}{7}$, $\cos^2\beta = \frac{4}{7}$, and the formula gives $y = \frac{1}{6}gt^2$. With $g = 9.8$ and $t = 1$ that is $y \approx 1.6333$. A direct step-by-step simulation of the wedge and particle (Newton's laws with the normal reaction found from the contact condition) gave the same value, and satisfied (i)–(iii) to within rounding in 2000 random cases.

Common slips

- Saying the particle moves at angle α to the horizontal. That is only true relative to the wedge.
- Conserving momentum in the direction of the slope, or vertically. The floor's reaction is an external vertical force, so only the horizontal component is conserved.
- Forgetting the wedge's kinetic energy $\frac{1}{2}MV^2$ in the energy equation.

Beyond STEP

A particle constrained to a moving surface is a classic test case for Lagrangian mechanics, which you meet in the first or second year at university. There you choose two coordinates (say, the wedge's position and the particle's distance down the face) and the constraint forces disappear from the equations altogether, so the constant accelerations drop out in a few lines. For Newtonian mechanics with constraints, and many problems of exactly this kind, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP II 1998, Question 11

Mechanics • Projectile Motion

STEP II 1998, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q11

This question has a full guided treatment in *Beyond STEP*, Section 11.11, *How Fielders Catch*: https://beyondstep.keithmcnulty.org/ch11/fielders.html

The key idea

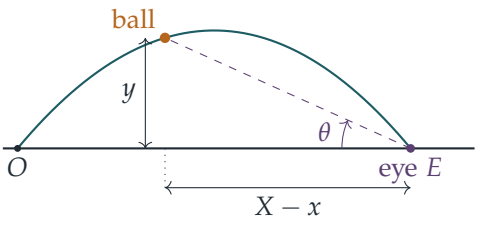
Put the origin at the point where the ball is struck, at eye level, and write $\tan\theta$ as "height of the ball over horizontal distance from the fielder's eye". The height of a projectile that starts and finishes at eye level factorises as $\frac{1}{2}g\,t(T-t)$, where T is the time of flight, and if the fielder ends up in the right place the horizontal gap is also a multiple of $T-t$. The factor $T-t$ cancels, leaving $\tan\theta$ proportional to t.

Solution

We model the ball as a particle moving under gravity alone (no air resistance), and the fielder's eye as a point at the same height as the point where the ball is struck. Take x horizontally from the striking point O towards the fielder and y vertically upwards. Let the ball's initial velocity have horizontal component $U>0$ and vertical component $W>0$. Then at time t

$$x = Ut, \qquad y = Wt - \tfrac{1}{2}gt^2 = \tfrac{1}{2}g\,t\,(T-t), \qquad \text{where } T = \frac{2W}{g}$$

is the time at which the ball returns to eye level.



If the fielder's eye is at horizontal position X (with $X > x$ while the ball is in flight), the line of sight rises at angle θ above the horizontal, where

$$\tan\theta = \frac{y}{X - x}.$$

The stationary fielder

"Perfectly placed" means the ball arrives at her eye: her eye is at the point where the ball returns to eye level, $X = UT$. Then for $0 < t < T$

$$\tan\theta = \frac{\frac{1}{2}gt(T-t)}{UT - Ut} = \frac{\frac{1}{2}gt(T-t)}{U(T-t)} = \frac{g}{2U}t,$$

cancelling $T - t$, which is non-zero while the ball is in flight. Hence

$$\frac{\mathrm{d}}{\mathrm{d}t}(\tan\theta) = \frac{g}{2U},$$

which is constant, as required. (Just before the catch, $\tan\theta \to gT/(2U) = W/U$: she sees the ball coming in at exactly the angle at which it was struck, as symmetry of the parabola suggests.)

The running fielder

Now let the next ball have velocity components U, W (so again $y = \frac{1}{2}gt(T-t)$ with $T = 2W/g$), and let the fielder start at distance D from O at the moment the ball is struck, running towards the batsman at constant speed v. Her eye is at $X = D - vt$, so

$$\tan\theta = \frac{\frac{1}{2}gt(T-t)}{D - (U+v)t}. \qquad (*)$$

She catches the ball exactly when her eye is where the ball returns to eye level at time T, that is, when

$$D - vT = UT \quad\Longleftrightarrow\quad D = (U+v)T.$$

We show that this happens if and only if $\frac{\mathrm{d}}{\mathrm{d}t}(\tan\theta)$ is constant, so "choose v to keep $\frac{\mathrm{d}}{\mathrm{d}t}(\tan\theta)$ constant" is the right rule.

If she has chosen the right speed, $D = (U+v)T$, and $(*)$ becomes

$$\tan\theta = \frac{\frac{1}{2}gt(T-t)}{(U+v)(T-t)} = \frac{g}{2(U+v)}t, \qquad \text{so} \qquad \frac{\mathrm{d}}{\mathrm{d}t}(\tan\theta) = \frac{g}{2(U+v)},$$

a constant, exactly as in the first part with U replaced by the ball's speed $U + v$ relative to her.

Ask yourself Isn't that only half of what's needed? Could some other v also give a constant rate?

No. Suppose $\frac{\mathrm{d}}{\mathrm{d}t}(\tan\theta) = k$, a constant, throughout the flight. At $t = 0$ the ball is at eye level, so $\tan\theta = 0$, and hence $\tan\theta = kt$. Putting this into $(*)$ and dividing by t (for $0 < t < T$),

$$\tfrac{1}{2}g\,(T - t) = k\big(D - (U + v)t\big) \quad \text{for all } t \in (0, T).$$

Two linear functions of t that agree on an interval have equal coefficients, so

$$\tfrac{1}{2}g = k(U + v) \qquad \text{and} \qquad \tfrac{1}{2}gT = kD.$$

The first gives $k \neq 0$; dividing the second by the first gives $D = (U + v)T$, which is precisely the catching condition.

So the speed $v = D/T - U$ that brings her to the ball is exactly the one for which $\frac{\mathrm{d}}{\mathrm{d}t}(\tan\theta)$ stays constant. In practice she cannot compute $D/T - U$ while the ball is in the air, but she can watch: if the ball seems to rise ever faster in her view she is too far back and should run faster, and if it seems to rise ever slower she should ease off. (Since she runs *towards* the batsman, the ball falls short of her starting point, $D > UT$, so $v > 0$.)

Checks

- Setting $v = 0$ in the second part recovers the first part: $D = UT$ and rate $g/(2U)$.
- Units: g/U has units $\mathrm{m\,s^{-2}}/\mathrm{m\,s^{-1}} = \mathrm{s^{-1}}$, right for the rate of change of a pure number.
- A numerical example: struck at 20 $\mathrm{m\,s^{-1}}$ at 45° with $g = 9.8\ \mathrm{m\,s^{-2}}$, $T \approx 2.886$ s and the range is 40.82 m; a fielder standing there sees $\tan\theta$ grow at $g/(2U) \approx 0.346\ \mathrm{s^{-1}}$. If she instead starts 49.47 m away and runs at 3 $\mathrm{m\,s^{-1}}$, the rate is $0.286\ \mathrm{s^{-1}}$. Differentiating $(*)$ numerically at many times confirms both rates are constant, and that a wrong v gives a rate that changes.

Common slips

- Measuring θ from the batsman rather than from the fielder's eye: the horizontal distance in $\tan\theta$ is $X - x$, not x.
- Proving only that the right v gives a constant rate, without showing that a constant rate forces the right v; it is the converse that makes this a usable rule.

Beyond STEP

This is the "optical acceleration cancellation" idea from the psychology of catching: fielders seem to run so that the ball's image rises at a steady rate, without ever predicting where it will land. It is an example of a feedback control law, where an observed quantity is held fixed rather than a trajectory being computed. For projectile motion, relative velocity and the beginnings of this way of thinking, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP II 1998, Question 12

Probability and Statistics • Bayes' Theorem

STEP II 1998, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q12

This question has a full guided treatment in *Beyond STEP*, Section 12.2, *The Base-Rate Fallacy*: https://beyondstep.keithmcnulty.org/ch12/base-rate.html

The key idea

This is Bayes' theorem, twice. The first part is the classic "rare disease" surprise: when the disease is rare, most positives can be false positives even for a fairly accurate test. For the second part, the trick is to see that the people who take STEP are not a random sample of the population. They are the AL-positive group, so the answer to the first part becomes the *prior* for the second test.

Solution

Let D be the event that a person has mathematitis, A the event that test AL is positive, and B the event that test STEP is positive. We are given

$$\mathrm{P}(D) = \tfrac{1}{100}, \qquad \mathrm{P}(A \mid D) = \tfrac{9}{10}, \qquad \mathrm{P}(A \mid D') = \tfrac{1}{11}.$$

Frankie's probability

By the law of total probability,

$$\begin{aligned}\mathrm{P}(A) &= \mathrm{P}(A \mid D)\,\mathrm{P}(D) + \mathrm{P}(A \mid D')\,\mathrm{P}(D') \\ &= \frac{9}{10}\cdot\frac{1}{100} + \frac{1}{11}\cdot\frac{99}{100} = \frac{9}{1000} + \frac{9}{100} = \frac{99}{1000}.\end{aligned}$$

Then by Bayes' theorem,

$$\mathrm{P}(D \mid A) = \frac{\mathrm{P}(A \mid D)\,\mathrm{P}(D)}{\mathrm{P}(A)} = \frac{9/1000}{99/1000} = \frac{1}{11}.$$

So even after a positive AL result, Frankie is a sufferer with probability only $\frac{1}{11}$. In a group of 1000 people, about 9 sufferers test positive but so do about 90 of the 990

non-sufferers, and the false positives swamp the true ones.

Finding p

Now restrict attention to the people for whom AL was positive. Within this group,

$$\mathrm{P}(D \mid A) = \tfrac{1}{11}, \qquad \mathrm{P}(D' \mid A) = \tfrac{10}{11}.$$

We read the given information about STEP as $\mathrm{P}(B \mid D \cap A) = 0.9$ and $\mathrm{P}(B \mid D' \cap A) = p$: STEP's behaviour depends only on whether the person is a sufferer, not on what AL said.

Ask yourself Why don't we start again from $\mathrm{P}(D) = \frac{1}{100}$?

Because STEP is only given to people who were AL-positive, and in that group the proportion of sufferers is no longer 1%; it is $\frac{1}{11}$, which is exactly what the first part found. The posterior from the first test becomes the prior for the second.

Within the AL-positive group,

$$\mathrm{P}(D \cap B \mid A) = \frac{1}{11} \cdot \frac{9}{10} = \frac{9}{110},$$
$$\mathrm{P}(D' \cap B \mid A) = \frac{10}{11} \cdot p = \frac{10p}{11}.$$

We are told that half of those who are positive on both tests are sufferers, that is $\mathrm{P}(D \mid A \cap B) = \frac{1}{2}$. This means the two probabilities above are equal:

$$\frac{9}{110} = \frac{10p}{11} \quad \Longrightarrow \quad p = \frac{9}{100} = 0.09.$$

Checks

- Odds form: the prior odds of disease are $1 : 99$. AL multiplies them by $\frac{0.9}{1/11} = 9.9$, giving $9.9 : 99 = 1 : 10$, which matches $\frac{1}{11}$. To reach odds $1 : 1$, STEP must multiply by 10, so $\frac{0.9}{p} = 10$ and $p = 0.09$.
- A simulation of two million people through both tests (with $p = 0.09$) gave 0.0909 for the proportion of sufferers among AL-positives and 0.5003 among those positive on both tests.

Common slips

- Using $\mathrm{P}(D) = 0.01$ as the prior for STEP, which gives $p = 0.0009$, a hundred times too small.
- Confusing $\mathrm{P}(A \mid D) = 0.9$ with $\mathrm{P}(D \mid A)$. The whole point of the first part is that they are very different.

Beyond STEP

Writing Bayes' theorem in odds form, with posterior odds equal to prior odds times the likelihood ratio, makes sequential testing almost mechanical: each independent piece of evidence just multiplies the odds by its own factor. This is the basis of Bayesian inference and of how medical screening programmes are actually designed. The assumption we made, that the tests are conditionally independent given disease status, is the same one behind "naive Bayes" classifiers. Blitzstein and Hwang, *Introduction to Probability*, has an excellent treatment of conditional probability and Bayes' rule with many examples like this one.

WORKED SOLUTION

STEP II 1998, Question 13

Probability and Statistics • Exponential Distribution

STEP II 1998, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q13

The key idea

The first result is the *memoryless property* of the exponential distribution: if a service has already lasted time t, the time still to go has the same distribution as a brand-new service. Once you have it, the shop story barely needs any calculation. Customers who are part-way through being served are, in terms of probability, just like customers starting afresh. So at every moment that matters, the race between the people still being served is a race between identical, independent exponential times.

Solution

The memoryless property

For $x \geq 0$,

$$\mathrm{P}(X > x) = \int_x^\infty \lambda \mathrm{e}^{-\lambda u}\,\mathrm{d}u = \Big[-\mathrm{e}^{-\lambda u}\Big]_x^\infty = \mathrm{e}^{-\lambda x}.$$

Take $s, t \geq 0$. The event $\{X > s+t\}$ is contained in $\{X > t\}$, so their intersection is just $\{X > s+t\}$. By the definition of conditional probability,

$$\begin{aligned}\mathrm{P}(X > s+t \mid X > t) &= \frac{\mathrm{P}(X > s+t \text{ and } X > t)}{\mathrm{P}(X > t)} = \frac{\mathrm{P}(X > s+t)}{\mathrm{P}(X > t)} \\ &= \frac{\mathrm{e}^{-\lambda(s+t)}}{\mathrm{e}^{-\lambda t}} = \mathrm{e}^{-\lambda s} = \mathrm{P}(X > s),\end{aligned}$$

which is the required result. (We could divide because $\mathrm{P}(X > t) = \mathrm{e}^{-\lambda t} > 0$.)

Both customers still being served

Model: the two service times are independent and each has the given exponential distribution. Let A and B be the two customers being served when I arrive.

Ask yourself Customer A has already been served for some unknown length of time. Doesn't that change the distribution of how much longer it takes?

No, and this is exactly what the first part tells us. Suppose A's service has lasted u so far. Then the probability that it lasts at least a further t is

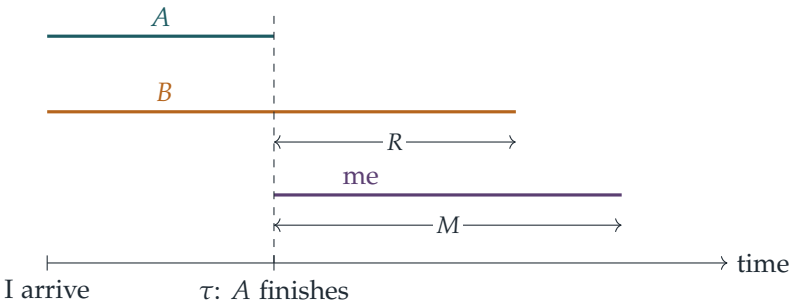
$$\mathrm{P}(X > u + t \mid X > u) = \mathrm{P}(X > t) = \mathrm{e}^{-\lambda t},$$

whatever u is. So A's *remaining* service time from the moment I walk in has the same exponential distribution with parameter λ, and the same goes for B. The two remaining times are independent because the full service times are. Hence

$$\mathrm{P}(\text{both still being served at time } t) = \mathrm{e}^{-\lambda t} \cdot \mathrm{e}^{-\lambda t} = \mathrm{e}^{-2\lambda t}.$$

Am I still being served when the other customer finishes?

Say A finishes first, at time τ after I arrive, and I go straight to A's assistant.



At time τ, let R be the time B still needs, and let M be my service time.

- M is exponential with parameter λ, since my service starts fresh at τ. It is independent of everything before, because service times for different customers are independent.
- R is also exponential with parameter λ. At time τ, B has already been served for some length of time (the time before I arrived plus τ), and by the memoryless property again the remaining time has the original distribution. It is independent of M.

I am still being served when B finishes exactly when $M > R$. Now M and R are independent with the *same* continuous distribution, so by symmetry $\mathrm{P}(M > R) = \mathrm{P}(R > M)$. Also $\mathrm{P}(M = R) = 0$ for continuous random variables, so these two probabilities add to 1. Hence

$$\mathrm{P}(\text{I am still being served when the other customer finishes}) = \tfrac{1}{2}.$$

If you prefer a calculation to the symmetry argument, condition on R:

$$\mathrm{P}(M > R) = \int_0^\infty \lambda \mathrm{e}^{-\lambda r}\, \mathrm{P}(M > r)\, \mathrm{d}r = \int_0^\infty \lambda \mathrm{e}^{-2\lambda r}\, \mathrm{d}r = \tfrac{1}{2}.$$

Checks

- At $t = 0$, $e^{-2\lambda t} = 1$: both customers are certainly being served when I arrive. As $t \to \infty$ the probability tends to 0, as it should.
- The answer $\frac{1}{2}$ doesn't depend on λ, which is what we'd expect, because changing λ only changes the unit of time.
- A simulation with $\lambda = 0.7$ and $t = 0.9$ gave 0.2831 for both customers still being served, against $e^{-1.26} \approx 0.2837$. In a simulation where both customers had already been in service for random lengths of time, I was still being served in a proportion 0.5000 of trials.

Common slips

- Thinking the customers already being served are "due to finish soon", so that their remaining times are shorter. For the exponential distribution, what has already happened makes no difference.
- Writing $P(X > s + t \text{ and } X > t)$ as $P(X > s + t)\,P(X > t)$. These events are not independent; one contains the other.

Beyond STEP

The exponential is the only continuous distribution with the memoryless property. That fact underpins the *Poisson process* and *queueing theory*, where a shop with c assistants and exponential service times is the classic M/M/c queue. The calculation for two customers extends to n: the minimum of n independent exponential variables with parameter λ is exponential with parameter $n\lambda$ (multiply the tail probabilities), and by symmetry each variable is equally likely to be the smallest. Grimmett and Welsh, *Probability: An Introduction*, covers exponential variables and the Poisson process at this level.

WORKED SOLUTION

STEP II 1998, Question 14

Probability and Statistics • Binomial and Poisson Expectation

STEP II 1998, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q14

The key idea

Despite the database label, this is a probability question: Dr Doom should take the new scheme if her *expected* salary under it beats the fixed salary A. The number of failures is binomial, so we need $\mathrm{E}\big[1/(1+X)\big]$. The awkward factor $\frac{1}{k+1}$ is absorbed by the identity $\frac{1}{k+1}\binom{N}{k} = \frac{1}{N+1}\binom{N+1}{k+1}$, which turns the sum into an almost complete binomial expansion of $(p+q)^{N+1}$. The same trick, with $\frac{\lambda^k}{(k+1)!}$, handles the Poisson model.

Solution

When to accept the new scheme

Let X be the number of Dr Doom's students who fail, and write $q = 1-p$. The students fail independently, each with probability p, so $X \sim \mathrm{B}(N,p)$:

$$\mathrm{P}(X=k) = \binom{N}{k} p^k q^{N-k}, \qquad k = 0,1,\ldots,N.$$

We assume Dr Doom is content to judge the schemes by expected salary. Under the new scheme her salary is the random variable $S = B/(1+X)$, so

$$\mathrm{E}(S) = B\sum_{k=0}^{N} \frac{1}{k+1}\binom{N}{k} p^k q^{N-k}.$$

Ask yourself How do we get rid of the $\frac{1}{k+1}$ inside the sum?

Use factorials:

$$\begin{aligned}\frac{1}{k+1}\binom{N}{k} &= \frac{N!}{(k+1)!\,(N-k)!} = \frac{1}{N+1}\cdot\frac{(N+1)!}{(k+1)!\,\big((N+1)-(k+1)\big)!} \\ &= \frac{1}{N+1}\binom{N+1}{k+1}.\end{aligned}$$

Substituting this, and multiplying and dividing by p so that the power of p matches the new binomial coefficient,

$$\begin{aligned}\mathrm{E}(S) &= \frac{B}{(N+1)p}\sum_{k=0}^{N}\binom{N+1}{k+1}p^{k+1}q^{N-k}\\ &= \frac{B}{(N+1)p}\sum_{j=1}^{N+1}\binom{N+1}{j}p^{j}q^{N+1-j}\\ &= \frac{B}{(N+1)p}\left[(p+q)^{N+1}-q^{N+1}\right]\end{aligned}$$

putting $j = k+1$ and then using the binomial theorem, since the sum is the full expansion of $(p+q)^{N+1}$ with only the $j=0$ term, q^{N+1}, missing. As $p+q=1$,

$$\mathrm{E}(S) = \frac{B\big(1-(1-p)^{N+1}\big)}{(N+1)p}.$$

She should accept if $\mathrm{E}(S) > A$. Multiplying through by $(N+1)p$, which is positive (we need $p>0$ for the formula to make sense; if $p=0$ she simply compares B with A), this is

$$A(N+1)p < B\big(1-(1-p)^{N+1}\big),$$

which is the required condition.

The Poisson model

A binomial distribution $\mathrm{B}(N,p)$ is well approximated by a Poisson distribution with mean $\lambda = Np$ when N is large and p is small, with Np of moderate size. So Dr Doom's failures could be modelled as Poisson if she has a large class, each student is unlikely to fail, and failures still occur independently. (More generally, a Poisson model suits failures that occur singly, independently and at a constant average rate.)

With $X \sim \mathrm{Po}(\lambda)$, $\lambda = Np$, the same idea works, now with $\frac{1}{k+1}\cdot\frac{1}{k!} = \frac{1}{(k+1)!}$:

$$\begin{aligned}\mathrm{E}\Big(\frac{B}{1+X}\Big) &= B\sum_{k=0}^{\infty}\frac{1}{k+1}\cdot\frac{\mathrm{e}^{-\lambda}\lambda^{k}}{k!} = \frac{B\mathrm{e}^{-\lambda}}{\lambda}\sum_{k=0}^{\infty}\frac{\lambda^{k+1}}{(k+1)!}\\ &= \frac{B\mathrm{e}^{-\lambda}}{\lambda}\big(\mathrm{e}^{\lambda}-1\big) = \frac{B\big(1-\mathrm{e}^{-\lambda}\big)}{\lambda},\end{aligned}$$

since the last sum is the exponential series with its first term, 1, missing. So Dr Doom's expected salary under this model is

$$\frac{B\big(1-\mathrm{e}^{-Np}\big)}{Np} \text{ pounds.}$$

Checks

- $N=1$: directly, $\mathrm{E}(S) = B\big(q+\frac{1}{2}p\big) = B\big(1-\frac{1}{2}p\big)$, and the formula gives $B\,\frac{1-q^2}{2p} = B\,\frac{1+q}{2} = B\big(1-\frac{1}{2}p\big)$.

- $p \to 0$: the binomial expression tends to B (no one fails), and so does the Poisson one as $\lambda \to 0$.
- The two answers agree when the approximation should be good: with $N = 1000$, $p = 0.003$, $\mathrm{E}(S)/B$ is 0.31655 from the binomial formula and 0.31674 from the Poisson one. Exact enumeration and a simulation with $N = 20$, $p = 0.3$ both give $\mathrm{E}(S)/B \approx 0.1586$.

Common slips

- Writing $\mathrm{E}\left(\frac{1}{1+X}\right) = \frac{1}{1+\mathrm{E}(X)} = \frac{1}{1+Np}$. Expectation does not pass through non-linear functions; in fact $\frac{1}{1+x}$ is convex, so by Jensen's inequality the true value is larger.
- Forgetting that the re-indexed sum misses the $j = 0$ term, giving $\mathrm{E}(S) = B/((N+1)p)$.

Beyond STEP

The trick here is a special case of computing $\mathrm{E}\big(1/(1+X)\big)$ via $\frac{1}{1+x} = \int_0^1 t^x \,\mathrm{d}t$, which links it to the probability generating function $G(t) = \mathrm{E}(t^X)$: the answer is $\int_0^1 G(t)\,\mathrm{d}t$. Try it with $G(t) = (q+pt)^N$ and $G(t) = \mathrm{e}^{\lambda(t-1)}$. Generating functions, and the convergence of binomial to Poisson, are covered well in Grimmett and Welsh, *Probability: An Introduction*.

WORKED SOLUTION

STEP III 1998, Question 1

Pure • Curve Sketching

STEP III 1998, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q1

The key idea

Write everything in terms of $\cos x$: with $\sin^2 x = 1 - \cos^2 x$, the function f becomes a quadratic in $\cos x$. Completing the square tells us the range of f at a glance, and this range is what controls the final part. For g, the quotient rule gives a derivative that is a constant multiple of $\mathrm{f}'(x)$, divided by a square.

Solution

Sketching $y = \mathrm{f}(x)$

Using $\sin^2 x = 1 - \cos^2 x$,

$$\mathrm{f}(x) = 2 + 2\cos x - \cos^2 x = 3 - (1 - \cos x)^2.$$

Since $0 \le 1 - \cos x \le 2$, we get $-1 \le \mathrm{f}(x) \le 3$. The value 3 is attained when $\cos x = 1$ and the value -1 when $\cos x = -1$.

Differentiating,

$$\mathrm{f}'(x) = -2(1 - \cos x)\sin x = 2\sin x\,(\cos x - 1).$$

This is zero when $\sin x = 0$ or $\cos x = 1$, which on $[0, 2\pi]$ means $x = 0,\ \pi,\ 2\pi$. The stationary points are

$$(0,\,3), \qquad (\pi,\,-1), \qquad (2\pi,\,3).$$

On $(0, \pi)$ we have $\sin x > 0$ and $\cos x - 1 < 0$, so $\mathrm{f}' < 0$ and f is decreasing. On $(\pi, 2\pi)$, $\sin x < 0$ while $\cos x - 1$ is still negative, so $\mathrm{f}' > 0$ and f is increasing. So $(\pi, -1)$ is a minimum, and the end points $(0, 3)$ and $(2\pi, 3)$ are maxima.

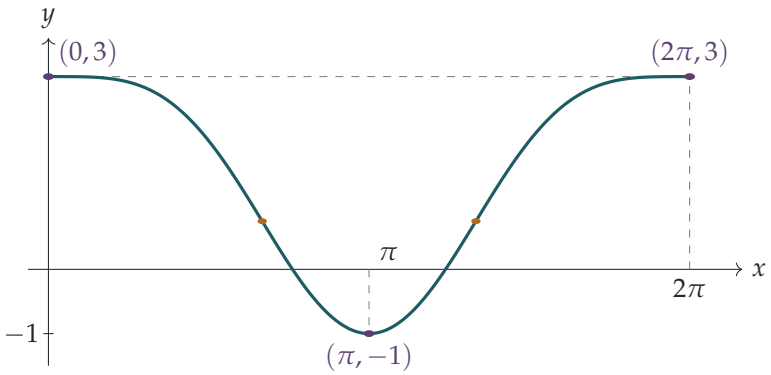
? **Ask yourself** How does the curve look near $x = 0$? Is it like a parabola?

No, it's flatter than a parabola. Near $x = 0$, $1 - \cos x \approx \frac{1}{2}x^2$, so $\mathrm{f}(x) \approx 3 - \frac{1}{4}x^4$. The curve is very flat at the top, which is consistent with $\mathrm{f}''(0) = 0$. Near $x = \pi$, write

$x = \pi + t$. Then $1 - \cos x = 1 + \cos t \approx 2 - \frac{1}{2}t^2$, so $\mathrm{f} \approx -1 + 2t^2$: an ordinary parabolic minimum.

Some other useful features for the sketch:

- The curve is symmetric about $x = \pi$, since $\cos(2\pi - x) = \cos x$.
- It crosses the x-axis where $(1 - \cos x)^2 = 3$, that is, $\cos x = 1 - \sqrt{3}$. This gives $x = \arccos(1 - \sqrt{3}) \approx 2.39$ and $x = 2\pi - \arccos(1 - \sqrt{3}) \approx 3.89$.
- $\mathrm{f}''(x) = 2\cos 2x - 2\cos x = 2(2\cos x + 1)(\cos x - 1)$. This changes sign where $\cos x = -\frac{1}{2}$, so the points of inflection are at $\left(\frac{2\pi}{3}, \frac{3}{4}\right)$ and $\left(\frac{4\pi}{3}, \frac{3}{4}\right)$.



The amber dots are the points of inflection.

Stationary points of g

By the quotient rule, wherever $c\,\mathrm{f}(x) + d \neq 0$,

$$\mathrm{g}'(x) = \frac{a\mathrm{f}'(x)\big(c\mathrm{f}(x) + d\big) - c\mathrm{f}'(x)\big(a\mathrm{f}(x) + b\big)}{\big(c\mathrm{f}(x) + d\big)^2} = \frac{(ad - bc)\,\mathrm{f}'(x)}{\big(c\mathrm{f}(x) + d\big)^2}.$$

Since $ad \neq bc$, we have $\mathrm{g}'(x) = 0$ exactly when $\mathrm{f}'(x) = 0$, at every point where g is defined. So g has no stationary points other than those of f.

We still need g to be defined at $x = 0, \pi, 2\pi$, which is where the other two conditions come in. At $x = 0$ and $x = 2\pi$ the denominator is $3c + d$, which is non-zero because $d \neq -3c$. At $x = \pi$ it is $d - c$, which is non-zero because $d \neq c$. So the stationary points of $y = \mathrm{g}(x)$ are at exactly $x = 0,\ \pi,\ 2\pi$, as required, with

$$\mathrm{g}(0) = \mathrm{g}(2\pi) = \frac{3a + b}{3c + d}, \qquad \mathrm{g}(\pi) = \frac{b - a}{d - c}.$$

When $|\mathrm{g}|$ is bounded

Suppose $d/c < -3$ or $d/c > 1$. In particular $c \neq 0$, so we can write

$$c\,\mathrm{f}(x) + d = c\left(\mathrm{f}(x) + \frac{d}{c}\right).$$

As $\mathrm{f}(x)$ takes values in $[-1,3]$, the bracket lies in $\left[\frac{d}{c}-1,\ \frac{d}{c}+3\right]$.

- If $d/c > 1$, the bracket is at least $\frac{d}{c}-1>0$.
- If $d/c < -3$, the bracket is at most $\frac{d}{c}+3<0$.

Either way there is a constant $k>0$ with $\left|c\,\mathrm{f}(x)+d\right| \geq |c|\,k$ for all x. In words, the denominator stays away from zero. The numerator is bounded, because $|a\mathrm{f}(x)+b| \leq 3|a|+|b|$. Hence

$$|\mathrm{g}(x)| \leq \frac{3|a|+|b|}{|c|\,k}$$

for every x in $[0,2\pi]$, so $|\mathrm{g}(x)|$ cannot be arbitrarily large.

Ask yourself What goes wrong if $-3 < d/c < 1$?

Then $-d/c$ lies strictly between -1 and 3. Since f is continuous and takes every value in $[-1,3]$, there is an x with $\mathrm{f}(x) = -d/c$, and the denominator vanishes there. The numerator does not vanish at the same point: if both did, we would have $a(-d/c)+b = 0$, that is, $ad = bc$. So g has a vertical asymptote, and $|\mathrm{g}|$ is unbounded. This shows that the condition in the final part is exactly the right one.

Checks

- Numerically, f over a fine grid on $[0,2\pi]$ has minimum -1 and maximum 3. Also $3-\mathrm{f}(0.1) \approx 2.496\times10^{-5}$, against $\frac{1}{4}(0.1)^4 = 2.5\times10^{-5}$, confirming the quartic flatness at $x=0$.
- A special case: $a=1$, $b=0$, $c=0$, $d=1$ gives $\mathrm{g}=\mathrm{f}$, and the formulas give $\mathrm{g}(0)=3$ and $\mathrm{g}(\pi)=-1$, as they should.
- For random a,b,c,d, a finite-difference derivative of g agrees with $(ad-bc)\mathrm{f}'/(c\mathrm{f}+d)^2$.

Common slips

- Missing the end points $x=0$ and $x=2\pi$ as stationary points. f' really is zero there.
- Forgetting to check that g is defined at the stationary points of f. That is exactly what the conditions $d \neq -3c$ and $d \neq c$ are for.
- Drawing a sharp, parabola-shaped top at $x=0$. The curve is much flatter there.

Beyond STEP

The map $t \mapsto \dfrac{at+b}{ct+d}$ with $ad \neq bc$ is a *Möbius transformation*. It is invertible, and its derivative $(ad-bc)/(ct+d)^2$ never vanishes, which is why composing with it cannot

create or destroy stationary points. Möbius maps are central in complex analysis and geometry, and they form a group under composition. For a beautiful picture-led account, see Needham, *Visual Complex Analysis*.

WORKED SOLUTION

STEP III 1998, Question 2

Pure • Beta Integral Recurrences

STEP III 1998, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q2

The key idea

This integral is the beta function in disguise. Each part needs one standard move. Part (i) uses the reflection $t \mapsto 1-t$, which swaps the roles of t and $1-t$. Part (ii) multiplies the integrand by $1 = t + (1-t)$. Part (iii) integrates by parts, which moves one power from $(1-t)$ onto t. Repeating part (iii) until the power of $(1-t)$ is zero leaves an integral we can do at once.

Solution

Part (i)

Substitute $u = 1-t$, so $\mathrm{d}t = -\mathrm{d}u$, and the limits $t = 0, 1$ become $u = 1, 0$:

$$\mathrm{I}(a,b) = \int_1^0 (1-u)^a u^b \,(-\mathrm{d}u) = \int_0^1 u^b (1-u)^a \,\mathrm{d}u = \mathrm{I}(b,a).$$

The minus sign from $\mathrm{d}t = -\mathrm{d}u$ is used up in swapping the limits back to 0 and 1.

Part (ii)

Since $t + (1-t) = 1$, we can write the integrand as

$$t^a(1-t)^b = t^a(1-t)^b\big(t + (1-t)\big) = t^{a+1}(1-t)^b + t^a(1-t)^{b+1}.$$

Integrating each term over $[0,1]$ gives $\mathrm{I}(a,b) = \mathrm{I}(a+1,b) + \mathrm{I}(a,b+1)$, as required.

Part (iii)

We integrate $\mathrm{I}(a,b)$ by parts, differentiating $(1-t)^b$ and integrating t^a:

$$\begin{aligned}\mathrm{I}(a,b) &= \left[\frac{t^{a+1}}{a+1}(1-t)^b\right]_0^1 - \int_0^1 \frac{t^{a+1}}{a+1}\cdot\left(-b(1-t)^{b-1}\right)\mathrm{d}t \\ &= \left[\frac{t^{a+1}}{a+1}(1-t)^b\right]_0^1 + \frac{b}{a+1}\int_0^1 t^{a+1}(1-t)^{b-1}\,\mathrm{d}t.\end{aligned}$$

Ask yourself Why does the boundary term vanish, and where do we need a and b to be positive?

At $t=1$ the factor $(1-t)^b$ is 0 because $b>0$. At $t=0$ the factor t^{a+1} is 0 because $a+1>0$. So the bracket is zero, and multiplying through by $a+1$ gives

$$(a+1)\,\mathrm{I}(a,b) = b\,\mathrm{I}(a+1,b-1),$$

which is the result. (If $0<b<1$, the last integral has a negative power of $(1-t)$. It still converges, because $b-1>-1$. For the integer case below we always have $b-1\geq 0$ anyway.)

Calculating $\mathrm{I}(a,b)$ for integer a,b. Write the result as $\mathrm{I}(a,b) = \dfrac{b}{a+1}\,\mathrm{I}(a+1,b-1)$ and apply it again and again. Each step raises the first argument by 1 and lowers the second by 1. We stop once the second argument reaches 0:

$$\begin{aligned}\mathrm{I}(a,b) &= \frac{b}{a+1}\,\mathrm{I}(a+1,b-1) = \frac{b}{a+1}\cdot\frac{b-1}{a+2}\,\mathrm{I}(a+2,b-2) = \cdots \\ &= \frac{b(b-1)\cdots 1}{(a+1)(a+2)\cdots(a+b)}\,\mathrm{I}(a+b,0).\end{aligned}$$

Every step is legitimate, since the second argument is a positive integer each time we use the formula. Now $\mathrm{I}(a+b,0) = \int_0^1 t^{a+b}\,\mathrm{d}t = \dfrac{1}{a+b+1}$. Also $(a+1)(a+2)\cdots(a+b) = \dfrac{(a+b)!}{a!}$, so

$$\boxed{\mathrm{I}(a,b) = \frac{a!\,b!}{(a+b+1)!}}$$

The formula is symmetric in a and b, as part (i) says it must be. It also holds when a or b is 0.

Checks

- $\mathrm{I}(1,1) = \int_0^1 (t-t^2)\,\mathrm{d}t = \frac{1}{2}-\frac{1}{3} = \frac{1}{6}$, and the formula gives $\frac{1!\,1!}{3!} = \frac{1}{6}$.
- The formula satisfies part (ii): $\frac{(a+1)!\,b!}{(a+b+2)!} + \frac{a!\,(b+1)!}{(a+b+2)!} = \frac{a!\,b!\,(a+b+2)}{(a+b+2)!} = \frac{a!\,b!}{(a+b+1)!}$.

- Numerically, Simpson's rule gives $\mathrm{I}(2,3) \approx 0.0166667$, which matches $\frac{2!\,3!}{6!} = \frac{1}{60}$. We also checked parts (i)–(iii) by quadrature at random non-integer a, b, with agreement to about 10^{-11}.

Common slips

- Integrating by parts the other way round. Differentiating t^a raises the power of $(1-t)$, which is the wrong direction.
- Forgetting to justify the boundary term. It vanishes only because $a+1 > 0$ and $b > 0$.
- Stopping the recursion one step early, or forgetting the final factor $\frac{1}{a+b+1}$ from $\mathrm{I}(a+b, 0)$.

Beyond STEP

$\mathrm{I}(a,b)$ is the beta function $B(a+1, b+1)$. For all real positive arguments it is linked to the gamma function by $B(x,y) = \Gamma(x)\Gamma(y)/\Gamma(x+y)$. Our answer is the special case $\Gamma(n+1) = n!$. In probability, $t^a(1-t)^b/\mathrm{I}(a,b)$ is the density of the beta distribution, which is central in Bayesian statistics. Blitzstein and Hwang, *Introduction to Probability*, has an attractive treatment, including a counting argument that explains why the answer is $a!\,b!/(a+b+1)!$.

WORKED SOLUTION

STEP III 1998, Question 3

Pure • Linear Recurrence Relations

STEP III 1998, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q3

The key idea

This is a first-order linear recurrence with a constant term, the discrete cousin of the differential equation $y' = cy - d$. The trick is the same as for the ODE: find the constant ("equilibrium") solution, then notice that the deviation from it simply gets multiplied by the same factor every day. For the limit, the only real work is understanding what $\big((1+c)^N - 1\big)/c$ does as $c \to 0$, and the binomial theorem settles it.

Solution

Solving for $c > 0$

We try $V_T = Ak^T + B$ for every T. Substituting into the recurrence,

$$Ak^{N+1} + B = (1+c)\big(Ak^N + B\big) - d \quad \Longleftrightarrow \quad Ak^N\big(k - (1+c)\big) = cB - d.$$

The right-hand side does not depend on N, while the left-hand side does unless it vanishes (we want $A \neq 0$ and $k \neq 0$, or the form is just a constant). So we need both sides to be zero for every N:

$$k = 1 + c, \qquad B = \frac{d}{c},$$

where dividing by c is fine because $c > 0$. With these values, $V_T = A(1+c)^T + d/c$ satisfies the recurrence for any constant A. Putting $T = 0$ gives $V_0 = A + d/c$, so $A = V_0 - d/c$ and

$$\boxed{V_N = \Big(V_0 - \frac{d}{c}\Big)(1+c)^N + \frac{d}{c}}$$

Ask yourself We guessed a form and it worked. How do we know there isn't some other solution?

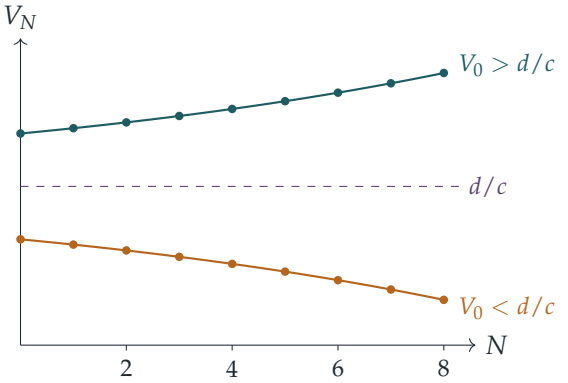
The recurrence determines V_1 from V_0, then V_2 from V_1, and so on, so there is exactly one sequence with a given V_0. Our formula satisfies the recurrence and has the right V_0, so it *is* that sequence. (If you prefer a direct argument, put $U_N = V_N - d/c$; the recurrence becomes $U_{N+1} = (1+c)U_N$, so $U_N = (1+c)^N U_0$, which is the same answer.)

It will be useful to rearrange the answer as

$$V_N = V_0(1+c)^N - d\,\frac{(1+c)^N - 1}{c}. \qquad (*)$$

This has a nice reading: the first term is V_0 growing at rate c per day, and the second is $d\big(1 + (1+c) + \cdots + (1+c)^{N-1}\big)$, the daily deductions each grown for the days remaining, summed as a geometric series.

The constant d/c is the equilibrium value. If $V_0 > d/c$ the bond grows without bound, if $V_0 = d/c$ it stays put, and if $V_0 < d/c$ it falls away ever faster:



(The picture uses $c = 0.1$ and $d/c = 5$, with $V_0 = 6$ and $V_0 = 4$.)

The case $c = 0$

Now the recurrence is $V_{N+1} = V_N - d$: the value drops by d each day, an arithmetic progression. So

$$\boxed{V_N = V_0 - Nd}$$

Notice our $c > 0$ formula cannot simply be evaluated at $c = 0$, because of the d/c; that is exactly why the question asks about the limit.

The limit as $c \to 0$

Fix N and use $(*)$. The first term tends to V_0, since $(1+c)^N \to 1$ for fixed N. For the second, expand by the binomial theorem:

$$(1+c)^N - 1 = Nc + \binom{N}{2}c^2 + \cdots + c^N,$$

$$\text{so} \quad \frac{(1+c)^N - 1}{c} = N + \binom{N}{2}c + \cdots + c^{N-1}.$$

Because N is fixed, this is a polynomial in c with finitely many terms, and every term after the first contains a positive power of c. Hence it tends to N as $c \to 0$, and

$$\lim_{c \to 0} V_N = V_0 - Nd,$$

which is the $c = 0$ solution, as required. (Equivalently, $\big((1+c)^N - 1\big)/c$ is the difference quotient for x^N at $x = 1$, so its limit is the derivative N.)

The words "for fixed N" matter: with c small but N large, terms like $\binom{N}{2}c$ need not be small, and the exponential growth eventually wins over the linear behaviour.

Checks

- $N = 1$: the formula gives $(V_0 - d/c)(1+c) + d/c = (1+c)V_0 - d$, which is the recurrence itself.
- $V_0 = d/c$ gives $V_N = d/c$ for all N, the equilibrium.
- Numerically, with $V_0 = 7$, $d = 2$, $N = 12$: the formula gives $V_{12} \approx -17.48$ for $c = 0.01$ and -17.005 for $c = 10^{-4}$, approaching $V_0 - 12d = -17$. The closed form also agrees exactly with direct iteration in many random rational cases.

Common slips

- Taking $B = -d/c$: the sign comes from $cB - d = 0$, so $B = +d/c$.
- Using $A = V_0$ instead of $A = V_0 - d/c$: the constant B also contributes at $T = 0$.
- Trying to set $c = 0$ directly in the $c > 0$ formula, which is undefined; the limit is needed.

Beyond STEP

Recurrences like this are *linear difference equations*, and their theory runs parallel to linear ODEs: the general solution is a particular (here constant) solution plus the general solution of the homogeneous equation $U_{N+1} = (1+c)U_N$. In finance this is the formula for a loan or annuity with fixed repayments, and the $c \to 0$ limit is the passage from compound to simple behaviour. Boyce and DiPrima, *Elementary Differential Equations*, includes a section on first-order difference equations alongside the ODE theory.

WORKED SOLUTION

STEP III 1998, Question 4

Pure • Polar Rose Curves

STEP III 1998, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q4

The key idea

Polar curves are easiest to understand by converting to Cartesian form when the equation is simple, and by tracking the sign of r when it isn't. The whole question hinges on the convention that a point with $r < 0$ is plotted at distance $|r|$ in the *opposite* direction, $\theta + \pi$. For $r = \cos k\theta$ that convention decides whether the negative loops land on fresh ground (giving new petals) or on top of petals already drawn.

Solution

The circle $r = \cos\theta$

Multiply both sides by r and use $x = r\cos\theta$, $y = r\sin\theta$, $r^2 = x^2 + y^2$:

$$\begin{aligned} r^2 = r\cos\theta \quad &\Longrightarrow \quad x^2 + y^2 = x \\ &\Longleftrightarrow \quad \left(x - \tfrac{1}{2}\right)^2 + y^2 = \tfrac{1}{4}. \end{aligned}$$

So every point of the curve lies on the circle with centre $(\frac{1}{2}, 0)$ and radius $\frac{1}{2}$.

Conversely, take any point (x, y) of this circle other than the origin. Since $x = x^2 + y^2 > 0$, its polar angle satisfies $-\frac{1}{2}\pi < \theta < \frac{1}{2}\pi$, and dividing $r^2 = x = r\cos\theta$ by $r > 0$ gives $r = \cos\theta$. The origin itself is reached at $\theta = \pm\frac{1}{2}\pi$. Hence the curve is exactly this circle, traced once as θ runs from $-\frac{1}{2}\pi$ to $\frac{1}{2}\pi$, and on this range $r = \cos\theta \geq 0$ throughout, so no sign convention is needed.

The curve $r = \cos 2\theta$

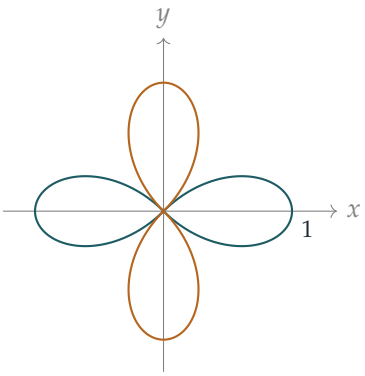
Now r changes sign, and we plot (r, θ) with $r < 0$ at distance $|r|$ in direction $\theta + \pi$.

- For $-\frac{1}{4}\pi < \theta < \frac{1}{4}\pi$, $r > 0$, rising from 0 to 1 at $\theta = 0$ and back to 0: a petal along the positive x-axis.
- For $\frac{1}{4}\pi < \theta < \frac{3}{4}\pi$, $r < 0$, so these points are drawn in directions $\frac{5}{4}\pi$ to $\frac{7}{4}\pi$: a petal

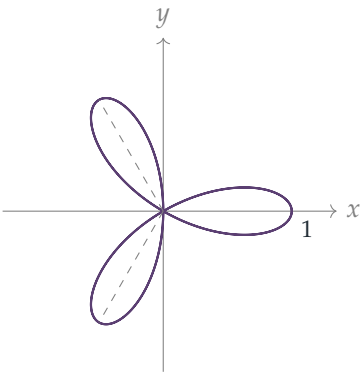
along the negative y-axis.

- Similarly $\frac{3}{4}\pi < \theta < \frac{5}{4}\pi$ $(r > 0)$ gives a petal along the negative x-axis, and $\frac{5}{4}\pi < \theta < \frac{7}{4}\pi$ $(r < 0)$ gives one along the positive y-axis.

The result is a four-leaved rose, each petal of length 1, symmetric about both axes and touching the others only at the origin. In the sketch, teal petals come from $r > 0$ and amber petals from $r < 0$.



$r = \cos 2\theta$



$r = \cos 3\theta$

The curves $r = \cos 2n\theta$

Since $\cos(-2n\theta) = \cos 2n\theta$, n and $-n$ give the same curve, so take $n \geq 1$ (we come back to $n = 0$ below).

Ask yourself When $r < 0$, do the loops land on top of the ones already drawn, or somewhere new?

We have $r > 0$ on intervals of length $\pi/(2n)$ centred at $\theta = k\pi/n = 2k \cdot \frac{\pi}{2n}$, for $k = 0, 1, \ldots, 2n-1$. Each gives a petal pointing in the direction $2k \cdot \frac{\pi}{2n}$, an *even* multiple of $\frac{\pi}{2n}$.

We have $r < 0$ on intervals centred at $\theta = (2k+1)\frac{\pi}{2n}$, and these points are plotted in direction $\theta + \pi$, so the petal points along

$$(2k+1)\frac{\pi}{2n} + \pi = (2k+1+2n)\frac{\pi}{2n},$$

an *odd* multiple of $\frac{\pi}{2n}$. So the negative loops land somewhere new, in the gaps between the positive petals.

Hence $r = \cos 2n\theta$ is a rose with $4n$ congruent petals, each of length 1, pointing along the directions $m\pi/(2n)$ for $m = 0, 1, \ldots, 4n-1$, each petal occupying an angle $\pi/(2n)$, and neighbouring petals meeting only at the origin. (For $n = 1$ this is the four-leaved rose above.) The petals have total angle $4n \cdot \frac{\pi}{2n} = 2\pi$, so they fill the full turn with no overlap.

For the degenerate case $n = 0$ the curve is $r = 1$, the unit circle.

The area is independent of n

Because the petals don't overlap, each part of the enclosed region is swept exactly once as θ goes from 0 to 2π, and the area formula $\frac{1}{2}\int r^2\,\mathrm{d}\theta$ (which only involves r^2, so the sign of r doesn't matter) gives, for $n \geq 1$,

$$\begin{aligned} A &= \frac{1}{2}\int_0^{2\pi} \cos^2 2n\theta\,\mathrm{d}\theta = \frac{1}{4}\int_0^{2\pi} (1+\cos 4n\theta)\,\mathrm{d}\theta \\ &= \frac{1}{4}\left[\theta + \frac{\sin 4n\theta}{4n}\right]_0^{2\pi} = \frac{\pi}{2}, \end{aligned}$$

since $\sin 8n\pi = 0$. This doesn't depend on n, as required: adding more petals makes each petal thinner in exact compensation. (The excluded case $n = 0$, the unit circle, has area π; the formula's $\sin 4n\theta/(4n)$ term is exactly where $n \neq 0$ is used.)

The curve $r = \cos 3\theta$

> **Ask yourself** Why does this curve have three petals, not six?

Now $r > 0$ on intervals centred at $\theta = 0, \frac{2}{3}\pi, \frac{4}{3}\pi$, giving petals in those directions. The negative loops are centred at $\theta = \frac{1}{3}\pi, \pi, \frac{5}{3}\pi$, and are plotted in directions $\frac{4}{3}\pi$, 2π and $\frac{8}{3}\pi \equiv \frac{2}{3}\pi$: exactly the directions of the positive petals. Indeed, replacing θ by $\theta + \pi$ gives

$$r = \cos(3\theta + 3\pi) = -\cos 3\theta$$

in the opposite direction, which is the same point. So $r = \cos 3\theta$ is a three-leaved rose, petals of length 1 along $\theta = 0, \frac{2}{3}\pi, \frac{4}{3}\pi$, each occupying an angle $\frac{1}{3}\pi$, and the whole curve is traced *twice* as θ runs from 0 to 2π (once for $0 \leq \theta \leq \pi$). This is the sketch on the right above.

The same argument shows that for any odd k, $r = \cos k\theta$ is a k-leaved rose traced twice, while for even k it is a $2k$-leaved rose traced once.

Checks

- Put $\theta = 0$ in $r = \cos\theta$: the point $(1, 0)$ is on $(x - \frac{1}{2})^2 + y^2 = \frac{1}{4}$, and so is the origin (at $\theta = \pm\frac{1}{2}\pi$).
- The circle $r = \cos\theta$ has area $\pi(\frac{1}{2})^2 = \frac{1}{4}\pi$, and $\frac{1}{2}\int_{-\pi/2}^{\pi/2} \cos^2\theta\,\mathrm{d}\theta = \frac{1}{4}\pi$ agrees. Similarly $\frac{1}{2}\int_0^{\pi} \cos^2 3\theta\,\mathrm{d}\theta = \frac{1}{4}\pi$ for the three-leaved rose. Using 0 to 2π there would double count and give $\frac{1}{2}\pi$.
- A numerical check: plotting $r = \cos 2n\theta$ for $n = 1, \ldots, 5$ finds $4n$ petal tips, and a Monte Carlo estimate of the enclosed area gave $1.57 \approx \pi/2 = 1.5708$ every time.

Common slips

- Ignoring negative r and drawing only two petals for $r = \cos 2\theta$.
- Saying that $r = \cos 3\theta$ has six petals because it has six loops of r: the negative loops land on top of the positive ones.
- Using $\frac{1}{2}\int_0^{2\pi} r^2 \, \mathrm{d}\theta$ for a curve traced twice (such as $r = \cos 3\theta$), which doubles the area.

Beyond STEP

These curves are the *rhodonea* or rose curves. Writing a point as the complex number $z = r\,\mathrm{e}^{\mathrm{i}\theta}$, the rose $r = \cos k\theta$ is $z = \frac{1}{2}\left(\mathrm{e}^{\mathrm{i}(k+1)\theta} + \mathrm{e}^{-\mathrm{i}(k-1)\theta}\right)$, the sum of two circular motions in opposite directions. This viewpoint links roses to epicycles and to Fourier series, and the area formula $\frac{1}{2}\int r^2 \, \mathrm{d}\theta$ is a special case of Green's theorem in vector calculus. Needham's *Visual Complex Analysis* develops this geometric way of seeing complex numbers beautifully.

WORKED SOLUTION

STEP III 1998, Question 5

Pure • Matrix Exponential

STEP III 1998, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q5

The key idea

An infinite series of matrices looks frightening, but both matrices here have powers that repeat in a very simple pattern: the powers of $\mathbf{M}$ cycle like the powers of i, and the powers of $\mathbf{N}$ vanish from the square onwards. Once you know every power, the series splits into familiar scalar series (or stops altogether), and you can read off the transformations. The final part is then a direct matrix calculation, and its geometric meaning comes from asking which lines each transformation leaves alone.

Solution

The matrix $\mathbf{M}$

Multiplying out,

$$\mathbf{M}^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -\mathbf{I},$$

as required. Hence, for every integer $k \geq 0$,

$$\mathbf{M}^{2k} = (\mathbf{M}^2)^k = (-1)^k\mathbf{I}, \qquad \mathbf{M}^{2k+1} = (-1)^k\mathbf{M}.$$

Splitting the series into even and odd powers of $\theta\mathbf{M}$,

$$\begin{aligned} \exp(\theta\mathbf{M}) &= \sum_{k=0}^{\infty} \frac{\theta^{2k}}{(2k)!}\mathbf{M}^{2k} + \sum_{k=0}^{\infty} \frac{\theta^{2k+1}}{(2k+1)!}\mathbf{M}^{2k+1} \\ &= \Big(\sum_{k=0}^{\infty} \frac{(-1)^k\theta^{2k}}{(2k)!}\Big)\mathbf{I} + \Big(\sum_{k=0}^{\infty} \frac{(-1)^k\theta^{2k+1}}{(2k+1)!}\Big)\mathbf{M} \\ &= \cos\theta\,\mathbf{I} + \sin\theta\,\mathbf{M} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}. \end{aligned}$$

(Splitting is legitimate because each entry of the matrix series is a series of real numbers whose terms are bounded in size by $|\theta|^r/r!$, so it converges absolutely and may be regrouped.)

Ask yourself Where have I seen $\cos\theta\,\mathbf{I} + \sin\theta\,\mathbf{M}$ before?

Since $\mathbf{M}^2 = -\mathbf{I}$, the matrix $\mathbf{M}$ behaves exactly like i, and the calculation is the matrix version of $e^{i\theta} = \cos\theta + i\sin\theta$. Geometrically, $\exp(\theta\mathbf{M})$ sends $(1,0)$ to $(\cos\theta, \sin\theta)$ and $(0,1)$ to $(-\sin\theta, \cos\theta)$: it is an **anticlockwise rotation of the plane about the origin through angle** θ. (Indeed $\mathbf{M}$ itself is the rotation through $\frac{\pi}{2}$, just as multiplication by i is.)

The matrix $\mathbf{N}$

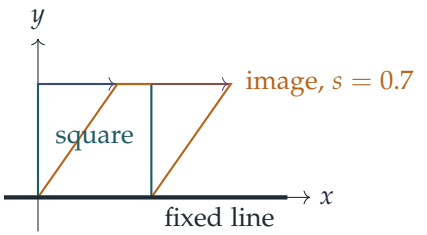
Here

$$\mathbf{N}^2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix},$$

so $\mathbf{N}^r = \mathbf{0}$ for all $r \geq 2$ and the series stops after two terms:

$$\exp(s\mathbf{N}) = \mathbf{I} + s\mathbf{N} = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}.$$

This sends (x, y) to $(x + sy, y)$. Every point moves parallel to the x-axis by a distance proportional to its height y; points on the x-axis do not move at all. So $\exp(s\mathbf{N})$ is a **shear parallel to the x-axis**, with the x-axis as its line of fixed points, sending the unit square to a parallelogram of the same area (the determinant is 1).



When do they commute?

Write $c = \cos\theta$ and $\sigma = \sin\theta$. Then

$$\exp(s\mathbf{N})\exp(\theta\mathbf{M}) = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}\begin{pmatrix} c & -\sigma \\ \sigma & c \end{pmatrix} = \begin{pmatrix} c + s\sigma & -\sigma + sc \\ \sigma & c \end{pmatrix},$$

$$\exp(\theta\mathbf{M})\exp(s\mathbf{N}) = \begin{pmatrix} c & -\sigma \\ \sigma & c \end{pmatrix}\begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} c & sc - \sigma \\ \sigma & s\sigma + c \end{pmatrix}.$$

Subtracting,

$$\exp(s\mathbf{N})\exp(\theta\mathbf{M}) - \exp(\theta\mathbf{M})\exp(s\mathbf{N}) = s\sin\theta\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

This is the zero matrix for every s if and only if $\sin\theta = 0$ (if $\sin\theta \neq 0$, take $s = 1$). So the two matrices commute for all s exactly when

$$\theta = n\pi, \qquad n \in \mathbb{Z}.$$

For these values $\exp(\theta\mathbf{M}) = (-1)^n\mathbf{I}$, which is either the identity or $-\mathbf{I}$.

Ask yourself Why should a rotation that commutes with this shear have to be a half-turn or nothing?

Let R be the rotation and S the shear (with $s \neq 0$), and suppose $SR = RS$. Take any point P on the x-axis, so that $S(P) = P$. Then

$$S\big(R(P)\big) = R\big(S(P)\big) = R(P),$$

so $R(P)$ is also a fixed point of the shear. But the only points the shear fixes are those on the x-axis. So the rotation must map the x-axis onto itself, and the only rotations about the origin that do this are through 0 (the identity) or π (the half-turn $(x, y) \mapsto (-x, -y)$). Conversely, these two commute with *every* linear map, since they are $\pm\mathbf{I}$: turning the plane through a half-turn and shearing it give the same result in either order. Any other rotation tilts the fixed line of the shear, so shearing before or after rotating gives different results.

Checks

- At $\theta = 0$ the rotation matrix is $\mathbf{I}$, and at $s = 0$ the shear is $\mathbf{I}$, as the series demands.
- Both determinants are 1 ($\cos^2\theta + \sin^2\theta$ and $1 \cdot 1$), as they should be for a rotation and a shear, which preserve area.
- Numerically, summing 80 terms of the series for $\theta = 1$ gives entries matching $\cos 1 \approx 0.5403$ and $\sin 1 \approx 0.8415$. With $\theta = 1$, $s = 1$ the commutator has diagonal entries $\pm\sin 1 \approx \pm 0.8415$, matching $s\sin\theta\,\mathrm{diag}(1, -1)$.

Common slips

- Assuming $\exp(\mathbf{A})\exp(\mathbf{B}) = \exp(\mathbf{A} + \mathbf{B})$ for matrices. That rule needs $\mathbf{AB} = \mathbf{BA}$, which is precisely what fails here unless $\sin\theta = 0$.
- Giving only $\theta = 0$ (or $\theta = 0, \pi$) instead of all integer multiples of π.
- Calling $\exp(s\mathbf{N})$ a stretch or an enlargement: it preserves area and fixes the x-axis pointwise, which is what makes it a shear.

Beyond STEP

The matrix exponential is the bridge between *Lie algebras* and *Lie groups*: $\mathbf{M}$ generates the group of rotations and $\mathbf{N}$ the group of shears, and $\exp(t\mathbf{A})$ solves the linear differential equation $\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{x} = \mathbf{A}\mathbf{x}$. The failure of $\exp(\mathbf{A})\exp(\mathbf{B}) = \exp(\mathbf{A} + \mathbf{B})$ for non-commuting matrices is measured by the commutator $\mathbf{AB} - \mathbf{BA}$, an idea you met in the last part. Strang's *Introduction to Linear Algebra* covers matrix exponentials and their use in differential equations.

WORKED SOLUTION

STEP III 1998, Question 6

Pure • Volumes of Polyhedra

STEP III 1998, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q6

The key idea

Put the cube on coordinate axes, so that its vertices are the points whose coordinates are all 0 or a. Distances then come from Pythagoras in three dimensions, and the awkward solids can be measured by cutting simple pieces off the cube. The regular tetrahedron is what is left of a cube once four corners are sliced off, and the regular octahedron is two square-based pyramids placed base to base.

Solution

Part (i)

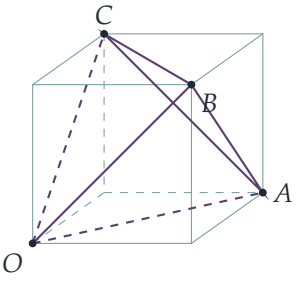
Take the cube $[0,a]^3$. Its vertices are the eight points (x,y,z) with each coordinate 0 or a, and two vertices are adjacent (joined by an edge) exactly when they differ in one coordinate.

Which four vertices can we choose? Call a vertex *even* or *odd* according to how many of its coordinates equal a. Moving along an edge changes one coordinate, so it always swaps even and odd. Suppose our four vertices included an even vertex P and an odd vertex Q. They aren't adjacent, so they differ in three coordinates: Q is the vertex opposite P. Every other vertex is adjacent to P or to Q (the three odd ones to P, the three even ones to Q), so no third vertex could be added. Hence the four vertices are all even,

$$O=(0,0,0),\quad A=(a,a,0),\quad B=(a,0,a),\quad C=(0,a,a),$$

or all odd (the reflection of this set in the plane $x=\frac{1}{2}a$, which gives a congruent solid).

Regularity. Any two of O,A,B,C differ in exactly two coordinates, each by a, so every one of the six distances is $\sqrt{a^2+a^2}=\sqrt{2}\,a$. Each of the four faces is therefore an equilateral triangle of side $\sqrt{2}\,a$, and $OABC$ is a regular tetrahedron. Each of its edges is a diagonal of one face of the cube.



Volume. Removing the tetrahedron from the cube leaves four corner pieces, one at each odd vertex. For instance, the corner at $(a,0,0)$ is the tetrahedron with that vertex and its three neighbours O, A, B. Its three edges at $(a,0,0)$ are mutually perpendicular and of length a, so it is a pyramid with a right-angled triangular base of area $\frac{1}{2}a^2$ and height a, of volume $\frac{1}{3}\cdot\frac{1}{2}a^2\cdot a=\frac{1}{6}a^3$. Hence

$$V_{\text{tet}}=a^3-4\cdot\tfrac{1}{6}a^3=\tfrac{1}{3}a^3.$$

For unit edges we need $\sqrt{2}\,a=1$, that is $a=1/\sqrt{2}$ and $a^3=1/(2\sqrt{2})$. So a regular tetrahedron of unit edge has volume

$$\frac{1}{3}\cdot\frac{1}{2\sqrt{2}}=\frac{1}{6\sqrt{2}}=\frac{\sqrt{2}}{12}.$$

Ask yourself Why do the four corner pieces and the tetrahedron fill the cube exactly, with no overlaps?

Each face of the tetrahedron, such as OAB, cuts off exactly one vertex of the cube, here $(a,0,0)$: the plane $x-y-z=0$ through O, A, B has $(a,0,0)$ strictly on one side and the other seven vertices on the other side or on it. So the cube splits into the four corner pieces, one beyond each face, and the tetrahedron in the middle.

Part (ii)

A regular octahedron of unit edge can be placed with its six vertices at

$$(\pm h,0,0),\quad (0,\pm h,0),\quad (0,0,\pm h),\qquad h=\tfrac{1}{\sqrt{2}},$$

where each vertex is joined to the four that are not opposite it. Two such vertices, for example $(h,0,0)$ and $(0,h,0)$, are at distance $\sqrt{2}\,h=1$. So all twelve edges have length 1, and the eight faces, one in each octant, are equilateral. The four vertices in the plane $z=0$ form a square of side 1. The octahedron is two square-based pyramids on this square, with apexes $(0,0,\pm h)$, each of height h. Hence

$$V_{\text{oct}}=2\cdot\tfrac{1}{3}\cdot 1^2\cdot\frac{1}{\sqrt{2}}=\frac{\sqrt{2}}{3}.$$

More generally, scaling lengths by e scales volumes by e^3, so a regular octahedron of edge e has volume $\frac{\sqrt{2}}{3}e^3$.

Part (iii)

It is now neater to centre the cube of side a at the origin, as $[-\frac{1}{2}a, \frac{1}{2}a]^3$. Its face centres are

$$(\pm\tfrac{1}{2}a, 0, 0), \quad (0, \pm\tfrac{1}{2}a, 0), \quad (0, 0, \pm\tfrac{1}{2}a).$$

This is exactly the configuration of part (ii) with $h = \frac{1}{2}a$. The centres of opposite faces are at distance a. The centres of adjacent faces are at distance $\sqrt{(\frac{1}{2}a)^2 + (\frac{1}{2}a)^2} = a/\sqrt{2}$. The convex solid with these six vertices has eight faces, one in each octant: for example, the face in the octant $x, y, z \geq 0$ is the triangle with vertices $(\frac{1}{2}a, 0, 0)$, $(0, \frac{1}{2}a, 0)$, $(0, 0, \frac{1}{2}a)$. Each face is equilateral with side $a/\sqrt{2}$, so the face centres form a regular octahedron of edge $e = a/\sqrt{2}$.

By part (ii), its volume is

$$\frac{\sqrt{2}}{3}\left(\frac{a}{\sqrt{2}}\right)^3 = \frac{\sqrt{2}}{3} \cdot \frac{a^3}{2\sqrt{2}} = \frac{a^3}{6}.$$

The tetrahedron of part (i), whose vertices are four of the vertices of the same cube, has volume $\frac{1}{3}a^3$. Since $\frac{1}{6}a^3 = \frac{1}{2} \cdot \frac{1}{3}a^3$, the octahedron has half the volume of the tetrahedron, as required.

Checks

- Using the base-times-height formula with base area $\frac{\sqrt{3}}{4}$ and height $\sqrt{2/3}$ gives the unit tetrahedron volume $\frac{1}{3} \cdot \frac{\sqrt{3}}{4} \cdot \sqrt{\frac{2}{3}} = \frac{\sqrt{2}}{12} \approx 0.11785$, which agrees with part (i).
- For equal edges, $V_{\text{oct}} = 4V_{\text{tet}}$, since $\frac{\sqrt{2}}{3} = 4 \cdot \frac{\sqrt{2}}{12}$. This matches the classical dissection of a tetrahedron of edge 2 into four tetrahedra of edge 1 (at its corners) and one octahedron of edge 1 (in the middle): measured in unit tetrahedra, $8 = 4 + 4$.
- Numerically, a determinant computation over all choices of four non-adjacent cube vertices gave volume $\frac{1}{3}$ for the unit cube, and a Monte Carlo estimate of the region $|x| + |y| + |z| \leq 1/\sqrt{2}$ gave 0.4715, against $\sqrt{2}/3 \approx 0.47140$.

Common slips

- Taking the cube's side to be 1 in part (i): the tetrahedron's edge is then $\sqrt{2}$, not 1, and its volume has to be rescaled by $(1/\sqrt{2})^3$.
- Subtracting the wrong corners: there are four corner pieces (one at each vertex not used), each of volume $\frac{1}{6}a^3$, not eight.
- In part (iii), using the cube's side as the octahedron's edge. Adjacent face centres are $a/\sqrt{2}$ apart.

Beyond STEP

The two tetrahedra inside a cube, and the octahedron of face centres, are the start of the story of the Platonic solids and their symmetry groups: the rotations of the cube that preserve one inscribed tetrahedron form a group of order 12, half of all 24 rotations of the cube. The cube and octahedron are *dual* (face centres of one give the vertices of the other), and volumes of such solids come out neatly from determinants, which measure how a linear map scales volume. Strang's *Introduction to Linear Algebra* treats determinants as volumes, and Armstrong's *Groups and Symmetry* studies the symmetry groups of these solids.

WORKED SOLUTION

STEP III 1998, Question 7

Pure • Curve Sketching and Integrals

STEP III 1998, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q7

The key idea

The first function is not a random warm-up: when we differentiate $\mathrm{b}(x)$ and set the derivative to zero, the equation that appears is exactly $\mathrm{f}(x/T) = 0$. So everything we learn about f (one positive root, lying between 2 and 3) tells us where b turns. The dependence of B on T comes from the natural substitution $x = Tu$, which strips T out of the integrand.

Solution

Sketching f and finding m

We have $\mathrm{f}(0) = 1 \cdot (-3) + 3 = 0$ and

$$\mathrm{f}'(s) = \mathrm{e}^s(s-3) + \mathrm{e}^s = \mathrm{e}^s(s-2).$$

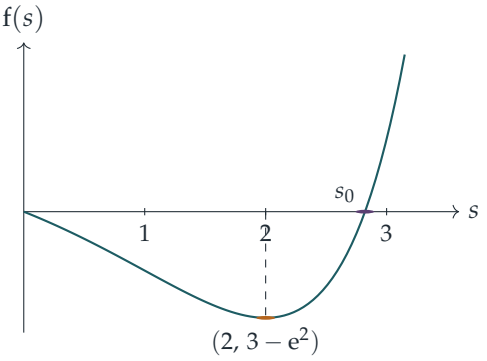
Since $\mathrm{e}^s > 0$, f is decreasing on $[0,2]$ and increasing on $[2,\infty)$, with a minimum $\mathrm{f}(2) = 3 - \mathrm{e}^2 \approx 3 - 7.29 = -4.29$. As $s \to \infty$, $\mathrm{e}^s(s-3) \to \infty$, so $\mathrm{f}(s) \to \infty$.

Because $\mathrm{f}(0) = 0$ and f is strictly decreasing on $[0,2]$, we have $\mathrm{f}(s) < 0$ for $0 < s \leq 2$. On $[2,\infty)$ it is strictly increasing from a negative value to $+\infty$, so it crosses zero exactly once there, at some $s_0 > 2$. Hence s_0 is the *only* positive root of f.

With $\mathrm{e} \approx 2.7$:

$$\mathrm{f}(1) = 3 - 2\mathrm{e} \approx -2.4, \qquad \mathrm{f}(2) \approx -4.29, \qquad \mathrm{f}(3) = 0 + 3 = 3.$$

So the smallest positive integer with $\mathrm{f}(m) > 0$ is $m = 3$, and $2 < s_0 < 3$ (in fact $s_0 \approx 2.82$).



The single turning point of b

Write $u = x/T$. By the quotient rule,

$$\mathrm{b}'(x) = \frac{3x^2(\mathrm{e}^{x/T}-1) - x^3\cdot\frac{1}{T}\mathrm{e}^{x/T}}{(\mathrm{e}^{x/T}-1)^2} = \frac{x^2\left(3\mathrm{e}^{u}-3-u\,\mathrm{e}^{u}\right)}{(\mathrm{e}^{u}-1)^2} = -\frac{x^2\,\mathrm{f}(u)}{(\mathrm{e}^{u}-1)^2}.$$

For $x > 0$ we have $x^2 > 0$ and $(\mathrm{e}^u - 1)^2 > 0$, so $\mathrm{b}'(x) = 0$ exactly when $\mathrm{f}(x/T) = 0$. By the first part this happens only at $x/T = s_0$, so b has a single turning point in $0 < x < \infty$, at

$$x = s_0 T \approx 2.82\,T.$$

Moreover $\mathrm{f}(u) < 0$ for $0 < u < s_0$ and $\mathrm{f}(u) > 0$ for $u > s_0$, so $\mathrm{b}' > 0$ before this point and $\mathrm{b}' < 0$ after it: it is a maximum.

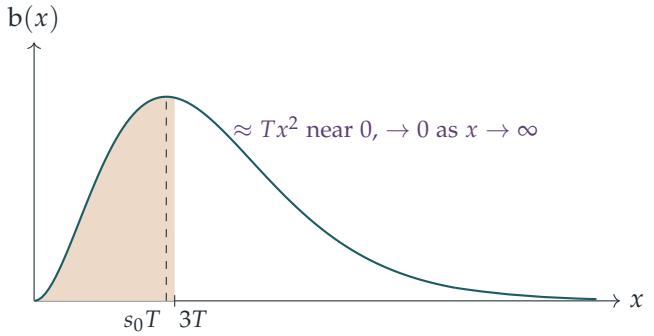
Ask yourself What does b look like near $x = 0$, where both top and bottom vanish?

For small x, $\mathrm{e}^{x/T} - 1 \approx x/T$, so

$$\mathrm{b}(x) \approx \frac{x^3}{x/T} = Tx^2,$$

and the graph leaves the origin like a parabola, with zero gradient. (So $\mathrm{b}(0) = 0$ is the natural value.) For large x the exponential dominates any power: $\mathrm{b}(x) \approx x^3\mathrm{e}^{-x/T} \to 0$, so the x-axis is an asymptote. Since $\mathrm{b}(x) > 0$ for all $x > 0$, the graph rises from the origin to a single hump at $x = s_0 T$ and then decays towards the axis.

Using the first part, the height of the hump is $\mathrm{b}(s_0 T) = T^3 s_0^3/(\mathrm{e}^{s_0} - 1) \approx 1.42\,T^3$, and $\mathrm{b}(3T) = 27T^3/(\mathrm{e}^3 - 1)$ is only slightly less.



(The sketch is drawn with $T = 1$; the shaded region is the one used in the final part.)

The form of B

Substitute $x = Tu$, so $\mathrm{d}x = T\,\mathrm{d}u$ and the limits $0, \infty$ are unchanged because $T > 0$:

$$B = \int_0^\infty \frac{T^3u^3}{\mathrm{e}^u - 1}\,T\,\mathrm{d}u = T^4 \int_0^\infty \frac{u^3}{\mathrm{e}^u - 1}\,\mathrm{d}u.$$

The last integral contains no T, so it is a constant K (finite, as we may assume). Hence $B = KT^4$, with $n = 4$.

Estimating K

With the same substitution and $m = 3$,

$$KT^4 = B \approx 2\int_0^{3T} \mathrm{b}(x)\,\mathrm{d}x = 2T^4 \int_0^3 \frac{u^3}{\mathrm{e}^u - 1}\,\mathrm{d}u, \qquad \text{so} \qquad K \approx 2\int_0^3 \frac{u^3}{\mathrm{e}^u - 1}\,\mathrm{d}u.$$

This is twice the shaded area in the sketch (with $T = 1$). Reading heights off the graph, using $\mathrm{e} \approx 2.7$:

$$\mathrm{b}(0) = 0, \quad \mathrm{b}(1) \approx \frac{1}{1.7} \approx 0.59, \quad \mathrm{b}(2) \approx \frac{8}{6.29} \approx 1.27, \quad \mathrm{b}(3) \approx \frac{27}{18.7} \approx 1.45.$$

The trapezium rule with strips of width 1 gives

$$\int_0^3 \approx \tfrac{1}{2}(0) + 0.59 + 1.27 + \tfrac{1}{2}(1.45) \approx 2.58,$$

so

$$K \approx 5.$$

An even rougher estimate treats the shaded region as a triangle of base 3 and height about 1.45, giving area about 2.2 and $K \approx 4.3$. Either way, K is roughly 4 to 5. (The triangle estimate is certainly low, because the curve lies above the straight line from the origin to $(3, \mathrm{b}(3))$.)

Checks

- A numerical bisection puts the root of f at $s_0 \approx 2.8214$, and a numerical scan of b′ with $T = 1.7$ finds exactly one sign change, at $x \approx 4.797 \approx 1.7 s_0$.
- Simpson's rule gives $B/T^4 \approx 6.4939$ for $T = 0.5$, 1 and 2.3: the same value each time, confirming $n = 4$. This matches the known exact value $K = \pi^4/15 \approx 6.494$.
- The integral $\int_0^3 u^3/(\mathrm{e}^u - 1)\,\mathrm{d}u \approx 2.552$, so the approximation the question allows gives $K \approx 5.10$. Our estimate of 5 is about right. The gap from 6.49 is the error in the question's approximation $B \approx 2\int_0^{3T}$, not in our arithmetic.

Common slips

- Forgetting that $\mathrm{f}(0) = 0$: $s = 0$ is a root of f, but it gives $x = 0$, which is outside $0 < x < \infty$, so there is still only one turning point.
- Changing variable in B but forgetting the factor T from $\mathrm{d}x = T\,\mathrm{d}u$, which gives $n = 3$.

Beyond STEP

The function b is Planck's law for black-body radiation, with x playing the role of frequency and T the temperature. The location of the peak, $x = s_0 T$, is Wien's displacement law, and $B \propto T^4$ is the Stefan–Boltzmann law. The exact value $K = \pi^4/15$ comes from expanding $1/(\mathrm{e}^u - 1) = \sum_{k\geq 1} \mathrm{e}^{-ku}$, which gives $K = 6\sum_{k\geq 1} 1/k^4 = 6\zeta(4)$. Justifying that swap of sum and integral is a typical first-year analysis question; see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 1998, Question 8

Pure • Vector Geometry of Spheres

STEP III 1998, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q8

The key idea

Everything here reduces to the same move: write "the point $\mathbf{r}$ lies on the sphere" as a scalar equation $|\mathbf{r}-\mathbf{c}|^2 = a^2$, expand it with the dot product, and read off what you need. For the line in (i) this gives a quadratic in λ, and the number of intersections is decided by its discriminant. For (ii) we need the distance from the centre to the plane, and for (iii) two sphere equations subtracted from each other give exactly the dot product that "perpendicular" asks about.

Solution

Part (i)

The point $\mathbf{b}+\lambda\mathbf{m}$ lies on the sphere exactly when $(\mathbf{b}+\lambda\mathbf{m})\cdot(\mathbf{b}+\lambda\mathbf{m}) = a^2$. Expanding, and using $\mathbf{m}\cdot\mathbf{m} = 1$ because $\mathbf{m}$ is a unit vector,

$$\lambda^2 + 2(\mathbf{b}\cdot\mathbf{m})\lambda + \mathbf{b}\cdot\mathbf{b} - a^2 = 0.$$

Each real root λ gives one point of intersection, and different values of λ give different points (since $\mathbf{m} \neq \mathbf{0}$). So there are two points of intersection exactly when this quadratic has two distinct real roots, that is, when its discriminant is positive:

$$4(\mathbf{b}\cdot\mathbf{m})^2 - 4(\mathbf{b}\cdot\mathbf{b} - a^2) > 0 \iff a^2 > \mathbf{b}\cdot\mathbf{b} - (\mathbf{b}\cdot\mathbf{m})^2,$$

which is the required condition.

There is precisely one point of intersection when the discriminant is zero:

$$a^2 = \mathbf{b}\cdot\mathbf{b} - (\mathbf{b}\cdot\mathbf{m})^2.$$

Then the repeated root is $\lambda = -\mathbf{b}\cdot\mathbf{m}$ (minus half the coefficient of λ), so the point is

$$\mathbf{p} = \mathbf{b} - (\mathbf{b}\cdot\mathbf{m})\,\mathbf{m}, \qquad \text{and} \qquad \mathbf{m}\cdot\mathbf{p} = \mathbf{b}\cdot\mathbf{m} - (\mathbf{b}\cdot\mathbf{m})(\mathbf{m}\cdot\mathbf{m}) = 0,$$

again using $\mathbf{m}\cdot\mathbf{m} = 1$. So a tangent line is perpendicular to the radius to its point of contact, as it should be.

Part (ii)

A plane perpendicular to $\mathbf{n}$ has equation $\mathbf{r}\cdot\mathbf{n} = k$ for a constant k, and since $\mathbf{n}$ is a unit vector its distance from the origin is $|k|$. Choosing the sign of $\mathbf{n}$ so that $k \geq 0$ (that is, $\mathbf{n}$ points from the origin towards the plane), the plane is

$$\mathbf{r}\cdot\mathbf{n} = l.$$

Ask yourself How far is the centre $\mathbf{d}$ from this plane?

Go from $\mathbf{d}$ along the normal: the point $\mathbf{d} + \mu\mathbf{n}$ lies in the plane when $\mathbf{d}\cdot\mathbf{n} + \mu = l$, so $\mu = l - \mathbf{d}\cdot\mathbf{n}$. This is the foot of the perpendicular from $\mathbf{d}$, so the distance from the centre to the plane is $|\mu| = |l - \mathbf{d}\cdot\mathbf{n}|$.

The plane touches the sphere exactly when this distance equals the radius (a smaller distance means the plane cuts the sphere in a circle, a larger one means they miss). So the condition is

$$|\mathbf{d}\cdot\mathbf{n} - l| = a, \qquad \text{equivalently} \qquad (\mathbf{d}\cdot\mathbf{n} - l)^2 = a^2,$$

i.e. $\mathbf{d}\cdot\mathbf{n} = l \pm a$. (If instead $\mathbf{n}$ points away from the plane, its equation is $\mathbf{r}\cdot\mathbf{n} = -l$ and the condition reads $|\mathbf{d}\cdot\mathbf{n} + l| = a$.)

Part (iii)

The first sphere has centre O and the second has centre D (position vector $\mathbf{d}$), both of radius a. A point $\mathbf{r}$ lies on both when

$$\mathbf{r}\cdot\mathbf{r} = a^2 \qquad \text{and} \qquad (\mathbf{r}-\mathbf{d})\cdot(\mathbf{r}-\mathbf{d}) = a^2.$$

Expanding the second and subtracting the first gives $-2\,\mathbf{r}\cdot\mathbf{d} + \mathbf{d}\cdot\mathbf{d} = 0$, so every point of intersection satisfies

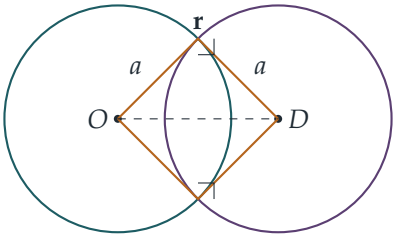
$$\mathbf{r}\cdot\mathbf{d} = \tfrac{1}{2}\,\mathbf{d}\cdot\mathbf{d}.$$

The two radii to the point $\mathbf{r}$ are the vectors $\mathbf{r}$ (from O) and $\mathbf{r}-\mathbf{d}$ (from D). They are perpendicular when

$$\mathbf{r}\cdot(\mathbf{r}-\mathbf{d}) = \mathbf{r}\cdot\mathbf{r} - \mathbf{r}\cdot\mathbf{d} = a^2 - \tfrac{1}{2}\,\mathbf{d}\cdot\mathbf{d}$$

is zero. The right-hand side does not depend on which intersection point we chose, so if $\mathbf{d}\cdot\mathbf{d} = 2a^2$ the radii are perpendicular at every point of intersection, which is what we had to show. (The argument also runs backwards: if the radii are perpendicular at even one common point, then $\mathbf{d}\cdot\mathbf{d} = 2a^2$.)

The spheres really do meet in this case: $|\mathbf{d}| = \sqrt{2}\,a$ lies strictly between 0 and $2a$, so they cross in a circle. The cross-section through O and D shows the right angles.



In the plane, two circles of radius a with perpendicular radii at a common point make OD the hypotenuse of an isosceles right-angled triangle, so $OD^2 = a^2 + a^2 = 2a^2$: the same answer by Pythagoras.

Checks

- In (i), take $\mathbf{b} = \mathbf{0}$: the line passes through the centre, the condition becomes $a^2 > 0$, and indeed such a line always meets the sphere twice.
- In (ii), with the sphere centred at the origin ($\mathbf{d} = \mathbf{0}$) the condition says $l = a$: a plane touches the sphere when it is at distance a from the centre.
- Numerically, with $a = 1$ and $|\mathbf{d}| = 1.2$ (not $\sqrt{2}$), an intersection point is $(0.6, 0.8, 0)$ and $\mathbf{r} \cdot (\mathbf{r} - \mathbf{d}) = 1 - 0.72 = 0.28 \neq 0$, matching $a^2 - \frac{1}{2}\mathbf{d} \cdot \mathbf{d}$. Random tests of all three parts agreed to within 10^{-13}.

Common slips

- Forgetting that $\mathbf{m} \cdot \mathbf{m} = 1$, and so carrying an unnecessary coefficient on λ^2.
- Writing the tangency condition in (ii) as $\mathbf{d} \cdot \mathbf{n} - l = a$ only: the centre can be on either side of the plane.

Beyond STEP

The quantity $\mathbf{b} \cdot \mathbf{b} - (\mathbf{b} \cdot \mathbf{m})^2$ is $|\mathbf{b} \times \mathbf{m}|^2$, the squared distance from the origin to the line, and $\mathbf{b} - (\mathbf{b} \cdot \mathbf{m})\mathbf{m}$ is the orthogonal projection of $\mathbf{b}$ away from $\mathbf{m}$, an idea at the heart of linear algebra. Spheres that cut at right angles are called orthogonal, and they are central to inversive geometry and to models of hyperbolic geometry. Strang, *Introduction to Linear Algebra*, covers projections; Needham, *Visual Complex Analysis*, shows orthogonal circles at work.

WORKED SOLUTION

STEP III 1998, Question 9

Mechanics • Moments of Inertia

STEP III 1998, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q9

The key idea

A solid cone is a stack of thin discs whose radius grows linearly with distance from the apex. If we know the moments of inertia of a thin disc (about its axis and about a diameter), we can add up the slices with a single integral. For the second axis we need the perpendicular axis theorem (to turn the disc's axial moment into a moment about a diameter) and the parallel axis theorem (to shift that axis up to the apex). The oscillation is then a compound pendulum, whose period needs exactly the moment of inertia about the apex.

Solution

Setting up the slices

Put the apex at the origin with the axis of the cone along the z-axis, so the base is the plane $z = h$. Let ρ be the (constant) density. The cross-section at height z is a disc of radius

$$r = \frac{az}{h},$$

by similar triangles. A slice of thickness δz is a thin disc of mass $\delta m = \rho\pi r^2\,\delta z = \rho\pi\dfrac{a^2z^2}{h^2}\,\delta z$. Integrating,

$$m = \int_0^h \rho\pi\frac{a^2z^2}{h^2}\,\mathrm{d}z = \tfrac{1}{3}\rho\pi a^2 h, \qquad \text{so} \qquad \rho\pi = \frac{3m}{a^2h}.$$

The moment of inertia about the axis

A uniform disc of mass M and radius r has moment of inertia $\frac{1}{2}Mr^2$ about its axis. (Split it into rings of radius s and mass $\frac{2Ms}{r^2}\,\delta s$; each contributes s^2 times its mass, and $\int_0^r \frac{2M}{r^2}s^3\,\mathrm{d}s = \frac{1}{2}Mr^2$.)

Every slice of the cone has its centre on the axis, so the slices' contributions simply add:

$$I_{\text{axis}} = \int_0^h \tfrac{1}{2}r^2\,\rho\pi r^2\,\mathrm{d}z = \tfrac{1}{2}\rho\pi\frac{a^4}{h^4}\int_0^h z^4\,\mathrm{d}z = \tfrac{1}{10}\rho\pi a^4 h$$
$$= \frac{1}{10}\cdot\frac{3m}{a^2h}\cdot a^4 h = \frac{3}{10}ma^2,$$

as required.

The moment of inertia about an axis through the apex

We use two theorems.

- **Perpendicular axis theorem.** For a lamina lying in a plane, the moment of inertia about an axis perpendicular to the plane equals the sum of the moments of inertia about any two perpendicular axes lying in the plane and meeting it at the same point.
- **Parallel axis theorem.** If a body of mass M has moment of inertia I_G about an axis through its centre of mass G, then its moment of inertia about a parallel axis a distance d away is $I_G + Md^2$.

For a thin disc, the two diameters are equivalent by symmetry, so the perpendicular axis theorem gives $2I_{\text{diam}} = \frac{1}{2}Mr^2$, that is, $I_{\text{diam}} = \frac{1}{4}Mr^2$.

Take the axis ℓ through the apex parallel to the base (say, along the x-axis). The slice at height z has its centre of mass on the z-axis, a distance z from ℓ, and its own diameter parallel to ℓ passes through that centre. By the parallel axis theorem, the slice contributes $\left(\frac{1}{4}r^2 + z^2\right)\delta m$. Hence

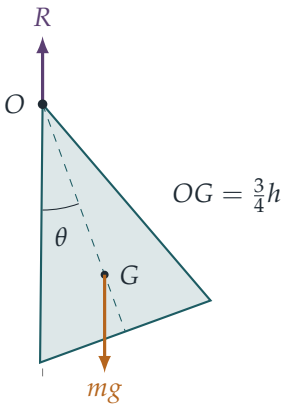
$$I_{\text{apex}} = \int_0^h \left(\frac{a^2z^2}{4h^2} + z^2\right)\rho\pi\frac{a^2z^2}{h^2}\,\mathrm{d}z = \rho\pi\frac{a^2}{h^2}\left(\frac{a^2}{4h^2}+1\right)\frac{h^5}{5}$$
$$= \frac{3m}{a^2h}\cdot\frac{a^2h^3}{5}\cdot\frac{a^2+4h^2}{4h^2} = \frac{3m}{20}\left(a^2+4h^2\right).$$

Ask yourself Why can't we apply the parallel axis theorem to the whole cone in one go, shifting from the centre of mass?

We could, but we don't yet know the moment of inertia about the parallel axis through G; that is what would need computing. Slicing avoids it, because each disc's moment about its own diameter is known, and each disc's centre is on the axis at a known distance z from ℓ. (As a by-product, $I_G = I_{\text{apex}} - m\left(\frac{3}{4}h\right)^2 = \frac{3m}{80}(4a^2+h^2)$.)

Small oscillations about the apex

Model the cone as a rigid body free to turn about a smooth horizontal axis through its apex, perpendicular to the plane of the motion. Such an axis is parallel to the base, so the moment of inertia about it is I_{apex}. Let θ be the angle between the cone's axis and the downward vertical.



The reaction at the axis has no moment about O, so the only torque is from the weight, acting at G, a distance $\frac{3}{4}h$ from O. Its moment about O is $mg \cdot \frac{3}{4}h\sin\theta$, tending to reduce θ. Taking moments about the axis,

$$I_{\text{apex}}\,\ddot{\theta} = -\tfrac{3}{4}mgh\sin\theta.$$

For small oscillations $\sin\theta \approx \theta$, so

$$\ddot{\theta} = -\omega^2\theta, \qquad \omega^2 = \frac{\frac{3}{4}mgh}{\frac{3m}{20}(a^2+4h^2)} = \frac{5gh}{4h^2+a^2}.$$

This is simple harmonic motion with period

$$\frac{2\pi}{\omega} = 2\pi\sqrt{\frac{4h^2+a^2}{5gh}},$$

as required.

Checks

- Units: $\frac{4h^2+a^2}{gh}$ has units of $\mathrm{m}^2/(\mathrm{m\,s^{-2}}\cdot\mathrm{m}) = \mathrm{s}^2$, as a period squared should.
- Thin cone ($a \to 0$): all the mass lies along the axis, with $\int z^2\,\mathrm{d}m = \frac{3}{5}mh^2$, matching $I_{\text{apex}} \to \frac{3}{5}mh^2$. The period becomes $2\pi\sqrt{4h/(5g)}$, the period of a simple pendulum of length $\frac{4}{5}h$, which is longer than $\frac{3}{4}h$, as it should be for a body whose mass is spread out along its length.
- Numerically, with $m = 1$, $a = 0.7$, $h = 2.3$, slicing with Simpson's rule and a Monte Carlo sample of points in the cone both gave $I_{\text{axis}} = 0.147$ and $I_{\text{apex}} \approx 3.2475$, matching the formulas; stepping $I\ddot{\theta} = -\frac{3}{4}mgh\sin\theta$ from a small amplitude with $g = 9.81$ gave a period of 2.7525 s, as the formula predicts.

Common slips

- Using $\frac{1}{2}r^2\,\delta m$ for each slice in the second part: about an axis parallel to the base, a disc turns about its diameter, contributing $\frac{1}{4}r^2\,\delta m$, not $\frac{1}{2}r^2\,\delta m$.
- Applying the parallel axis theorem from the apex rather than from a centre of mass: the theorem only holds when one of the two axes passes through the centre of mass.
- Treating the cone as a simple pendulum of length $\frac{3}{4}h$; the moment of inertia, not just the position of G, sets the period.

Beyond STEP

The moment of inertia about an arbitrary axis through a point is encoded in a single symmetric 3×3 matrix, the inertia tensor; the perpendicular and parallel axis theorems are special cases of how it behaves under rotation and translation. Its eigenvectors, the principal axes, explain why spinning objects are stable about some axes and wobble about others. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops rigid-body rotation and the compound pendulum at this level.

WORKED SOLUTION

STEP III 1998, Question 10

Mechanics • Oblique Collisions

STEP III 1998, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q10

The key idea

With equal masses, the mass cancels from both conservation laws, leaving two clean vector statements: the *sum* of the velocities is unchanged, and the *sum of the squared speeds* is unchanged. Comparing $|\mathbf{a}+\mathbf{b}|^2$ with $|\mathbf{a}|^2+|\mathbf{b}|^2$ isolates the scalar product $\mathbf{a}\cdot\mathbf{b}$, and that single quantity drives every part of the question. In the last part, momentum fixes the first ball's velocity once we know the second's, so the energy equation becomes a quadratic in one unknown speed, and a quadratic with a real root has a non-negative discriminant.

Solution

Model the balls as smooth spheres of equal mass m moving on the smooth table, so that during the collision the only horizontal forces are the ones the balls exert on each other.

Conservation laws and relative speed

Momentum and kinetic energy give

$$m\mathbf{v}_1 + m\mathbf{v}_2 = m\mathbf{v}_1' + m\mathbf{v}_2',$$
$$\tfrac{1}{2}m|\mathbf{v}_1|^2 + \tfrac{1}{2}m|\mathbf{v}_2|^2 = \tfrac{1}{2}m|\mathbf{v}_1'|^2 + \tfrac{1}{2}m|\mathbf{v}_2'|^2,$$

that is,

$$\mathbf{v}_1 + \mathbf{v}_2 = \mathbf{v}_1' + \mathbf{v}_2' \quad (1), \qquad |\mathbf{v}_1|^2 + |\mathbf{v}_2|^2 = |\mathbf{v}_1'|^2 + |\mathbf{v}_2'|^2 \quad (2).$$

Take the scalar product of each side of (1) with itself:

$$|\mathbf{v}_1|^2 + |\mathbf{v}_2|^2 + 2\,\mathbf{v}_1\cdot\mathbf{v}_2 = |\mathbf{v}_1'|^2 + |\mathbf{v}_2'|^2 + 2\,\mathbf{v}_1'\cdot\mathbf{v}_2'.$$

Subtracting (2) leaves

$$\mathbf{v}_1\cdot\mathbf{v}_2 = \mathbf{v}_1'\cdot\mathbf{v}_2'. \qquad (3)$$

Now the squared relative speed is

$$|\mathbf{v}_1 - \mathbf{v}_2|^2 = |\mathbf{v}_1|^2 + |\mathbf{v}_2|^2 - 2\,\mathbf{v}_1\cdot\mathbf{v}_2,$$

and by (2) and (3) each term on the right is the same after the collision as before. So $|\mathbf{v}_1 - \mathbf{v}_2|^2 = |\mathbf{v}_1' - \mathbf{v}_2'|^2$, and since speeds are non-negative, $|\mathbf{v}_1 - \mathbf{v}_2| = |\mathbf{v}_1' - \mathbf{v}_2'|$: the relative speed is conserved.

One ball initially at rest

Say $\mathbf{v}_2 = \mathbf{0}$. Then (3) gives

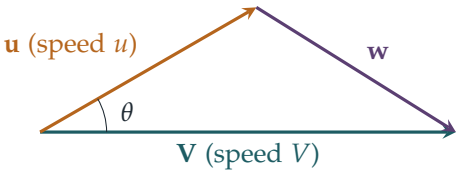
$$\mathbf{v}_1' \cdot \mathbf{v}_2' = \mathbf{v}_1 \cdot \mathbf{0} = 0.$$

Since both balls are moving afterwards, $\mathbf{v}_1' \neq \mathbf{0}$ and $\mathbf{v}_2' \neq \mathbf{0}$, so a zero scalar product means that the two final velocities are perpendicular. (Without that condition the scalar product could vanish only because one velocity is zero, as in a head-on collision, where the balls simply swap velocities.)

The collision that loses energy

Now (1) still holds but (2) doesn't. Call the ball at rest A and the moving ball B. Let B start with velocity $\mathbf{V}$, where $|\mathbf{V}| = V$, and finish with velocity $\mathbf{u}$ of speed u, at angle θ to $\mathbf{V}$. Let A finish with velocity $\mathbf{w}$.

Momentum gives $\mathbf{V} = \mathbf{u} + \mathbf{w}$, so the three velocities form a triangle:



Hence $\mathbf{w} = \mathbf{V} - \mathbf{u}$ and, since $\mathbf{V} \cdot \mathbf{u} = Vu\cos\theta$,

$$|\mathbf{w}|^2 = V^2 - 2Vu\cos\theta + u^2$$

(the cosine rule in the triangle). The final kinetic energy is a fraction $1 - k$ of the initial kinetic energy. Cancelling $\frac{1}{2}m$,

$$u^2 + |\mathbf{w}|^2 = (1-k)V^2 \quad \Longrightarrow \quad 2u^2 - 2Vu\cos\theta + V^2 = (1-k)V^2,$$

so the final speed u of the second ball satisfies

$$\boxed{2u^2 - 2Vu\cos\theta + kV^2 = 0.}$$

Ask yourself The question only asks us to show $k \leq \frac{1}{2}$. Where does an inequality come from, when all we have is an equation?

The speed u is a real number that actually occurs, so this quadratic in u *has* a real root. Its discriminant must therefore be non-negative:

$$4V^2\cos^2\theta - 8kV^2 \geq 0 \quad \Longrightarrow \quad k \leq \tfrac{1}{2}\cos^2\theta \leq \tfrac{1}{2},$$

where we divided by $8V^2 > 0$ (the second ball is moving) and used $\cos^2\theta \leq 1$. So $k \leq \frac{1}{2}$, as required.

Equality needs $\cos^2\theta = 1$ and a double root $u = \frac{1}{2}V\cos\theta$. As $u \geq 0$, this means $\theta = 0$ and $u = \frac{1}{2}V$, and then $\mathbf{w} = \frac{1}{2}\mathbf{V}$ too: the balls move off together at half speed, which is the perfectly inelastic head-on collision. At most half the kinetic energy can be lost.

Checks

- $k = 0$ gives $2u^2 = 2Vu\cos\theta$, so $u = V\cos\theta$ (or $u = 0$). Then $\mathbf{w} = \mathbf{V} - \mathbf{u}$ is perpendicular to $\mathbf{u}$, since $\mathbf{u}\cdot(\mathbf{V} - \mathbf{u}) = Vu\cos\theta - u^2 = 0$. This agrees with the second part.
- A head-on collision with coefficient of restitution e gives $k = \frac{1}{2}(1 - e^2)$, which lies between 0 and $\frac{1}{2}$. With $e = 0.3$ this is $k = 0.455$.
- Numerically, we simulated 10^5 random elastic collisions of equal smooth spheres. Momentum, kinetic energy, $\mathbf{v}_1\cdot\mathbf{v}_2$ and the relative speed all agreed to within 10^{-13}. In 2×10^5 random energy-losing outcomes the quadratic held to 10^{-14}, and the largest k found was 0.49999.

Common slips

- Treating "relative speed" as $\big||\mathbf{v}_1| - |\mathbf{v}_2|\big|$. It is the magnitude of the relative *velocity*, $|\mathbf{v}_1 - \mathbf{v}_2|$.
- In the second part, forgetting to use the fact that both balls are moving. A zero scalar product means perpendicular only if neither vector is zero.
- In the last part, using (2) or (3) again. They came from conservation of kinetic energy, which no longer holds. Only momentum survives.

Beyond STEP

The scalar-product argument is a frame-free version of a standard trick: in the zero-momentum frame, an elastic collision simply rotates the two velocities, keeping their sizes. The maximum loss of $\frac{1}{2}$ is the general fact that only the kinetic energy relative to the centre of mass can be lost, here $\frac{1}{2}\cdot\frac{1}{2}mV^2$. These ideas lead on to scattering theory and to the collisions of special relativity. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers collisions in the centre-of-mass frame carefully.

WORKED SOLUTION

STEP III 1998, Question 11

Mechanics • Large-Amplitude Pendulum

STEP III 1998, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q11

This question has a full guided treatment in *Beyond STEP*, Section 11.3, *The Real Pendulum*: https://beyondstep.keithmcnulty.org/ch11/real-pendulum.html

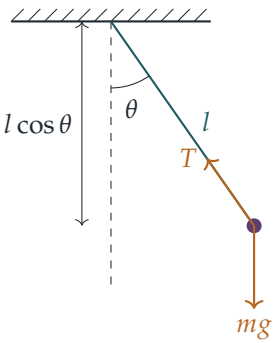
The key idea

Energy conservation gives the angular speed at every angle, and the period is four times the time for a quarter swing, $\int \mathrm{d}t = \int \mathrm{d}\theta/\dot{\theta}$. The half-angle identity $\cos\theta = 1 - 2\sin^2(\theta/2)$ turns the energy equation into the form of the integrand. The suggested substitution removes the awkward infinity at $\theta = \gamma$ and leaves an integrand of the form $(1-x)^{-1/2}$ with x small, which is exactly what the binomial expansion is for.

Solution

The energy equation

Model the pendulum as a particle of mass m on a light inextensible string of length l, swinging in a vertical plane with no air resistance. Measure θ from the downward vertical.



The bob is a depth $l\cos\theta$ below the pivot. The tension acts along the string, perpendicular to the bob's velocity (which is along the arc), so it does no work; only gravity does work, and mechanical energy is conserved. The bob's speed is $l\dot\theta$.

At the extreme angle $\theta=\gamma$ the bob is momentarily at rest. Between there and a general angle θ it falls a height $l\cos\theta - l\cos\gamma$, so

$$\tfrac12 ml^2\dot\theta^2 = mgl(\cos\theta-\cos\gamma) \quad\Longrightarrow\quad \tfrac12 l\left(\frac{\mathrm{d}\theta}{\mathrm{d}t}\right)^2 = g(\cos\theta-\cos\gamma),$$

which is the first result.

The period

Using $\cos\alpha = 1-2\sin^2(\alpha/2)$ on both cosines,

$$\cos\theta-\cos\gamma = 2\big(\sin^2(\gamma/2)-\sin^2(\theta/2)\big),$$

so the energy equation becomes

$$\left(\frac{\mathrm{d}\theta}{\mathrm{d}t}\right)^2 = \frac{4g}{l}\big(\sin^2(\gamma/2)-\sin^2(\theta/2)\big).$$

By symmetry (the motion is reversible, and the energy equation is unchanged under $\theta\mapsto-\theta$), the four quarter swings $0\to\gamma\to0\to-\gamma\to0$ take equal times. During the quarter swing from $\theta=0$ to $\theta=\gamma$, θ is increasing, so we take the positive square root:

$$\frac{\mathrm{d}t}{\mathrm{d}\theta} = \frac12\sqrt{\frac{l}{g}}\,\big(\sin^2(\gamma/2)-\sin^2(\theta/2)\big)^{-1/2}.$$

Integrating from 0 to γ and multiplying by 4,

$$P = 2\sqrt{\frac{l}{g}}\int_0^\gamma \big(\sin^2(\gamma/2)-\sin^2(\theta/2)\big)^{-1/2}\,\mathrm{d}\theta,$$

as required. (The integrand blows up at $\theta=\gamma$, where the bob is at rest, but the integral converges; the substitution below shows this explicitly.)

The small-amplitude approximation

Write $k=\sin(\gamma/2)$, with $0<\gamma<\pi$ so that $0<k<1$, and put $\sin(\theta/2)=k\sin\phi$. As θ runs from 0 to γ, $\sin\phi$ runs from 0 to 1, so ϕ runs from 0 to $\pi/2$. Differentiating,

$$\tfrac12\cos(\theta/2)\,\mathrm{d}\theta = k\cos\phi\,\mathrm{d}\phi.$$

Since $0\le\theta/2<\pi/2$ and $0\le\phi\le\pi/2$, both cosines are non-negative, so

$$\cos(\theta/2)=\sqrt{1-k^2\sin^2\phi}, \qquad \big(k^2-k^2\sin^2\phi\big)^{1/2} = k\cos\phi.$$

Ask yourself What happens to the troublesome factor $(\sin^2(\gamma/2) - \sin^2(\theta/2))^{-1/2}$ under this substitution?

It becomes $1/(k\cos\phi)$, and the $k\cos\phi$ in $\mathrm{d}\theta$ cancels it exactly:

$$\frac{\mathrm{d}\theta}{\left(k^2 - \sin^2(\theta/2)\right)^{1/2}} = \frac{1}{k\cos\phi}\cdot\frac{2k\cos\phi\,\mathrm{d}\phi}{\sqrt{1-k^2\sin^2\phi}} = \frac{2\,\mathrm{d}\phi}{\sqrt{1-k^2\sin^2\phi}}.$$

Hence

$$P = 4\sqrt{\frac{l}{g}}\int_0^{\pi/2}\left(1-k^2\sin^2\phi\right)^{-1/2}\mathrm{d}\phi,$$

an integral with a perfectly well-behaved integrand. (This much is exact.)

For small γ, $x = k^2\sin^2\phi$ satisfies $0 \le x \le k^2$, which is small, so the binomial expansion $(1-x)^{-1/2} = 1 + \frac{1}{2}x + \frac{3}{8}x^2 + \cdots$ gives

$$\left(1-k^2\sin^2\phi\right)^{-1/2} \approx 1 + \tfrac{1}{2}k^2\sin^2\phi,$$

with an error of order k^4. Since $\int_0^{\pi/2}\sin^2\phi\,\mathrm{d}\phi = \int_0^{\pi/2}\frac{1}{2}(1-\cos 2\phi)\,\mathrm{d}\phi = \frac{\pi}{4}$,

$$P \approx 4\sqrt{\frac{l}{g}}\left(\frac{\pi}{2} + \frac{k^2}{2}\cdot\frac{\pi}{4}\right) = 2\pi\sqrt{\frac{l}{g}}\left(1 + \frac{k^2}{4}\right).$$

Finally, for small γ, $k = \sin(\gamma/2) \approx \gamma/2$ (the error is of order γ^3), so $k^2 \approx \gamma^2/4$ up to terms of order γ^4, and

$$P \approx 2\pi\sqrt{\frac{l}{g}}\left(1 + \frac{\gamma^2}{16}\right),$$

as required. The leading term is the familiar small-oscillation period; the correction says that larger swings take slightly longer.

Checks

- As $\gamma \to 0$ we recover $2\pi\sqrt{l/g}$, the period from the linearised equation $\ddot{\theta} = -(g/l)\theta$.
- Dimensions: $\sqrt{l/g}$ is a time, and γ is dimensionless.
- Numerically, with $l/g = 1.7/9.81$ and $\gamma = 0.5$, stepping the equation $\ddot{\theta} = -(g/l)\sin\theta$ with a fine Runge–Kutta scheme gives $P = 2.65705$, matching the exact ϕ-integral to all digits shown; the approximation gives 2.65646, a relative error of about 2×10^{-4}. That error scales like γ^4 (it is about $0.0036\gamma^4$), confirming that the γ^2 term is right.

Common slips

- Forgetting the factor 4: the integral from 0 to γ is only a quarter of a period.

- Using $\cos(\theta/2) \approx 1$ in the substitution before expanding. That discards a $k^2 \sin^2\phi$ term of the same order as the correction you are trying to find; keep $\sqrt{1 - k^2 \sin^2\phi}$ exact until the binomial step.
- Stopping at $1 + k^2/4$ without converting $k = \sin(\gamma/2)$ into $\gamma/2$.

Beyond STEP

The exact integral $\int_0^{\pi/2} (1 - k^2 \sin^2\phi)^{-1/2}\, \mathrm{d}\phi$ is the *complete elliptic integral of the first kind*, $K(k)$, so the exact period is $4\sqrt{l/g}\, K(\sin(\gamma/2))$. Continuing the binomial series gives $1 + \frac{1}{16}\gamma^2 + \frac{11}{3072}\gamma^4 + \cdots$, and Gauss found that K can be computed astonishingly fast with the arithmetic–geometric mean. The fact that the period depends on amplitude is the simplest example of nonlinear oscillation. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats the pendulum and oscillations at first-year level.

WORKED SOLUTION

STEP III 1998, Question 12

Probability and Statistics • Network Reliability

STEP III 1998, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q12

The key idea

There are only six roads, so there are only $2^6 = 64$ patterns of open and blocked roads, and the cleanest route is to count the *bad* patterns: those in which the villages split into groups that can't reach each other. Sorting the bad patterns by how many roads are still open makes the count very short, because a network with four or more open roads on four villages can never be disconnected.

A warning before we start: the formula as printed in the paper has p^2 in front of the bracket, but the correct factor is p^3. The printed version can't be right, as we show in the Checks. We prove the corrected result

$$P = 1 - p^3(6p^3 - 12p^2 + 3p + 4).$$

Solution

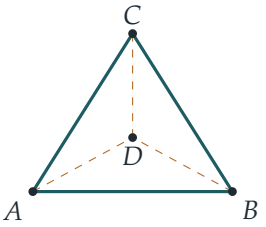
Write $q = 1 - p$ for the probability that a given road is open. Since the roads behave independently, any particular pattern with k roads open and $6 - k$ blocked has probability $q^k p^{6-k}$. Let D be the event that the network is *disconnected*, meaning some village can't reach some other village. Then $P = 1 - \mathbb{P}(D)$, and $\mathbb{P}(D)$ is the sum of $q^k p^{6-k}$ over all disconnected patterns. We count these according to k, the number of open roads.

Four or more open roads: never disconnected. If the network is disconnected, the villages split into at least two groups with no open road between groups. The only possible splits are $1 + 3$ (a lone village plus the other three) and $2 + 2$. All open roads lie inside the groups. In a $1 + 3$ split the group of three has only 3 roads; in a $2 + 2$ split each pair has only 1 road, so 2 in total. Either way at most 3 roads can be open. So every pattern with $k \geq 4$ open roads is connected.

Exactly three open roads.

Ask yourself With three open roads, which patterns fail to connect all four villages?

By the argument above, a disconnected pattern with three open roads must be a $1+3$ split in which all three roads among the group of three are open: a *triangle*, with the fourth village cut off. There are 4 choices of the isolated village, so 4 such patterns. (The other $\binom{6}{3} - 4 = 16$ patterns are the spanning trees of the tetrahedron, and they are connected.)



open roads AB, BC, CA (solid);
blocked roads to D (dashed):
D is cut off

Two or fewer open roads: always disconnected. To link four villages we need at least three roads (each road can join at most one new village to the group already reachable from A). So all $\binom{6}{2} = 15$ patterns with two open roads, all 6 with one, and the 1 with none are disconnected.

Adding up. Therefore

$$\mathbb{P}(D) = 4p^3q^3 + 15p^4q^2 + 6p^5q + p^6 = p^3\left(4q^3 + 15pq^2 + 6p^2q + p^3\right).$$

Now expand the bracket with $q = 1 - p$:

$$\begin{aligned} 4(1-p)^3 &= 4 - 12p + 12p^2 - 4p^3, \\ 15p(1-p)^2 &= 15p - 30p^2 + 15p^3, \\ 6p^2(1-p) &= 6p^2 - 6p^3, \\ p^3 &= p^3. \end{aligned}$$

Adding, the constant term is 4, the p terms give $-12p + 15p = 3p$, the p^2 terms give $12 - 30 + 6 = -12$, and the p^3 terms give $-4 + 15 - 6 + 1 = 6$. So

$$\mathbb{P}(D) = p^3\left(6p^3 - 12p^2 + 3p + 4\right), \qquad \text{hence} \qquad P = 1 - p^3\left(6p^3 - 12p^2 + 3p + 4\right).$$

Checks

- $p = 0$ (no road is ever blocked) gives $P = 1$, and $p = 1$ gives $P = 1 - (6 - 12 + 3 + 4) = 0$. Both are right, but both hold for the p^2 version too, so they can't tell the two apart.

- Small p separates them. Disconnection needs at least three blocked roads (the three at one village, say), so $\mathbb{P}(D)$ should be about $4p^3$ when p is small. Our formula gives $\mathbb{P}(D) \approx 4p^3$; the printed one would give $4p^2$, which is impossible.
- At $p = \frac{1}{3}$ our formula gives $P = 1 - \frac{1}{27} \cdot \frac{105}{27} = \frac{208}{243} \approx 0.856$. A computer enumeration of all 64 patterns at 21 values of p agrees exactly with the p^3 formula, and a simulation of 200 000 snowfalls at $p = 0.4$ gave 0.767 against the formula's 0.7655.

Common slips

- Counting only the cases with an isolated village and forgetting to check the $2+2$ splits. Here they add nothing new: a $2+2$ split leaves at most two open roads, so those patterns are already in the "two or fewer" count. You still need to say so.
- Adding up probabilities of overlapping events such as "A is cut off" and "B is cut off" without correcting for the overlap. Counting whole patterns avoids double counting.
- Trusting the printed p^2 and forcing the algebra to fit. If honest working disagrees with a "show that", test a special case.

Beyond STEP

This is a small case of *network reliability*: each edge of a graph fails independently, and we want the probability that the surviving graph is connected. The answer is always a polynomial in p whose coefficients count connected spanning subgraphs (here $1, 6, 15, 16$ for $6, 5, 4, 3$ open roads). For large graphs it is computationally hard to find, and closely tied to the Tutte polynomial and to percolation theory. Grimmett and Welsh, *Probability: An Introduction*, is a good place to build the probability side; Bollobás, *Modern Graph Theory*, covers random graphs and the Tutte polynomial.

WORKED SOLUTION

STEP III 1998, Question 13

Probability and Statistics • Maximum Likelihood Estimation

STEP III 1998, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q13

The key idea

Here the unknown is n, and it is an integer, so calculus is no use: we can't differentiate with respect to n. The standard tool for maximising a sequence is to look at the *ratio* of consecutive terms. The sequence rises while the ratio exceeds 1 and falls once it drops below 1, and a tie (two MLEs) happens exactly when the ratio equals 1 at some point.

Solution

The number of heads in n tosses of a fair coin is binomial, so

$$\mathrm{P}(k \text{ heads in } n \text{ tosses}) = \binom{n}{k}\frac{1}{2^n}, \qquad n \geq k,$$

and the probability is 0 if $n < k$. Throughout we take $k \geq 1$ (the friend has seen at least one head), so only $n \geq k$ is a candidate.

The general fair-coin calculation

Write $L(n) = \binom{n}{k}2^{-n}$ for $n \geq k$, thought of as a function of n with k fixed. For $n \geq k$,

$$\frac{L(n+1)}{L(n)} = \frac{\binom{n+1}{k}}{\binom{n}{k}} \cdot \frac{1}{2} = \frac{n+1}{n+1-k} \cdot \frac{1}{2},$$

using $\binom{n+1}{k} = \frac{n+1}{n+1-k}\binom{n}{k}$. Since $n+1-k > 0$, we can clear the denominator without reversing anything:

$$\frac{L(n+1)}{L(n)} \geq 1 \iff n+1 \geq 2(n+1-k) \iff n \leq 2k-1,$$

with equality exactly when $n = 2k-1$. So

$$L(k) < L(k+1) < \cdots < L(2k-1) = L(2k) > L(2k+1) > L(2k+2) > \cdots.$$

? **Ask yourself** How does the ratio tell us where the maximum is?

For $n < 2k-1$ the ratio is greater than 1, so each term is bigger than the one before: the sequence strictly increases up to $n = 2k-1$. At $n = 2k-1$ the ratio is exactly 1, so $L(2k) = L(2k-1)$. For $n > 2k-1$ the ratio is below 1 and the sequence strictly decreases from then on. Hence the largest value is attained at exactly two values of n.

One head

With $k = 1$, the two MLEs are $n = 2k-1 = 1$ and $n = 2k = 2$. Directly,

$$L(1) = \tfrac{1}{2}, \quad L(2) = 2 \cdot \tfrac{1}{4} = \tfrac{1}{2}, \quad L(3) = 3 \cdot \tfrac{1}{8} = \tfrac{3}{8},$$

and the argument above shows $L(n)$ keeps decreasing after $n = 2$. So there are two MLEs, 1 and 2, as required.

k heads

By the general calculation, the two MLEs of the number of tosses are

$$n = 2k-1 \quad \text{and} \quad n = 2k.$$

Biased coin

Now let $0 < p < 1$ and $q = 1-p$, so that

$$L(n) = \binom{n}{k} p^k q^{n-k}, \qquad n \geq k.$$

For $n \geq k$, the factor p^k cancels and one extra factor q appears:

$$\frac{L(n+1)}{L(n)} = \frac{(n+1)q}{n+1-k}.$$

Again $n+1-k > 0$, so

$$\frac{L(n+1)}{L(n)} \gtreqless 1 \iff (n+1)(1-p) \gtreqless n+1-k \iff k \gtreqless (n+1)p \iff n+1 \lesseqgtr \frac{k}{p}.$$

So L increases while $n+1 < k/p$, is level when $n+1 = k/p$, and decreases once $n+1 > k/p$. Note that $k/p > k$, since $p < 1$, so the turning point lies inside the range $n \geq k$.

- If k/p is *not* an integer, let $m = \lfloor k/p \rfloor$. Then $n+1 < k/p$ exactly when $n+1 \leq m$, so $L(k) < \cdots < L(m)$, and $L(m+1) < L(m)$ with the decrease continuing. There is a single MLE,
$$n = \left\lfloor \frac{k}{p} \right\rfloor.$$

- If $k/p = N$ is an integer, then L increases up to $L(N-1)$, has $L(N) = L(N-1)$, and then decreases. There are two MLEs,

$$n = \frac{k}{p} - 1 \quad \text{and} \quad n = \frac{k}{p}.$$

  Both are at least k: since N is an integer with $N > k$, we have $N - 1 \geq k$.

So the MLE is unique **if and only if k/p is not an integer**. For a fair coin $k/p = 2k$ is always an integer, which is why the earlier parts always gave two MLEs, $2k-1$ and $2k$.

Checks

- The answer makes sense: with k heads and head probability p, we'd expect about k/p tosses, and the MLE is k/p rounded down (or the tie either side when it is an integer).
- Take $p = \frac{1}{3}$, $k = 1$, so $k/p = 3$. Then $L(2) = 2 \cdot \frac{1}{3} \cdot \frac{2}{3} = \frac{4}{9}$ and $L(3) = 3 \cdot \frac{1}{3} \cdot \frac{4}{9} = \frac{4}{9}$, a tie, while $L(4) = 4 \cdot \frac{1}{3} \cdot \frac{8}{27} = \frac{32}{81} < \frac{4}{9}$.
- A computer computed $L(n)$ exactly with fractions and found the maximisers for $k = 1, \ldots, 39$ with a fair coin (always $\{2k-1, 2k\}$) and for 924 pairs (p, k) with p a fraction of denominator up to 12 and $k \leq 14$. Every case agreed with the rule above.

Common slips

- Differentiating with respect to n. Here n is an integer and $\binom{n}{k}$ isn't defined for other values of n without extra machinery; the ratio test is the right tool.
- Getting the inequality the wrong way round, or forgetting that equality of the ratio with 1 is what produces the second MLE.
- Answering $\lfloor k/p \rfloor$ in every case and missing the tie when k/p is an integer.

Beyond STEP

Maximum likelihood is the central method of estimation in statistics: choose the parameter value that makes the observed data most probable. Usually the parameter is continuous and we maximise the log-likelihood by calculus. Estimating an integer like n (the "number of trials" problem) is a classic case where it fails, and this question shows that the MLE need not even be unique. Blitzstein and Hwang, *Introduction to Probability*, is good on the binomial distribution, and Grimmett and Welsh, *Probability: An Introduction*, covers the background probability.

WORKED SOLUTION

STEP III 1998, Question 14

Probability and Statistics • Unbiased Estimators

STEP III 1998, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q14

This question has a full guided treatment in *Beyond STEP*, Section 13.3, *The German Tank Problem*: https://beyondstep.keithmcnulty.org/ch13/german-tanks.html

The key idea

Z_2 is a sum of independent pieces, so its mean and variance come straight from those of a single uniform variable. Z_1 depends on the maximum, and the natural way to handle a maximum is through its distribution function: the largest of the X_i is at most m exactly when *every* X_i is at most m. That turns into a product of probabilities, and differentiating gives the density.

Solution

The mean of Z_2

Each X_i has density $1/N$ on $[0, N]$, so

$$\mathrm{E}(X_i) = \frac{N}{2}, \qquad \mathrm{E}(X_i^2) = \frac{1}{N}\int_0^N x^2\,\mathrm{d}x = \frac{N^2}{3}, \qquad \mathrm{Var}(X_i) = \frac{N^2}{3} - \frac{N^2}{4} = \frac{N^2}{12}.$$

By linearity of expectation,

$$\mathrm{E}(Z_2) = \frac{2}{n}\sum_{i=1}^{n}\mathrm{E}(X_i) = \frac{2}{n}\cdot n\cdot\frac{N}{2} = N.$$

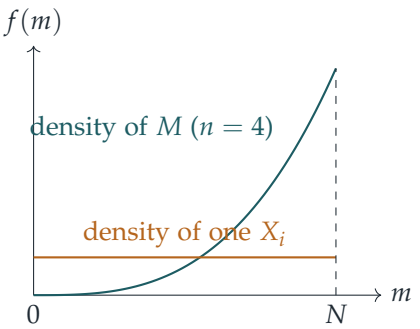
The distribution of the maximum

Let $M = \max\{X_1, \ldots, X_n\}$. For $0 \le m \le N$, the event $M \le m$ happens exactly when $X_i \le m$ for every i, and the X_i are independent, so

$$\mathrm{P}(M \le m) = \prod_{i=1}^{n}\mathrm{P}(X_i \le m) = \left(\frac{m}{N}\right)^n.$$

Differentiating, M has density

$$f(m) = \frac{n\,m^{n-1}}{N^n}, \qquad 0 \le m \le N.$$



The maximum piles up near N, which is why it needs only a small correction factor to become unbiased. Its moments are

$$\mathrm{E}(M) = \int_0^N m \cdot \frac{n\,m^{n-1}}{N^n}\,\mathrm{d}m = \frac{n}{N^n}\cdot\frac{N^{n+1}}{n+1} = \frac{n}{n+1}N,$$
$$\mathrm{E}(M^2) = \int_0^N m^2 \cdot \frac{n\,m^{n-1}}{N^n}\,\mathrm{d}m = \frac{n}{n+2}N^2.$$

The mean of Z_1

$$\mathrm{E}(Z_1) = \frac{n+1}{n}\,\mathrm{E}(M) = \frac{n+1}{n}\cdot\frac{n}{n+1}N = N.$$

So both estimators have mean N, as required.

The variances

For Z_1, first

$$\mathrm{Var}(M) = \frac{n}{n+2}N^2 - \frac{n^2}{(n+1)^2}N^2 = \frac{n\big[(n+1)^2 - n(n+2)\big]}{(n+2)(n+1)^2}N^2 = \frac{n}{(n+2)(n+1)^2}N^2,$$

since $(n+1)^2 - n(n+2) = 1$. Scaling by $\left(\frac{n+1}{n}\right)^2$,

$$\boxed{\mathrm{Var}(Z_1) = \frac{(n+1)^2}{n^2}\cdot\frac{n}{(n+2)(n+1)^2}N^2 = \frac{N^2}{n(n+2)}.}$$

For Z_2, the X_i are independent, so their variances add:

$$\boxed{\mathrm{Var}(Z_2) = \frac{4}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{4}{n^2}\cdot n\cdot\frac{N^2}{12} = \frac{N^2}{3n}.}$$

Ask yourself Which is smaller, and is it always the same one?

The ratio is

$$\frac{\mathrm{Var}(Z_1)}{\mathrm{Var}(Z_2)} = \frac{3n}{n(n+2)} = \frac{3}{n+2}.$$

This equals 1 when $n = 1$ (and then $Z_1 = Z_2 = 2X_1$, so the two estimators are actually the same), and is less than 1 for every $n \geq 2$.

Which to prefer

Both are unbiased, so we compare their spreads. We prefer Z_1:

- For $n \geq 2$ it has strictly smaller variance, and the advantage grows with n: $\mathrm{Var}(Z_1)$ falls like $1/n^2$, while $\mathrm{Var}(Z_2)$ falls only like $1/n$. With $n = 10$, for example, the standard deviation of Z_1 is about $0.09N$ against about $0.18N$ for Z_2.
- Z_1 is always at least the largest identification number seen, which N must be. Z_2 can come out smaller than a number we have actually observed (for example, with readings $1, 2, 30$ we get $Z_2 = 22$), giving an estimate we know to be impossible.

Checks

- When $n = 1$ both formulae give $N^2/3$, as they must since $Z_1 = Z_2 = 2X_1$, and $\mathrm{Var}(2X_1) = 4 \cdot N^2/12$.
- A simulation with $N = 10$ and 200 000 trials gave sample means within 0.01 of 10 for both estimators, and for $n = 4$ sample variances 4.16 and 8.33, against $\frac{100}{24} \approx 4.17$ and $\frac{100}{12} \approx 8.33$.

Common slips

- Trying to work with the density of the maximum directly. Go through $\mathrm{P}(M \leq m)$ first; the independence makes it a simple product.
- Forgetting to square the constant: $\mathrm{Var}(cY) = c^2 \mathrm{Var}(Y)$, so the factors are $\left(\frac{n+1}{n}\right)^2$ and $\frac{4}{n^2}$.

Beyond STEP

This is the classic "German tank problem", which Allied statisticians used in the Second World War to estimate German tank production from serial numbers, with remarkable accuracy. The maximum is a *sufficient statistic* here: once you know it, the rest of the sample tells you nothing more about N. The Lehmann–Scheffé theorem then shows that Z_1 has the smallest variance of all unbiased estimators. Sufficiency and

efficient estimators are treated in first-year statistics; Blitzstein and Hwang, *Introduction to Probability*, is a good place to start on order statistics.

Beyond STEP

STEP is the hardest mathematics examination most school students ever sit. It is also one of the best: its questions are short, unfamiliar and deep, and the best of them are small pieces of university mathematics in disguise.

This book takes the most interesting STEP questions, from the specimen papers that preceded the first examination in 1987 to the papers of 2025, and uses each one as a doorway. Every question is solved carefully, with the questions a good student asks along the way, and then the book steps through the door: where the idea comes from, what it becomes at university, and what to read next.

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- 13 chapters across pure mathematics, mechanics, probability and statistics
- 136 past STEP questions, each with a guided solution
- The university mathematics behind every question
- Follow-up exercises with answers, and further reading

1998 Worked Solutions. The whole book, every chapter and all the worked solutions are free to read, download and share at **beyondstep.keithmcnulty.org**

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